A

Project Report on

Investigations of Misalignment and Inner Race Defects of Rotor-Bearing Systems Based on Machine Learning Algorithms

Submitted

in partial fulfillment of the requirements for the degree of

Bachelor of Technology

in

Mechanical Engineering

by

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CERTIFICATE

This is to certify that the project work titled **"Investigations of Misalignment and Inner Race Defects Response of Rotor- Bearing Systems Based on Machine Learning Algorithm"** is the bonafide work submitted by the following students, to the Rajarambapu Institute of Technology, Rajaramnagar during the academic year 2023 - 2024, in partial fulfillment for the award of the degree of B. Tech in Mechanical Engineering under our supervision. The contents of this report, in full or in parts, have not been submitted to any other Institution or University for the award of any degree.

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DECLARATION

We declare that this report reflects our thoughts about the subject in our own words. We have sufficiently cited and referenced the sources, referred to or considered in this work. We have not plagiarized or submitted the same work for the award of any other degree. We also declare that we have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea/data/fact/source in my submission. We understand that any violation of the above will be cause for disciplinary action by the Institute.

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ABSTRACT

The vibration in the rotor bearing system is caused due to mainly the misalignment in the rotor but also it can happen due to these eccentricities in the journals, misalignment in the rotating shaft, defects in the bearing, misalignment of the shaft and use of bad belts. The misalignment in the rotor bearing system is one of the reasons which produce the vibration in the system which can degrade the life of the bearing. For example rotor-bearing system could involve a rotating machine such as a motor or turbine, where the rotor is improperly aligned, causing excessive vibrations in the system. Then it is necessary to predict that how much time turbine bearing can sustain under this vibration as the failure in the bearing lead to the downtime of the machine and the few hours of the downtime may cost huge economic loss. The machine learning techniques like Decision tree, Random forest and Support vector machine can be used to detect this bearing defect more accuratelyand in less time. In this experimental study we aim to develop one support vector machine model for the detection of bearing defects and misalignment by using the vibration signature as the input data. This model can detect the misalignment as well as bearing defect in very less time by using the vibration signature.

Content

LIST OF FIGURES

LIST OF TABLES

LIST OF TABLES

LIST OF SYMBOLS, ABBREVIATIONS AND NOMENCLATURE

Symbol	Details
RMS	Root Mean Square
a_{peak}	Peak Acceleration Amplitude
a_{RMS}	Root Mean Square Acceleration Amplitude
l	Width of Bearing
w	Defect Width
h	Defect Depth
BSF	Ball Spin Frequency
FTF	Fundamental Train Frequency
BPFI	Ball Pass Frequency Inner Race
BPFO	Ball Pass Frequency Outer Race
N	Rotating Frequency
D	Average Bearing Diameter
d	Roller Diameter
n	Number of Roller
Ө	Contact Angle

INTRODUCTION

Background

The rolling element bearings are widely used in rotating machines for a broad range of applications. They are used to give support for a rotating shaft. These are used in commercial applications such as automobiles, electric motors. They are also used in complex mechanisms such as gas turbines, rolling mills, dental drills, power transmissions, etc. So, rolling element bearings are considered vital components, because any defect in these components leads to malfunction of the complete system.

Therefore, detection of defects in the bearing is significant for the smooth functioning of various machines and to avoid undue stoppages that have a high economic impact. Different methods used for detection and diagnosis of bearing defects are given below,

- Vibration and acoustic measurements.
- Temperature measurement.
- Wear analysis.

Among these, the vibration measurement is the most common method used for fault detection [2]. To study the vibration response of faulty bearing, two methods have been used as given below. Run the bearing until failure and see the changes in the vibration response in intervals of time. Create the defects in the bearing by different techniques like acid etching, spark erosion, electrical discharge machining, scratching, or mechanical indentation, and then measure the vibration response and compare it with that of good bearings.

The very first method is time-consuming on the other hand testing bearing by creating the defects is a quick process. Therefore, creating artificial defects in the bearing components to study the vibration response using experimental analysis is based on condition monitoring of bearings. The results are then comparing with the vibration response of healthy bearing. We can also compare the experimental results with results from the industry. A case study is used for condition monitoring and inspection purpose.

Machine condition monitoring systems also provide information about the smooth running of roller element bearings, which is vital information for the proper functioning of the machine. To avoid sudden failures, it is important to monitor the condition of the bearings. Several techniques are currently available for the detection of defects in roller bearing elements. These include vibration and acoustics measurements like overall level, kurtosis, spectral analysis, high-frequency resonance, shock pulse, sound pressure, sound intensity, and spectrographic oil analysis (oil monitoring). Vibration and acoustic measurement systems are quite effective for detecting the defects in roller bearing elements. One approach is to detect roller bearing defects through measurements, and comparing the response with that of healthy bearings.

Problem statement

Vibrations induced due to Misalignment and Inner race Defects are one of the major causes to reduce the life of bearing. In this investigation, the experimental study of the rotor-bearing system with different Misalignments, Bearing Defects and speeds will be performed. In this context, the design and fabrication of an experimental test rig can be developed. The vibration amplitudes under operational fluctuations of the rotating machine will be obtained. Faults in the rotor-bearing system will be detected by using the machine learning algorithms.

Objective

The following objectives have been finalized for this dissertation work:

1. To analyses and highlight the limitations and challenges inherent in traditional diagnostic techniques when identifying and classifying defects, including misalignment, in rotor bearings.
2. To evaluate the accuracy, sensitivity, and reliability of novel diagnostic methods proposed for detecting and classifying diverse types of defects, specifically focusing on misalignment and bearing faults, in rotor bearings.
3. To offer insights and recommendations for the development of robust and efficient diagnostic systems tailored to address the unique challenges associated with rotor bearing defects and misalignment.
4. To contribute to the reduction of downtime and enhance overall machinery reliability and productivity in industries relying on rotor bearings, through the implementation of advanced diagnostic systems capable of timely and accurate detection and classification of defects, including misalignment, in rotor bearings.

Condition Based Monitoring

Condition Based Monitoring (CBM) is a maintenance program that recommends maintenance actions based on the information collected through monitoring of the condition. When evidence of abnormal behaviors of a physical asset occurs, the CBM attempts to avoid unnecessary maintenance by taking unnecessary actions. When the CBM program is properly established and effectively implemented, it can significantly reduce maintenance costs by reducing the number of unnecessary scheduled preventive maintenance operations.

CBM is usually an automatic process that uses instrumentation such as machine vibration analysis and thermal imaging equipment. In automatic monitoring mode, when any predefined condition limit is exceeded, the CBM system emits a signal or another form of output. This signal or output is then sent directly to a Computerized Maintenance Management System (CMMS) so that a work order is generated automatically. This is particularly suited to continuous process plants where plant failure and downtime are extremely costly, such as in the case of high- speed machinery like turbines.

Vibration analysis is the most commonly used technology to monitor the condition of bearings in rotating machinery. The presence of certain frequencies of vibrations can also be mapped or represented. The conditions thus indicate an impending defect of that system. Comparison of the vibration spectra of new equipment with the equipment that has been used will enable the maintenance personnel to decide the requirement of maintenance.

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Vibration-based monitoring has been in practice for the past 50 years or so. Many researchers have tried different approaches and different descriptors under different environmental conditions and tried to investigate the relationship between a tested bearing and changes in vibration responses under operating conditions. Displacement, velocity, acceleration, peak acceleration, root meansquare (RMS) of the overall acceleration, crest factor, frequency spectrum analysis, spike energy, shock e-pulse, acoustics emission, cepstrum, and statistical descriptors such as mean, standard deviation, skew, and kurtosis, have all been reported as useful in evaluating the condition of bearings. The following techniques are mostly used in the analysis of bearings. Vibration Measurement Techniques for condition monitoring can be classified into time-domain techniques, frequency domain techniques, and time-frequency techniques

Time Domain Techniques

Signal analysis in the time domain has been used to monitor the rotating machine conditions. It is difficult to analyses complex signals if frequently encountered in industrial equipment. Some of the time domain techniques for condition monitoring are root mean square (RMS), mean, peak value, crest factor, etc.

The Root mean square (RMS), measures the overall level of a discrete signal.

$$RMS = \sqrt{\frac{1}{N}\sum_{K=1}^{N} a_k^2} \quad \text{(1.1)}$$

Where N is the number of discrete points and a_k represents the signal from each sampled point. The RMS is a powerful tool to estimate the average power in system vibrations. A substantial amount of research has employed RMS to successfully identify the bearing defects using accelerometer and acoustic emission (AE) sensors.

The mean acceleration signal is the standard statistical mean value. Unlike RMS, the mean is reported only for rectified signals since for raw time signals, the mean remains close to zero. As the mean increases, it appears to deteriorate the bearing.

$$x' = \frac{1}{N}\sum_{i=1}^{N} x_i \quad \text{---------(1.2)}$$

Peak value can be measured in the time domain or frequency domain. The peak value is the maximum displacement (amplitude).

The crest factor is the ratio of peak acceleration and RMS value. This factor detects acceleration bursts, even if the RMS signal has not changed.

$$crest\ factor = \text{Peak/RMS} \quad \text{………..(1.3)}$$

However, the crest factor can be counterintuitive. At advanced stages of material wear, bearing damage propagates, RMS increases, and crest factor decreases. But the crest factor is unreliable to locate defects in rolling elements.

Frequency Domain Techniques

Another conventional approach is the processing of vibration signals in the frequencydomain. The basic indicator is the characteristic defect frequencies, which depend on the rotational speed and the location of the defect in a bearing. The presence of the defect frequencies in the director processed frequency spectrum is a powerful sign of the fault. The signature of the defected bearingis spread across a wide frequency band and It can be easily masked with low-frequency machinery vibrations and noise. The consecutive impact between the defect and rolling elements may excite theresonances of the structure and the resonant frequencies dominate the frequency spectrum. Therefore, the characteristic defect frequencies cannot be noticed easily because of their low amplitudes concerning resonant amplitudes. Frequency domain or spectrum analysis is applied more often to monitor the machine condition. However, a simple Fourier transformation at low frequency has limitations in bearing defect detection. Researchers have developed signal processing techniques to enhance the defect frequencies over mechanical noise. There are three standard techniques used for monitoring the machine condition [8], namely, signal average method, cepstrum analysis, and high-frequency resonance technique.

Signal Averaging improves statistical accuracy, but it does not improve the signal-to-noise ratio. A signal average is used to develop an average scheme. When the vibration signature is a generalized periodic function, then spectrum analysis detects peaks at its characteristic defect frequencies.

Cepstrum Analysis technique groups, the patterns of sidebands and harmonics from the spectrum into one signature. Cepstrum analysis detects periodicities in the spectrum while being insensitive to the transmission path. The technique works well if the defect frequency signal has not been convoluted with other signals.

The High-Frequency Resonance Technique (HFRT), also known as Demodulated Resonance Analysis or Envelope Power Spectrum, exploits the large amplitude of defect signals around a resonance. HFRT, while being highly sensitive to initial bearing damage, provides an envelope signal with a high signal-to-noise ratio, free of mechanical noise. The following steps are involved in this technique. The signal is band passed around a system resonant frequency. The canter frequency coincides with the largest amplitude resonant frequency.

A nonlinear rectification of the band passed signal demodulates the band passed signal resulting in the signature. The signal is passed through a low pass filter cancelling high-frequency components. The output signal of the low pass filter contains mainly the harmonics of the defect frequencies.

HFRT a highly sensitive method for easy identification of an impending bearing failure can be used to detect the location of even the smallest spall in a bearing (inner or outer race, or roller). HFRT is more popular for bearing fault detection. However, to find the bearing resonance frequency requires many computations and also several runs of impact tests to be performed. Hence extra instrumentation such as impact hammers or vibration exciters and their controllers are needed for HFRT [9, 10].

Types of Defects in Bearings

Bearing defects may be categorized as localized defects and distributed defects. Localized defects include cracks, pits, and spalls on the rolling surfaces. The mode of failure of rolling element bearings is spalling of the races or the rolling elements, caused when a fatigue crack begins below the surface of the metal and propagates towards the surface until a piece of metal breaks away to leave a small pit or spall. Fatigue failure may be due to overloading or shock loading. Pitting or crackoccurs due to excessive shock loading. Whenever a local defect on an element interacts with its mating element, sharp changes in the contact stresses at the interface generate a pulse of very short duration. This pulse produces both vibration and noise which can be monitored for the detection of the presence of a defect in the bearing. Distributed defects include surface roughness, waviness, misaligned races, and off-size rolling elements. If the surface feature of the wavelength of the bearing is less than the Hertzian contact width of the rolling element raceway contact, then it is known as roughness. If the feature of the wavelength of the bearing is longer than the Hertzian contact width of the rolling element raceway contact, then it is known as waviness, shown in Figure 1.1(a). Distributed defects are caused by a manufacturing error, improper installation, or abrasive wear. Due to distributed defects, the variation in contact force between rolling elements and raceways results in an increased vibration level. Some of the distributed defects are shown in Figure 1

Figure 1.1 Distributed Defects (Waviness at inner and outer race)

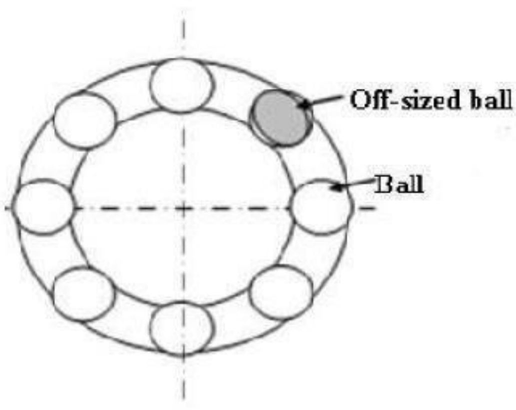


Figure 1.2 Distributed Defects (Off-sized ball)

Bearing Failure Modes

The normal service life of a rolling element bearing rotating under a load is determined by material fatigue and wear at the running surfaces. Premature bearing failures can be caused by several factors, the most common of which are fatigue, wear, plastic deformation, corrosion, brinelling, poor lubrication, faulty installation, and incorrect design [11].

Fatigue: Every rolling elements of the bearing is subjected to the load once in every revolution when it comes below the shaft. Hence, all the elements of the bearing undergo fatigue loading. Fatigue failure initiates with the formation of a minute crack which propagates due to stress concentration effects and leads to total failure.

Wear: Wear is another causes the bearing failure. It is caused mainly by dirt and foreign particles entering into the bearing due to inadequate sealing or contaminated lubricant. The abrasive natures of the foreign particles roughen the contact surfaces. Severe wear changes the profiles of raceways and the rolling elements, which may increase the bearing clearance. The rolling friction increases considerably and can lead to high levels of slip and skidding, finally leading to complete breakdown.

Plastic Deformation: Plastic deformation of the bearing contact surfaces can be the result of excessive loading while being stationary or undergoing small movements, resulting in the indentation on the raceways. During operation, the deformed bearing would rotate unevenly producing excessive vibrations, and would not be suitable for further service.

Corrosion: Corrosion damage occurs when water, acids, or other contaminants in the oil enter the bearing. This can be caused by damaged seals, use of acidic lubricants, or condensation which occurs when bearings operating at high temperatures are suddenly cooled in a humid environment. This results in rust on the running surfaces which produces unevenness and noisy operation. The rust particles thus present in the lubricant have an abrasive effect and generate wear. The rust pits also form the initiation sites for subsequent flaking and spalling.

Brinelling: Brinelling manifests itself as regularly spaced indentations distributed over the entire raceway circumference, resembling the Hertzian contact area. Three possible causes for brinelling are,

- Static overloading leading to plastic deformation of the raceways,
- Being subjected to vibration and shock loads and
- Formation of the loop for the passage of electric current.

All these causes will result in repetitive indentations on the raceways. In some instances, a large number of indentations may occur when the bearing is used occasionally. The bearing operation will be noisy and uneven in the presence of brinelling. Each indentation acts like a small fatigue site, producing sharp impacts. The continuous operation will lead to the development of spalling at the indentation sites and increase the distributed damage.

Lubrication: Inadequate lubrication is one of the common causes of premature bearing failure as it leads to skidding, slip, increased friction, heat generation, and sticking. At a highly stressed region of Hertzian contact, when there is an insufficient lubricant, the contact surfaces will weld together for an instant, it should tear apart as the rolling element moves on. The three critical points of bearing lubrication occur at the cage-roller interface, the roller race interface also the cage-race interface. Lubricant starvation and improper lubricant selection lead to severe consequences as the increased temperature can anneal the bearing elements reducing surface hardness and fatigue life as well as degrading the lubricant.

Faulty Installation: The faulty installation can occur due to excessive preloading in any radial or axial direction, misalignment, loose fits, and damage due to excessive force. Radial preloading results in noisy operation, and also increases the temperature differences between the inner and outer races. The increased temperature difference is likely to produce undesirable loading, causing higher contact pressures, which leads to premature fatigue, heavy wear of rolling elements, overheating, and the eventual seizure. Over preloading may also be due to the out-of-round shape of the shafts or housings, causing deformation of the inner or outer raceways. Excessive axial preload may be due to tightening forces produced during installation. Premature fatigue, heavy rolling element wear, and overheating will be the result. If improper mounting methods are used in the assembly of the bearing in the rotating machine, indentation or scoring damage of the raceways and rolling elements can be caused.Even if the damage is small, it can develop into premature spalling.

Incorrect Design: Incorrect design includes a poor selection of bearing type or size for the required operating conditions and inadequate support by the mating parts. Incorrect bearing selection may result in reduced fatigue life and may lead to premature failure. Inadequate support may give rise to excessive clearance of the mating parts of the bearing resulting in slippage of the inner race on the shaft. Fretting can occur if the slippage is slight and continuous, generating abrasive metal particles. The looseness will be self-perpetuating and will lead to increased friction and temperature resulting in catastrophic failure.

Chapter 2

LITERATURE REVIEW

Introduction

Many researchers have studied the correlation between bearing vibration and defects on the races of bearing, unbalance. Due to unbalance and bearing defects the vibrations are formed which leads to fatigue failure of the rotor-bearing system like a turbine, pump, etc. To get prior information about the failure and cause of failure of the rotor-bearing system which is known as condition monitoring which is used for predictive maintenance. Therefore, it is necessary to develop a system that can easily and accurately diagnose the fault. To use machine learning tools like a neural network, a support vector machine is important for fault diagnosis.

Literature Survey

1. **Wang et al. (2018)** proposed a fault diagnosis method based on multi-scale permutation entropy (MSPE) and support vector machine (SVM). The authors applied MSPE to analyze vibration signals andextract informative features at different scales. These features were then fed into an SVM classifier for fault diagnosis. The experimental results demonstrated the effectiveness of the proposed method in accurately identifying and classifying outer race defects.
2. **Li et al. (2019)** presented an intelligent fault diagnosis method for deep groove ball bearings based on time-frequency image analysis and extreme learning machine (ELM). The authors utilized the time-frequency image of vibration signals and employed ELM as the classification model.The study showed promising results in achieving high accuracy in diagnosing outer race defects.
3. **Li and Lim (2018)** proposed an enhanced fault diagnosis method for deep groove ball bearings based on kernel Fisher discriminant analysis (KFDA) and ensemble empirical mode decomposition (EEMD). The authors used KFDA to reduce the dimensionality of the extracted features obtained from EEMD. The reduced features were then employed for fault classification. The study demonstrated the effectiveness of the proposed method in achieving improved fault diagnosis performance for outer race defects.
4. **Yong Eun Lee, Bok-Kyung Kim, Jun-Hee Bae, Kyung Chun Kim (2021)**

Fault diagnosis For rotating machinery is important for reliability and safety, and shaft misalignment is one of the most \common causes of mechanical vibration or shaft failure. In this study, we detected defects caused by misalignment of the shaft axis in rotating machinery using an artificial intelligence machine learning technique for fault recognition.

5. Youcef Khodja, A., Guersi, N., Saadi, M. N., & Boutasseta, N. (2020).
In this paper, we propose a novel method for the classification of bearing faults using a convolutional neural network (CNN) and vibration spectrum imaging (VSI).The normalized amplitudes of the spectral content extracted from segmented temporal vibratory signals using a time-moving segmentation window are transformed into spectral images for training and testing of the CNN classifier.

6. N. Bachschmid P. Peennacchi and A. Vania ***et al*** identified the different types of faults in the rotor shaft and analyzed by frequency domain and also by numerical simulation.

7. Pennacchi ***et al.*** gave the use of a modal model for fault identification in the large rotating machinery.

8. Lees ***et al.*** estimated the rotor unbalance and misalignment which is mainly focused on the parallel and angular misalignment.

9. Kumbhar S. G., et al present a study on defects in taper roller bearing i.e rolling element bearingby using vibration signature analysis. It further explains the types of defects and causes of defects. They carried out experimental as well as a theoretical study on defected bearings.

10. Desavale R. G., et al has presented an effective empirical model for fault diagnosis of roller bearing by using the Buckingham p-theorem using FLT0 system has been derived.

11. Desavale R. G., et al presented an experimental and numerical study on spherical roller bearingby using the multivariable regression analysis method.

12. Patel V. N., et al have explained the nature of vibrations generated by rolling element bearing having multiple localized defects on bearing races. The study was carried out in Time and Frequency Domain. They also performed the experimental study on vibration behaviors of deep groove ball bearings under dynamic radial load for local defects.

13. Saruhan H., et al explained Vibration Analysis of Rolling Element Bearings Defects Sham Kulkarni and Wadkar S.B. performed Experimental Investigation for Distributed Defects in Ball Bearing using Vibration Signature Analysis.

14. Shrikant S. N., et al. [10] performed an Experimental Investigation of Rolling Element Bearing withand without Unbalanced Rotor by using an FFT analyzer.

15. Sakshi Kokil, et al. [11] performed an experimental study on the detection of a fault in rolling element bearing by using condition monitoring methodology. It also explains the numerical methodology for the same.

16.H. Arslan and N. Aktürk et. Al [12] carried out an Investigation of vibration in Rolling Element bearings caused due to local defects on different parts of bearing. They experimented on angular contact ball bearing with and without defects to obtain some mathematical relations.

17. Zeki Kiral and Hira Karagulle et al [13] performed a Vibration analysis of rolling element bearings with various defects under the action of an unbalanced force created during the working of the system. They performed the finite element vibration analysis on roller element bearings for theoretical results.

18. Desavale et al [14] presented the MVRA and ANN model to describe the vibration characteristics of rolling element bearings. The vibration signals collected from rolling element bearings were used to demonstrate the proposed EDBM method compared with the MVRA. The results showed the effectiveness of the EDBM method in extracting fault characteristics and diagnosing faults of rolling element bearings. Good correlations between EDBM, MVRA, and ANN have been observed.

19.J.S.Rapur and Rajiv Tiwari *et al* [15] mechanically developed faults and operationally developed faults have been extensively diagnosed with help of artificial neural network (ANN) and support vector machine (SVM).

20. Purushottam Gangsar *et al* [16] had done a comparative analysis of the time, frequency, and time-frequency domain-based features of vibration and current signal for identifying various faults in inductive motors using SVM.

21. Purushottam Gangsar and Rajiv Tiwari *et al* [17] had done the developed a new support vector machine (SVM) based fault diagnosis methodology for the inductive motor in the practical situation of limited data or information case. Vibration as well as current signals have been studied.

22.G.N.D.S. Sudhakar and A.S. Shekhar *et al* [18] had compared the equivalent load minimization method, frequency minimization method and vibration minimization method to identify unbalance in the rotor-bearing system.

Gaps and scope of present work

In various applications Misalignment in shaft of the rotor in the rotor is the main cause of the vibration. This can also induce defects in bearing and which finely shut down the machine. To avoid this, it is becoming necessary to detect unbalance in the rotating system. The unbalance hasvibrations of the frequency 1X of rotating frequency and the rotating frequency. Also, there is a needto detect the type of defects in the bearing which causes the vibration at defect frequency. From the literature, it is noticed that very few works have been done on the fault diagnosis of the rotor- bearing system. Hence the present experimental investigation will be carried out to fabricate the setup for detection of rotor unbalance and bearing defect of different sizes on the outer race.

Closure

This chapter gives detailed information about the work done in the field of fault diagnosis of the rotor-bearing system, various models of fault diagnosis. Also, it deals with previous research work in the fault diagnosis using the vibration signature.

Chapter 3

VIBRATION THEORY

Introduction

Vibration is the motion of a particle or a body or a system of connected bodies displaced from a position of equilibrium. Most vibrations are undesirable in rotating machines and structures, they produce increased stresses, energy losses and causes wear, increase bearing loads, induce fatigue. In this chapter, the basic concept, industrial accelerometer, and causes of vibrations are discussed. Vibration is simply the oscillation about a reference point (i.e., a shaft vibrates relative to the casing of a piece of machinery and a bearing vibrates relative to a bearing housing.) Vibration exists when a system responds to some internal or external excitation. The amplitude of vibration depends on the magnitude of the excitation force, the mass and stiffness of the system, and its damping. The vibration occurs because we can't build a perfect piece of machinery or install it perfectly. If we could build a perfect piece of machinery, the Centre of the mass of the rotating element would be located exactly at its center of gravity. When the center of mass and center of gravity does not coincide then there is some degree of unbalance. Other sources of vibration are machine tolerances, machine structure, bearing

design, loading and lubrication, machine mounting and rolling and rubbing between moving parts. The analysis of vibration requires an understanding of the terminology used to describe the components of vibration [5].

Vibration is an effect caused by machine condition. Vibration Data has high information content. Vibration measurements contain a lot of useful information that will help to determine the health of the machine for example:

- It provides information for safe machine operation.
- It can detect the condition of a machine change.
- It can be used to diagnose the cause of change.
- It can be used to classify the condition of a machine.

Vibration measurement is normally a non-intrusive measurement procedure; it can be carried out with the machine running in its normal operating condition.

Vibration Analysis – Basic Parameters

Basic parameters that are used in the theory of vibration are described below. Frequency- The cyclic movement in a given unit of time. The units of frequency are Hz or RPM.

RPM = revolutions or cycles per minute

Hertz (Hz) = revolutions or cycles per second.

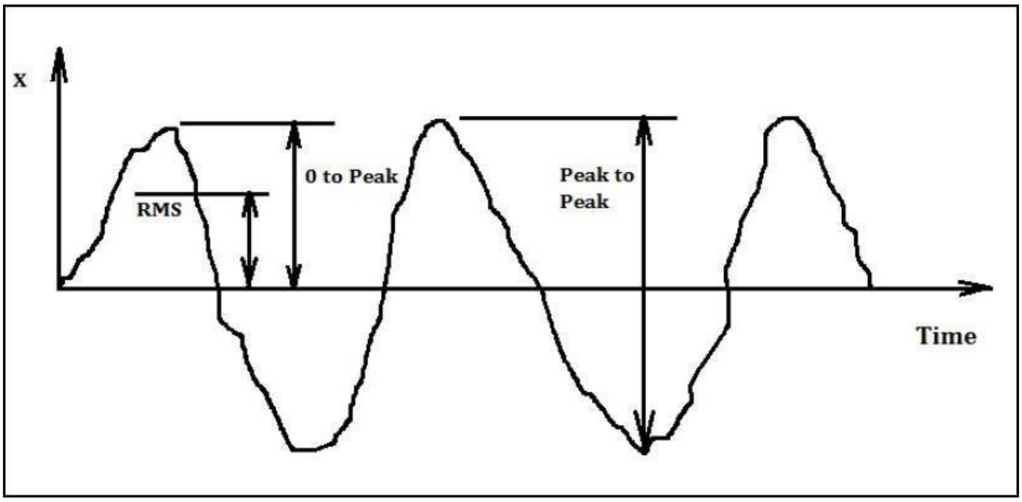


Figure 3.1 Time V/s Displacement graph

Amplitude: The magnitude of the dynamic motion of vibration. Amplitude is typically expressed in terms of either Peak to Peak or 0 to Peak or RMS (Root Mean Square).

Fundamental Frequency – Fundamental frequency is the primary rotating speed of the machine or shaft being monitored and is usually referred to as the running speed of the machine. The fundamental frequency is important because many machinery faults such as misalignment or unbalance occur at some multiple of the fundamental frequency, for example, misalignment at 1 x fundamental frequency.

Harmonics - These are the vibration signals having frequencies that are an exact multiple of the fundamental frequency (i.e., 1 X , 2 X , 1 x BPFI, 2 X BPFI, etc).

Displacement - Displacement is the actual physical movement of a vibrating surface. Displacement is usually expressed in microns. When measuring displacement, we are interested in the Peak displacement which is the total distance from the upper limit to the lower limits of travel. Also takes from 0 to peak displacement.

Velocity- Velocity is the speed at which displacement occurs. Also, velocity is the rate of change in the relative position. Velocity is usually measured in mm/sec.

Acceleration - Acceleration is the rate of change of velocity. This we can simply define as the change of velocity in a period of time or change in the rate of velocity. Acceleration is usually measured in g's of gravitational force.

Vibration frequency spectrum - Machinery vibration consists of various frequency components as illustrated below. The amplitude of each frequency component indicates the condition of a particular rotating element within the machine.

Simply stated, vibration frequency spectrum converts vibration signal into true amplitude representation of the individual frequency components. Since most machinery faults are displayed at or near frequency components associated with running speed, the ability to display and analyse the spectrum as components of frequency is extremely important. The advantage of frequency spectrum analysis is the ability to normalize each
vibration component so that the complex machine spectrum can be divided into discrete components. This ability of analysis simplifies the mechanical degradation within the machine.

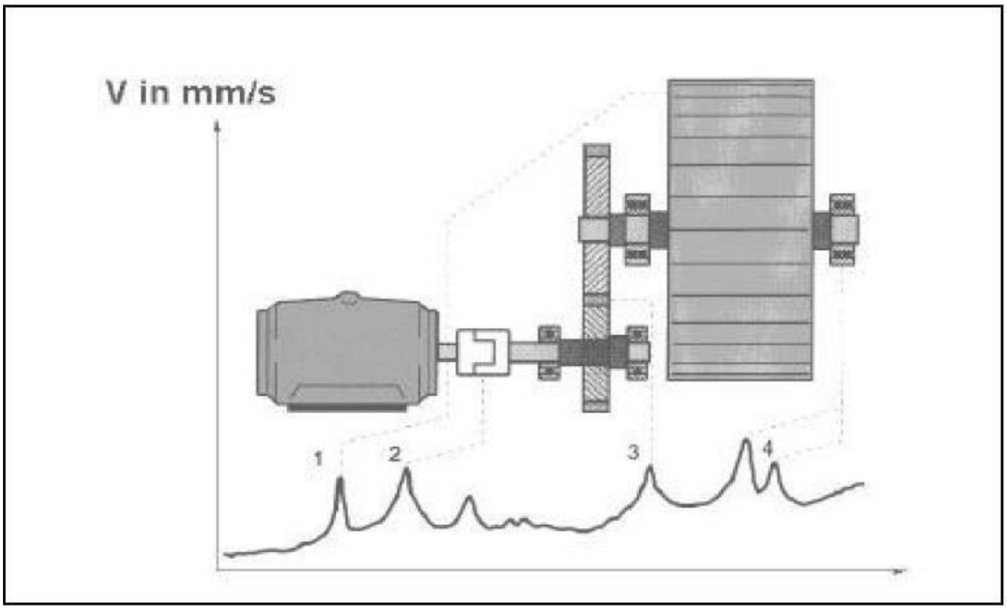


Figure 3.2 Frequency spectrum for machine

FFT (Fast Fourier Transform*)* - FFT is the most used tool in the analysis of spectral data with respectto vibration analysis of machine components. Fourier transform is a mathematical operation that decomposes the time-domain function into its frequency domain components.[3]

Signal Processing - The raw data collected by the transducers must be enhanced to provide useful information. For vibration data, this raw data must be "conditioned" to prevent errors. Such conditioning includes-

- Amplifying to enhance the resolution of low energy signals.
- Data averaging to remove spurious data and conversion to frequency domain [3]
- Filtering to remove unwanted or spurious signals.

Industrial Accelerometer

Accelerometers are the most widely used transducer in the vibration monitoring program. A typical accelerometer contains a piezoelectric crystal element which is pre-loaded by a mass of some type and the entire assembly is enclosed in a rugged protective housing. The piezoelectric crystal produces electrical output when it is physically stressed by either pressure or tension effect. The variable vibration force exerted by the mass on the crystal produces an electrical output proportional to the acceleration.

Accelerometers have a broad frequency range typically from 2 Hz to 10 KHz the accelerometer is also easily mounted, either with a stud, a magnet, and adhesive or by handholding the device onto the machinery surface. Accelerometers also have good temperature and environmental responses and are usually of rugged construction. The position and how data are collected is very important to a successful vibration monitoring program. To properly diagnose a fault, data must be collected in the right plane and must be repeatable. Some faults show the high amplitude in the radial direction and some in the axial direction. Measure location should usually be on exposed parts of the machine that are normally accessible and that reflect the vibration of the bearing housing. Vertical and horizontal mounting directions are the most usual transducer locations for horizontally mounted machines; any angular position is acceptable provided that the location reasonably represents the dynamic forces present in the machine. For vertically mounted machines the location giving the maximum vibration reading should be used as a future monitoring reference point. The data collection points should be marked to ensure that data is collected at the same point every time. When analyzing a machine for changes, the analysis can be inaccurate if data is not collected at the same point each time. Measurements should be carried out when the rotor bearing reached its normal steady-state operating temperature, speed, load, voltage, and pressure. Where machine speeds vary measurement should be taken at all conditions at which the machine operates for a prolonged period.

Vibration analysis –Fault Detection

Many causes create vibration in the machine. Some of them are discussed below.

Unbalance- Vibration caused by unbalancing occurs at a frequency equal to1 RPM of the unbalanced part, and the amplitude of vibration is proportional to the amount of unbalance present. Normally, the largest amplitude will be measured in the radial (vertical or horizontal) directions.

Misalignment- Generally, misalignment can exist between shafts that are connected with a coupling, gearbox, or other intermediate drives. Three types of misalignment are:

- Angular – where the centerline of the two shafts meets at an angle.
- Offset – where the shaft centerlines are displaced from one another.
- A combination of angular and offset misalignment.

The misalignment conditions produce vibration at the fundamental (1 x RPM) frequency components since they create an unbalanced condition in the machine. Also, mechanical looseness is caused by loose rotating components and loose machine foundations. Mechanical looseness causes vibration at a frequency of twice the rotating speed (2 x RPM) and higher orders of the loose machine part. In most cases, vibration at fundamental (1 x RPM) frequency will also be produced.

Bearing Problems - One of the results of damage to rolling element bearings that the natural frequencies of the bearing component are excited. The resonant vibration or ringing occurs at high frequencies. This vibration is most effectively measured at a level of acceleration. Measuredvibration gives an effective indication of the condition of rolling element bearings. Rotational frequencies related to the motion of the rolling elements, cage, and races are also produced by mechanical degradation of the bearing. These frequencies are dependent on bearing geometry and shaft speed and can be found typically, in the 3–10 x RPM range. When a bearing defect exists in a rolling element bearing the vibration signature will show high-frequency vibration generated each time a damaged roller or damaged race makes contact. These repetition rates are known as the naturalbearing defect frequencies. Each rolling element bearing has its defect frequency. These are given as

- Rolling element passing defect Outer race Frequency.
- Rolling element Passing defect Inner race Frequency.
- Rolling element Passing Frequency.

Shock pulses occur during bearing operation when the rolling element passes over an irregularity in the surface of the bearing race. Of course, there is no such thing as a perfectly smooth surface in real life, and so even new bearings emit a signal of weak shock pulses in rapid succession. This level rises when the lubrication film between rolling elements and races becomes depleted. A defect (pit or crack) on the surface of a rolling element or race produces a strong shock pulse with up to 1000 times the intensity of the initial level. The irregular peak shows a clear indication of bearing damage.

Closure

In this chapter, the various causes of bearing vibrations are discussed. The selection of vibration pickup for vibration measured is presented. The main important vibration parameters are misalignments and unbalance are presented. According to Machinery Classification ISO 2372, for high-frequency vibration (above about 1000Hz) it is necessary to measure acceleration because displacement and velocity drop off rapidly. For rotating machines operating at frequencies less than 50 Hz, the stiffness of the bearing assembly plays an important role in dynamic response. Hence, in such cases, vibration displacement may be appropriate. Since in the present project work the attention is focused on bearing which runs at less than 1440 rpm (24 Hz), the acceleration of vibrations may be considered for the studies. That is why; attempts are initiated in this chapter to perform the studies using amplitude of vibrations.

Chapter 4

Algorithms Information & Key Features

Decision Trees:

A model used for the classification.

A decision tree is composed of the following components:

- **Internal Nodes :** Each internal nodes represents the features in the dataset and in the corresponds to a decision based on the features.
- **Branches :** Arising form the internal node, branches represents the possible outcomes (or decision) based on the feature value.
- **Leaf Nodes :** These nodes represents the final outcomes of the decision.
- **For Fin :** The model predict the fin correctly for the 100% of the time.1.000 in the first row first column.
- **For Flap:** The model predict the flap correctly for 88.9% of the time, 0.889 for the second row second column but misclassifies the Flap as Fin 11.1% of the time i.e. 0.111 in the second row frist column.
- **For Jet :** The model predicts jet correctly 100% i.e. 1.00 in the third row, third column.
- **Process :** The algorithm recursively splits the data based on the feature that best separate the classes. This continues until pure nodes are reached .Decision trees are prone to overfitting, especially with deep trees and can be sensitive to small changes in the data.

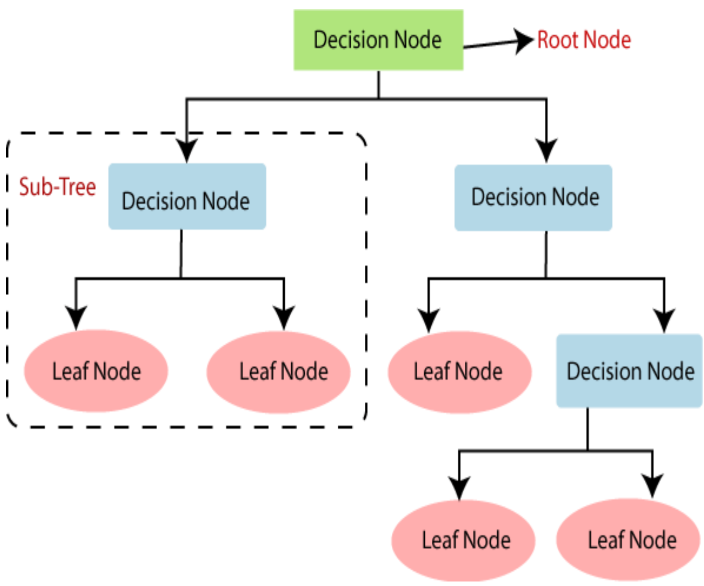


Figure 4.1 Decision Trees

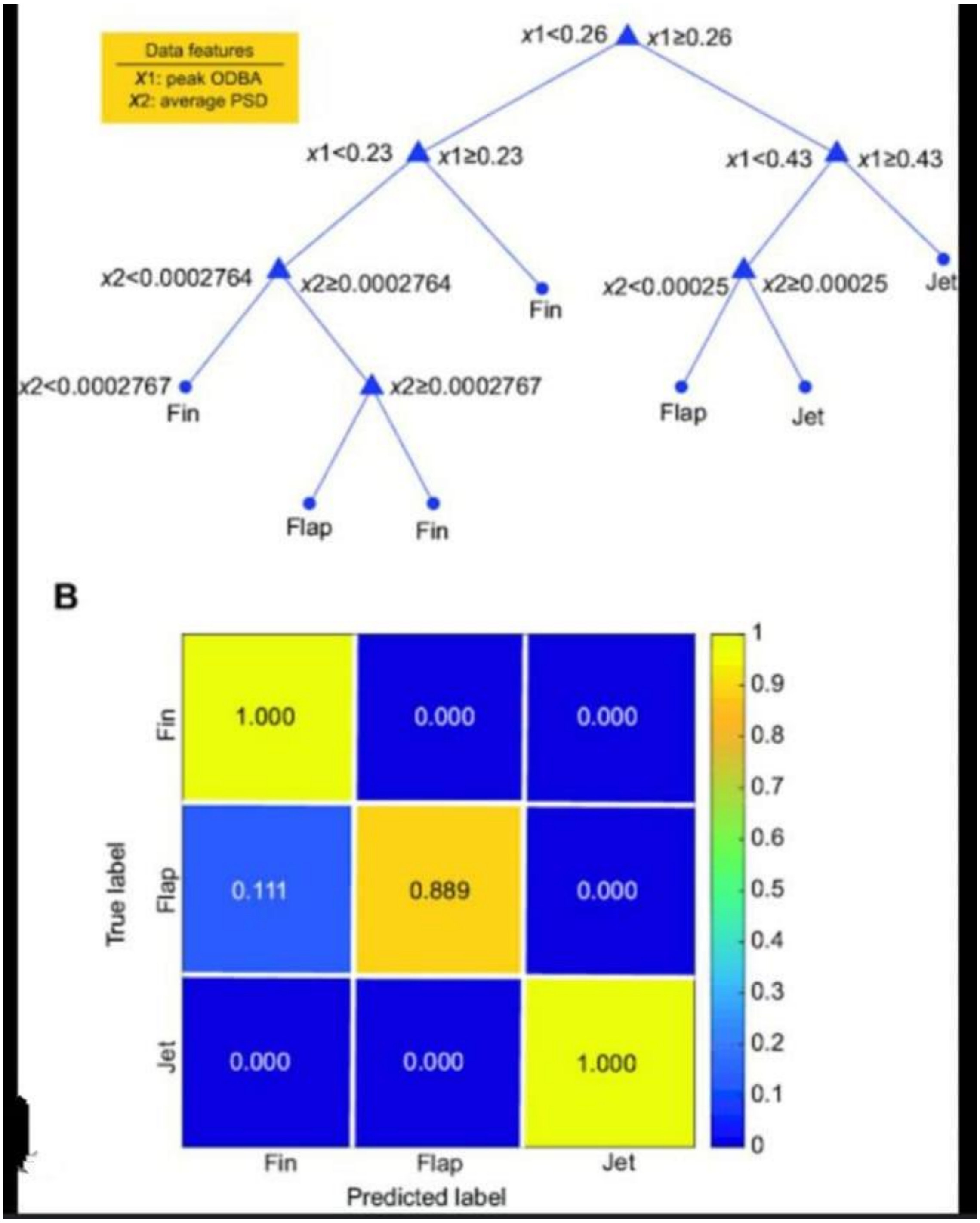


Figure 4.2:Confusion Matrix of Decision Trees

Random Forest:

- Random Forest is a machine learning algorithm that operates by constructing multiple decision trees during training and outputs either the class (for classification) or mean prediction (for regression) of the individual trees. It is widely used due to its simplicity, flexibility, and effectiveness.
- Random Forest predicts the class label for data points by majority voting from individual decision trees.
- **Regression :**
  What it does: Random forest predict the continuous output i.e. numerical value by averaging the output of individual decision trees.
- Reduce overfiting as compared to individual decision trees, improved generalization and robust to noise in the data
- **Ensemble :** An ensemble of the decision trees, where each tree is set on random subset of the data and features
- **Aggregation :** The final prediction is made by the combining the prediction of the all the trees

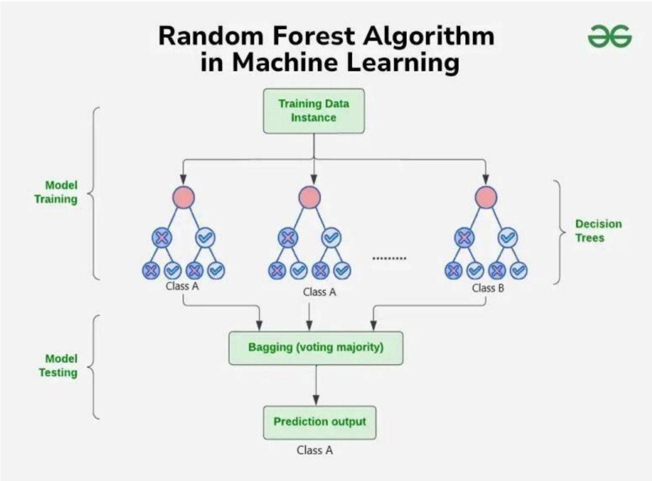


Figure 4.3 Random Forest

Support Vector Machine

"Support Vector Machine" (SVM) is a supervised machine learning algorithm which can be used for both classification or regression challenges. However, it is mostly used in classification problems. In the SVM algorithm, it is possible to plot each data item as a point in n-dimensional space (where n is number of features you have) with the value of each feature being the value of a particular coordinate. Then, classification is done by finding the hyper-plane that differentiates the two classes very well.

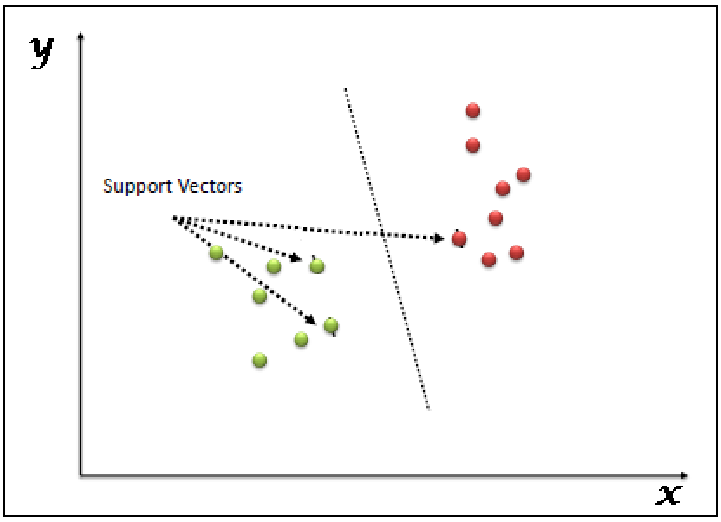


Figure 4.4: Support vector

Support Vectors are simply the co-ordinates of individual observation. The SVM classifier is a frontier which best segregates the two classes (hyper-plane/ line).

4.1 Identify the right hyper-plane

- Hyper planes are decision boundaries that help classify the data points.
- Data points falling on either side of the hyperplane can be attributed to different classes.
- Also, the dimension of the hyperplane depends upon the number of features. If the number of input features is 2, then the hyperplane is just a line.
- If the number of input features is 3, then the hyperplane becomes a two-dimensional plane. It becomes difficult to imagine when the number of features exceeds 3.

Here, consider three hyper-planes (A, B and C). Now, identify the right hyper-plane to classify star and circle. It is required to remember a thumb rule to identify the right hyper-plane: "Select the hyper-plane which segregates the two classes better". In this scenario, hyper-plane "B" has excellently performed this job.

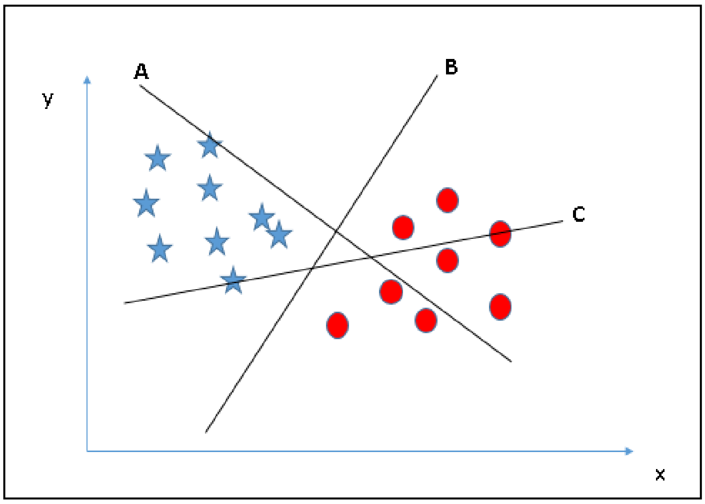


Figure 4.5: Hyper planes 1

Consider three hyper-planes (A, B and C) and all are segregating the classes well. Now, how can the right hyper-plane is to be identified?

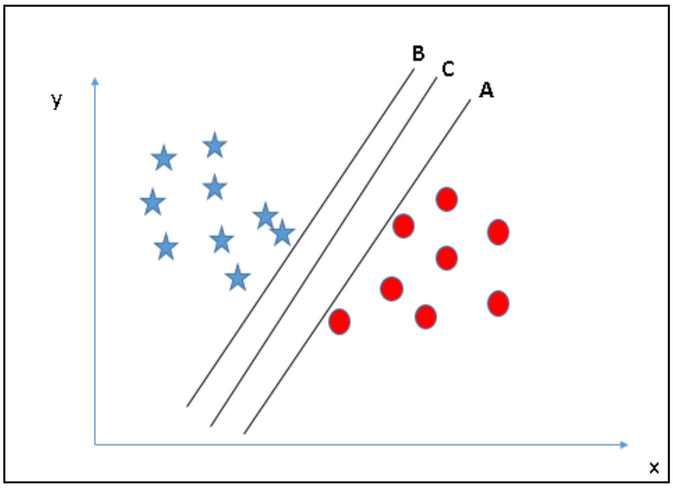


Figure 4.6: Hyper planes 2

Here, maximizing the distances between nearest data point (either class) and hyper-plane will help to decide the right hyper-plane. This distance is called as Margin. The margin for hyper- plane C is high as compared to both A and B. Hence, name the right hyper-plane as C. Another reasonfor selecting the hyper-plane with higher margin is robustness. If hyper-plane having low margin is selected, then there is high chance of mis-classification.

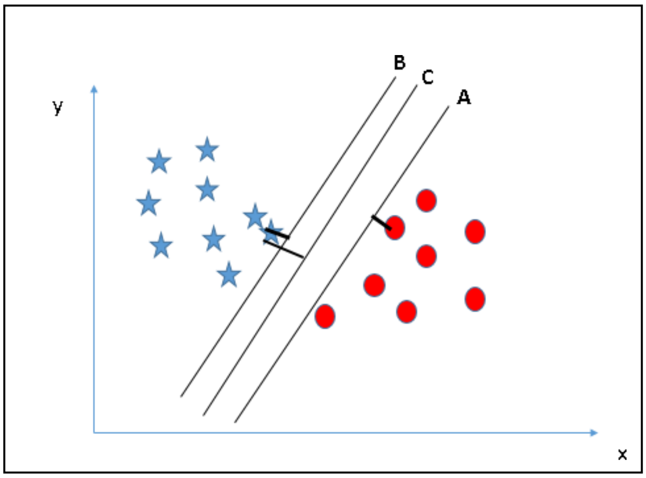


Figure 4.7: Margin

Using the rules as discussed in previous section to identify the right hyper-plane. Some might have selected the hyper-plane B as it has higher margin compared to A. But here is the catch, SVM selects the hyper-plane which classifies the classes accurately prior to maximizing margin. Here, hyper-plane B has a classification error and A has classified all correctly. Therefore, the right hyper-plane is A.

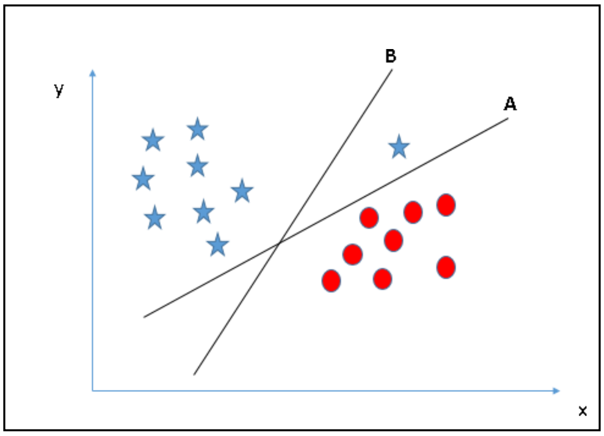


Figure 4.8: Classification error

Below, it is not viable to segregate the two classes using a straight line, as one of the stars lies in the territory of other (circle) class as an outlier.

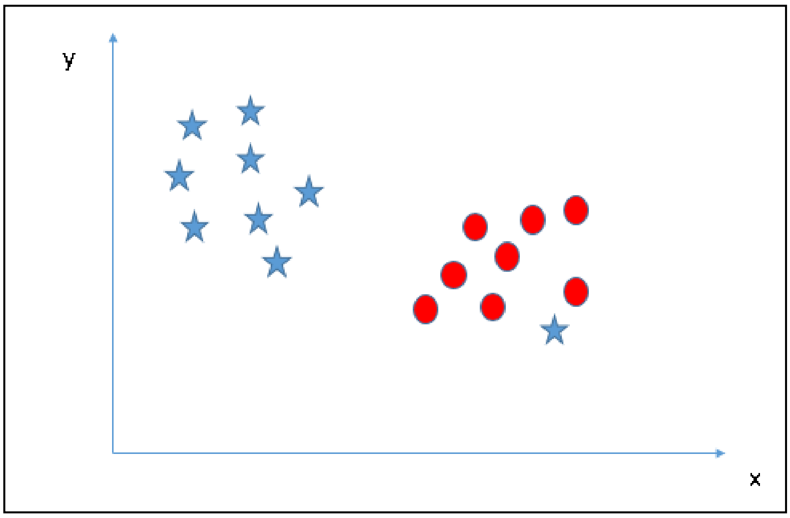


Figure 4.9: Desegregated data

As it is already mentioned, one star at another end is like an outlier for star class. The SVM algorithm has a feature to ignore outliers and find the hyper-plane that has the maximum margin. Hence, SVM classification is robust to outliers.

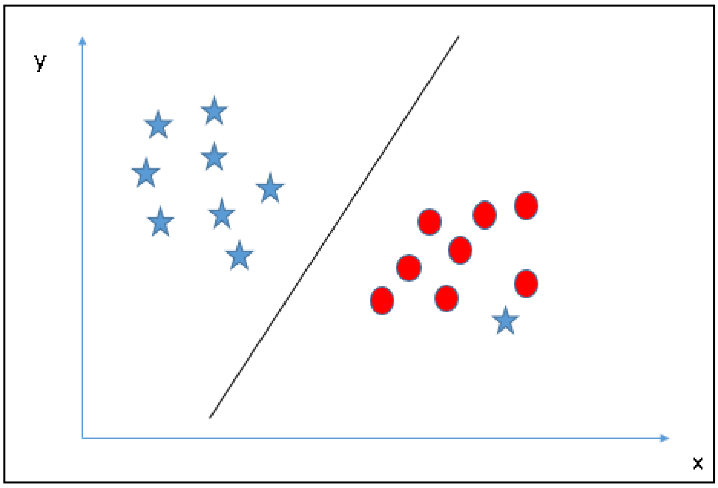


Figure 4.10: Ignoring outliers

In the scenario below, linear hyper-plane between the two classes is not possible, so how does SVM classify these two classes?

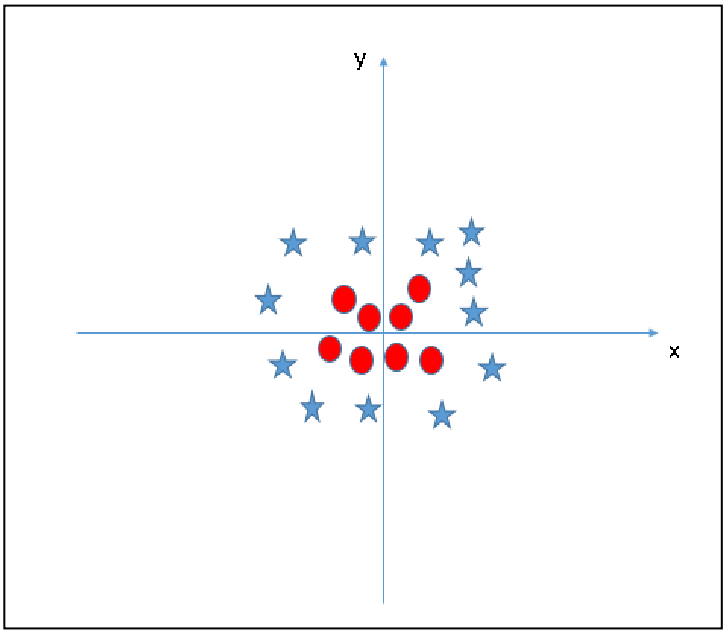


Figure 4.11: Segregated data by circular hyper-plan

SVM can solve this problem. Easily! It solves this problem by introducing additional feature. Here, adds a new feature z=$x^2 + y^2$. Now, plot the data points on axis x and z:

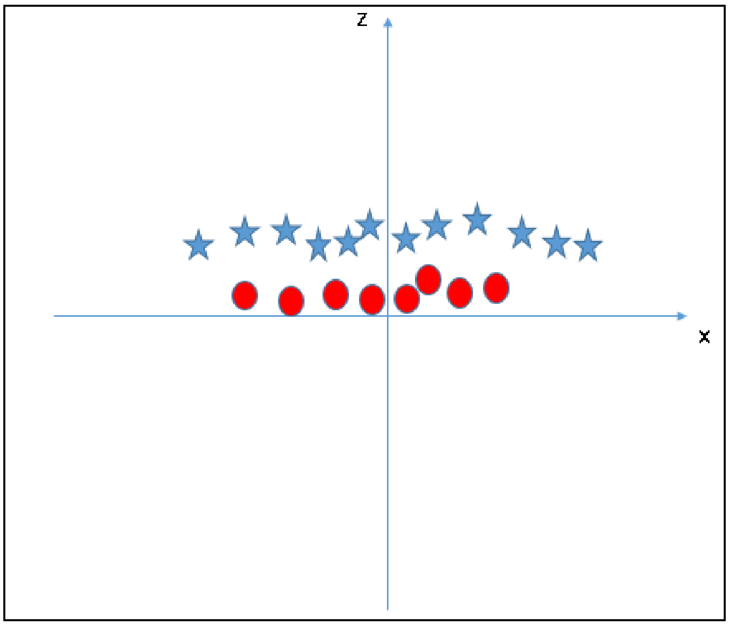


Figure 4.12: Destitution of data in X-Z plan

In above plot, points to consider are:

- All values for z would be positive always because z is the squared sum of both x and y
- In the original plot, red circles appear close to the origin of x and y axes, leading to lower value of z and star relatively away from the origin result to higher value of z.

In the SVM classifier, it is easy to have a linear hyper-plane between these two classes. But another burning question which arises is, is it needed to add this feature manually to have a hyper-plane. No, the SVM algorithm has a technique called the kernel trick. The SVM kernel is a function that takes low dimensional input space and transforms it to a higher dimensional space i.e. it converts non separable problem to separable problem. It is mostly useful in non-linear separation problem. Simply put, it does some extremely complex data transformations, then finds out the process to separate the data based on the labels or outputs that has been defined. The hyper-plane in original input space looks like a circle:

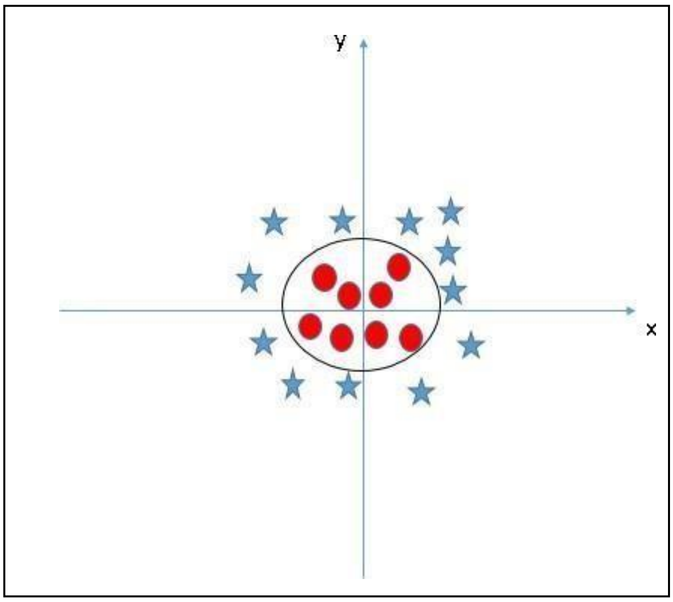


Figure 4.13: Hyper-plan in original input space

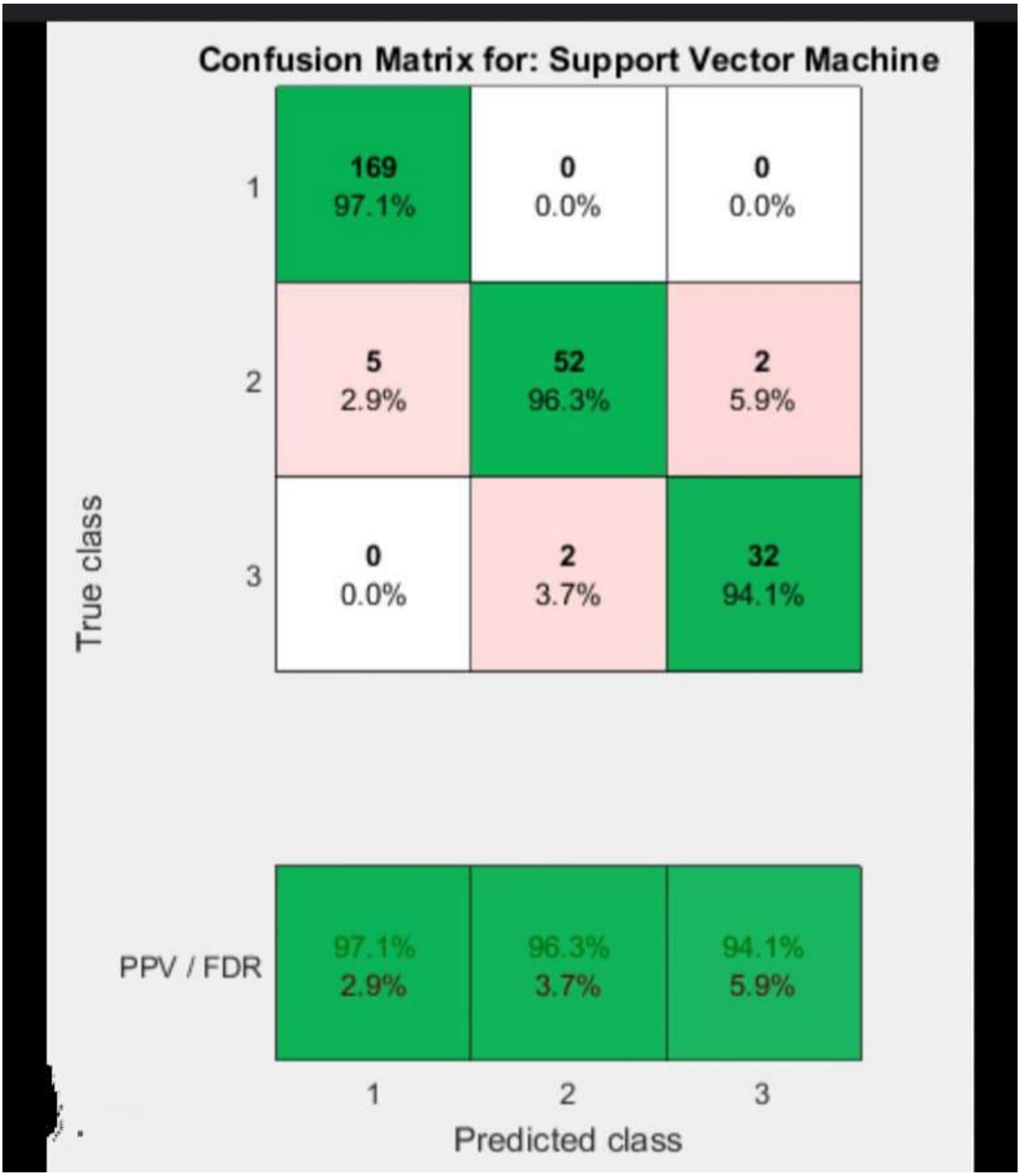


Figure 4.14: Confusion Matrix of SVM

Chapter 5

EXPERIMENTATION

Introduction

In Chapter 2, various causes of vibrations were discussed. Some of the bearing's defects are presented and are considered through equations. A functional relationship was proposed for the bearings defect frequencies in terms of geometrical parameters. It is intended to obtain these constants based on experimental observations. In this chapter, the plan of experimentation isdiscussed.

Experimental Setup

Figure 4.1 shows the experimental setup designed and fabricated for the present study. It consists of a rotor supported on two ball bearings, bearing number 1 and bearing number 2. The rotor is of diameter 25 mm and is of length 750 mm. The rotor is driven by an electric motor. There is a speed control unit and the speed of the rotor may be varied as desired. A Rectifier is used to convert AC Supply voltage into DC Voltage as a DC motor is used. The whole setup is builton a chassis. The motor is connected to the rotor through a rubber flexible coupling. The purpose ofthis coupling is to prevent the transfer of vibrations of the motor to the rotor. Accelerometer pickupsare kept on the bearing housings to measure the vibration amplitudes. The output of the accelerometers is fed to the FFT analyzer for further analysis. The experimental setup has a provisionto change the bearings. Several experiments are performed to study the influence of various variableson the amplitude of vibrations. For studying the effect of load, a provision is made to add loads on the rotor shaft. The weights are added in a hole provided on the disc of the shaft.

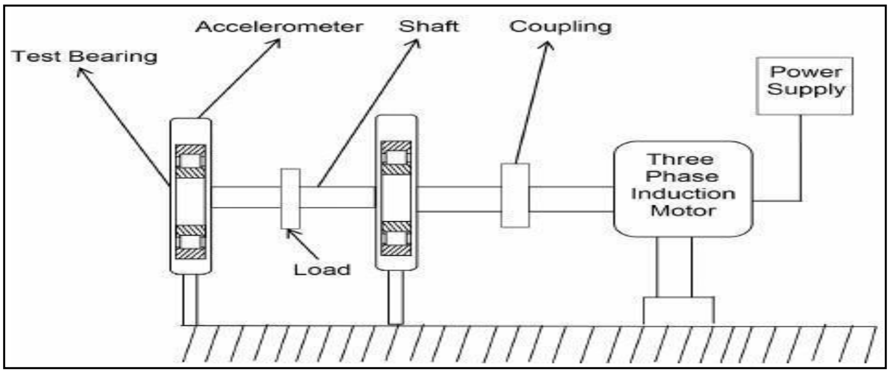


Figure 5.1: Schematic arrangements

Figure 5.2: Experimental Setup

Defect Creation

The following size of defects is created on Inner.

Table 5.1: Size of Defect

Defect No.	Length (l) in mm	Width (w) in mm	Depth (h) in mm
1	16	3.2,5.1	1mm
2	16	3.2,8.2	1mm
3	16	5.1	1mm
4	16	8	1mm

Electro Discharge Machining (EDM)

Electro Discharge Machining (EDM) is an electro-thermal non-traditional machining process, where electrical energy can be used to generate an electrical spark and material removal mainly occurs when thermal energy is induced by the spark. EDM is mainly used for materials whichare difficult-to-machine and also for high strength temperature resistant alloys. EDM is used to machine difficult geometries in small batches or even on a job-shop basis. Work material is to be machined by EDM has to be electrically conductive.

In EDM, a potential difference is applied between the tool and the workpiece. Both the tool and the work material are to be conductors of electricity. The tool and the work material are immersed in a dielectric medium. Generally, kerosene or deionized water is used as the dielectric medium. A gap is maintained between the tool and the workpiece. Depending upon the applied potential difference and the gap between the tool and workpiece, an electric field would be established. Generally, the tool is connected to the negative terminal of the generator and the workpiece is connected to the positive terminal. As the electric field is established between the tool and the job, the free electrons on the tool are subjected to electrostatic forces. If the work function or the bonding energy of the electrons is less, electrons would be emitted from the tool (assuming it to be connected to the negative terminal). Such emission of electrons is called or termed as cold emission.

The high-speed electrons then impinge on the job and ions on the tool. The kinetic energy of the electrons and ions on impact with the surface of the job and tool respectively would be converted into thermal energy or heat flux. Such intense localized heat flux leads to an extreme instantaneous confined rise in temperature which would be more than 10,000°C. Such localized extreme rise in temperature leads to material removal. Material removal occurs due to instant vaporization of the material as well as due to melting. The molten metal is not removed completely but only partially

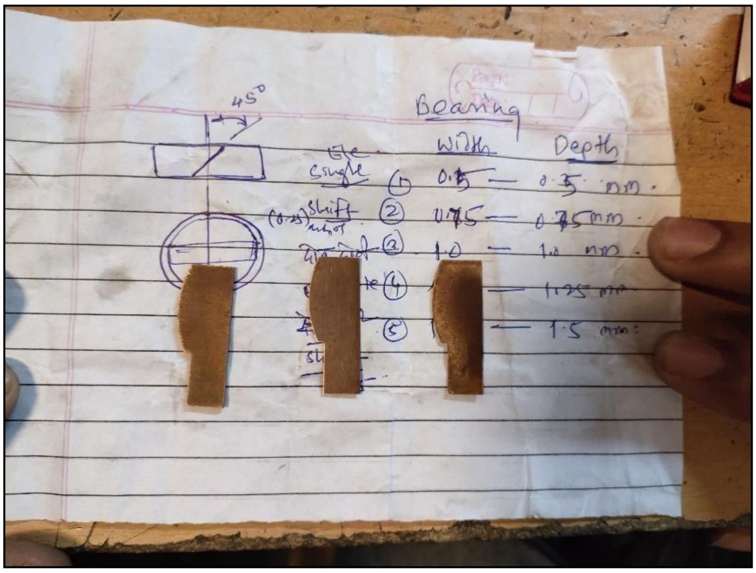


Figure 5.3: Electrodes for Defect Creation

Figure 5.4: Defect Creation on Inner Race using EDM

Test Procedure

Healthy Bearing

1. Place the healthy bearing in the housing, make sure the bearing fits properly in the housing, and fasten the housing.

2. Place the accelerometer on the housing and make sure the FFT is ON.

3. Start the motor and take the readings by varying the speed and weight.

Defective Bearing

1. Place the bearing with 3.2 mm,5,1mm defect on Inner Race or Outer Race in the housing, make sure the bearing fits properly in the housing, and fasten the housing.
2. Place the accelerometer on the housing and make sure the FFT is ON.
3. Start the motor and take the readings by varying speed and weight.
4. Change the bearing with a different defect and repeat step (2) and step (3)

Table 5.2: Conditions for the Experimentation

Bearing	Misalignment (mm)	Speed (rpm)
Healthy	0,0.25,0.5,0.75, 1	720,1000,1440,2160
3.2,5.1 (mm)	0,0.25,0.5,0.75,1	720,1000,1440,2160
3.2,8.2(mm)	0,0.25,0.5,0.75,1	720,1000,1440,2160
5.1(mm)	0,0.25,0.5,0.75,1	720,1000,1440,2160
8(mm)	0,0.25,0.5,0.75,1	720,1000,1440,2160

Effect of Various Parameters on Vibration Amplitude

Effect of speed on vibration amplitude

The speed of the rotor is varied from 720 rpm, 1000 rpm to 1440 rpm, 2160 rpm. For each speed, the amplitude of vibration is measured. Figure 5.8 shows the variation of vibration amplitude with speed. It is observed that the amplitude of vibration is increasing with the speed of the rotor. The trend of increasing amplitude with speed may be true as long as the speed of the rotor is less than the natural frequency.

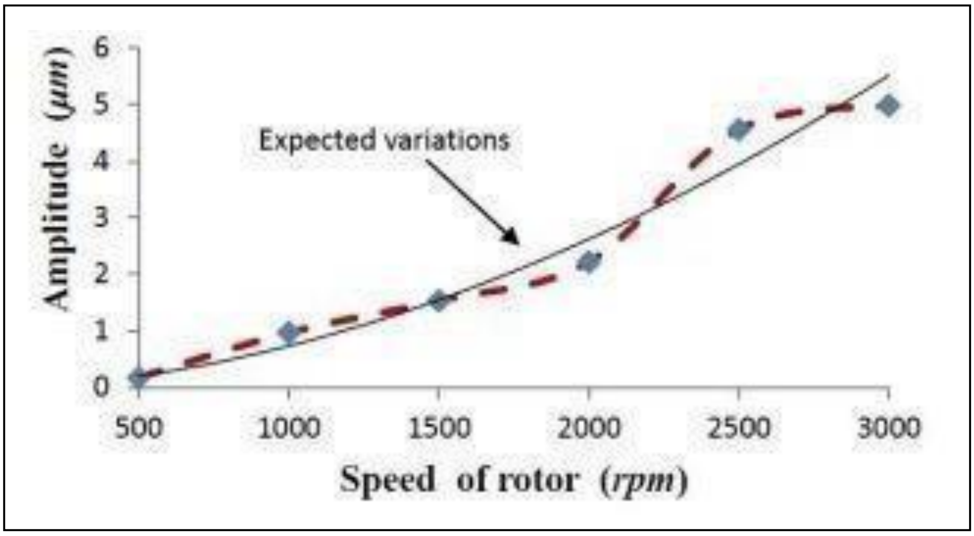


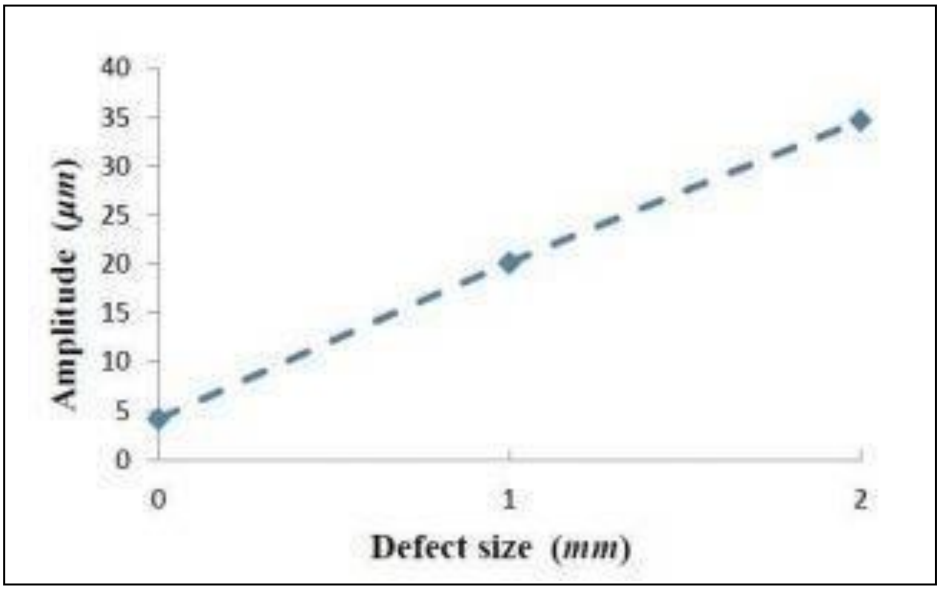
Figure 5.8: Effects of speed on vibration amplitude

Effect of defect size on vibration amplitude

To study the effect of defects, experiments are conducted on defective bearings. In the experimentation, a defect is created by EDM on the inner and outer race of bearing. The depth of the defect is termed as the size of the defect. The defects of sizes (0),(3.2 mm, 5.1 mm), (3.2 mm, 8.2 mm),(5.1 mm),(8mm) are tried for experimentation. Further, in the experiments, the defects are created only on one of the two bearings which support the shaft.

The results are plotted as in Figure 5.10. The amplitude is observed to increase with theincrease in the size of the defect. The reason could be, a defect reduces the stiffness which in turn reduces the natural frequency. Hence, when the defect size is increased the operating point moves towards the resonance.

Figure 5.10: Effect of defect size on vibration amplitude



Closure

In this chapter, the experimental setup is first described. Preliminary experiments are performed to observe the effect of various parameters such as speed, load, and defect size. Next, it is planned to perform a large number of experiments with varieties of combinations of the parameters.

VALIDATION OF EXPERIMENTAL RESULTS

Introduction

The experimentation to be carried out for analyzing the bearing vibrations was highlighted in Chapter 4. The experimental setup along with the instrumentation required was also described. The effects of parameters such as speed, load, and defects on the bearing vibrations were first studied. In this chapter, it is proposed to perform case studies on a bearing.

Characteristics Frequencies of Faulty Bearing

The rolling element of bearing fails in different ways. Following are some conditions that occur due to bearing failure:

Mechanical looseness: If the bearing is mounted on the shaft with the help of the sleeve, over some time interference fit between shaft and sleeve may change during operation. This causes theloosening of the sleeve over the shaft. This results in a higher vibration of those bearings as comparedto the vibrations of other normal working bearings of the machine. For such vibration's peaks occurat twice the speed of the shaft.

Unbalance in rotor: Sometimes, sleeve looseness may create an unbalance in the rotor; frequency analysis reveals the existence of a peak at the speed of the shaft as well as at twice the speed of the shaft. Similar looseness conditions may occur if the sleeve taper and bore taper of a bearing do not match properly. Even when mounting of such bearings achieves sufficient grip between the sleeve and inner race, dynamic loading between the sleeve and bore can diminish the grip over some time and the assembly becomes loose.

Seizure of the shaft: Another typical characteristic feature of this type of failure is increased bearing temperature. The looseness permits relative motion between a bearing pedestal and the outer race of a bearing which is supposed to operate in a condition of interference fit. The relative motion causes heavy friction under working conditions and creates a subsequent temperature rise at mating surfaces. The temperature rise may result in the severe seizure of the shaft and inner race, and the removal of the bearing may require gas cutting.

Generally, vibration frequencies due to bearing faults lie in the range of 2 to 10 times the rotor speed. In essence, when defects exist in bearings, vibration signatures show that the frequencyof bearing component vibrations correlate with the type of bearing fault. By identifying the type of bearing characteristic frequency (BCF), the cause of the defect may be diagnosed. In most of the applications, namely fibrizer rotor, ID fans, blowers, etc., the outer race is always fixed. In such cases, the BCF of the bearing defects namely defects on outer-race, defects on the inner race, defects on the roller, and defects on cage may be estimated using the following equations

The frequency at which the balls or rollers are spinning is caused by the rotating movement of the balls or rollers around their axis

$$BSF = 0.5 * N * (D/d) * [1-(d \cos\Theta /D)^2] \quad (6.1)$$

The defect frequency of body rotation or ball support (FTF) can be calculated with the following expression

$$FTF = 0.5 * N * [1 - (d \cos\Theta /D)] \quad (6.2)$$

The defect frequency produced in the inner race or ball passage (BPFI) is calculated with the following expression

$$BPFI = 0.5 * N * n * [1 + (d \cos\Theta /D)] \quad (6.3)$$

The Defect frequency produced in the outer race or ball passage (BPFO) is calculated with the following expression

$$BPFO = 0.5 * N * n * [1 - (d \cos\Theta /D)] \quad (6.4)$$

Where,

N: Rotating Frequency (Hz)

D: Average bearing diameter (mm)

d: Diameter of the rolling circumference of the balls or rollers (mm)

n: Number of rollers or balls forming the bearing

However, in reality, some sliding motion may also occur, which may cause a slight deviation of the BCF's.

Dynamic Response of Bearing

The present work analyzes the dynamic responses of healthy as well as faulty bearings, supporting a horizontal rotor with self-aligning ball bearings. The excitation induced by rotation produces varying compliance frequencies, first by a healthy bearing and subsequently by actually distributed defects in the outer race and inner race of a bearing. The present study characterizes the vibration frequencies resulting from self-aligning spherical roller bearings with distributed defects.

Table 5.1 gives the geometry and operating parameters of the bearings used for studies on the rotor.

Table 6.1: Parameters of bearing analyzed

Parameters	**Value**
Deep groove ball bearing	6206
Bearing width, mm	16
Inner race diameter, mm	52
Outer race diameter, mm	62
Pitch diameter, mm	42
Ball diameter, mm	9.3
Number of Balls	9
Shaft speed, rpm	720, 1000, 1440, 2160

Effect of Vibration Response of Healthy Bearing

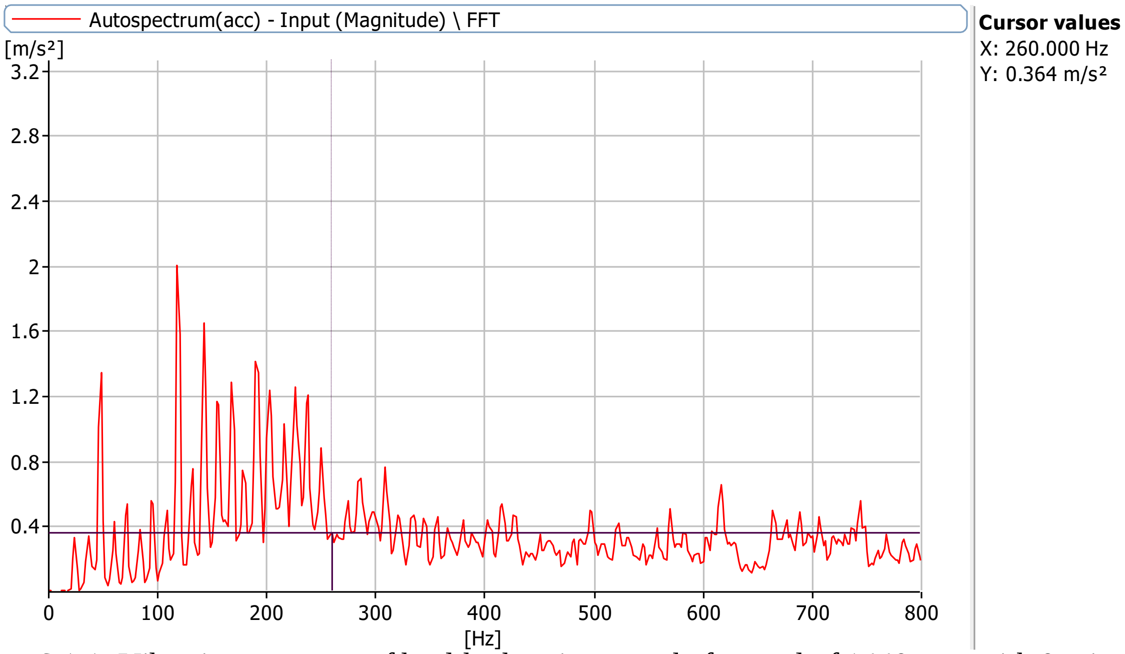


Figure 6.1.1: Vibration response of healthy bearing at a shaft speed of 1440 rpm with 0 misalignment at X axis

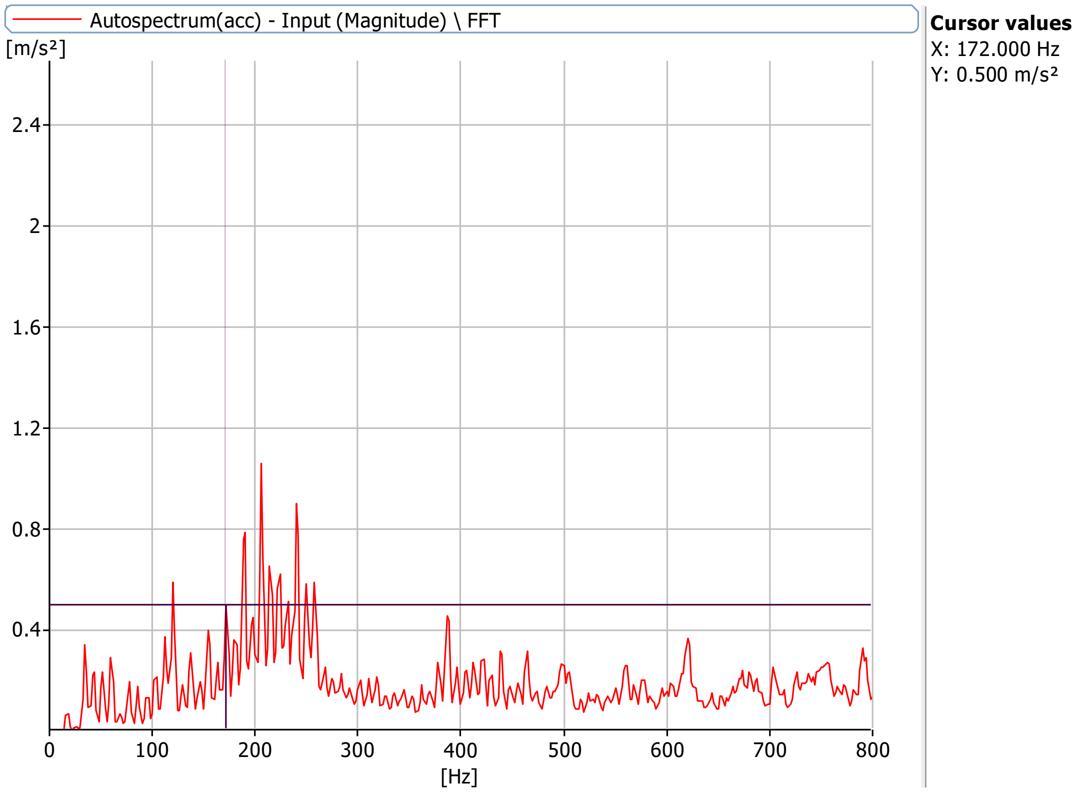


Figure 6.1.2: Vibration response of healthy bearing at a shaft speed of 1000 rpm with 0 misalignment at X axis

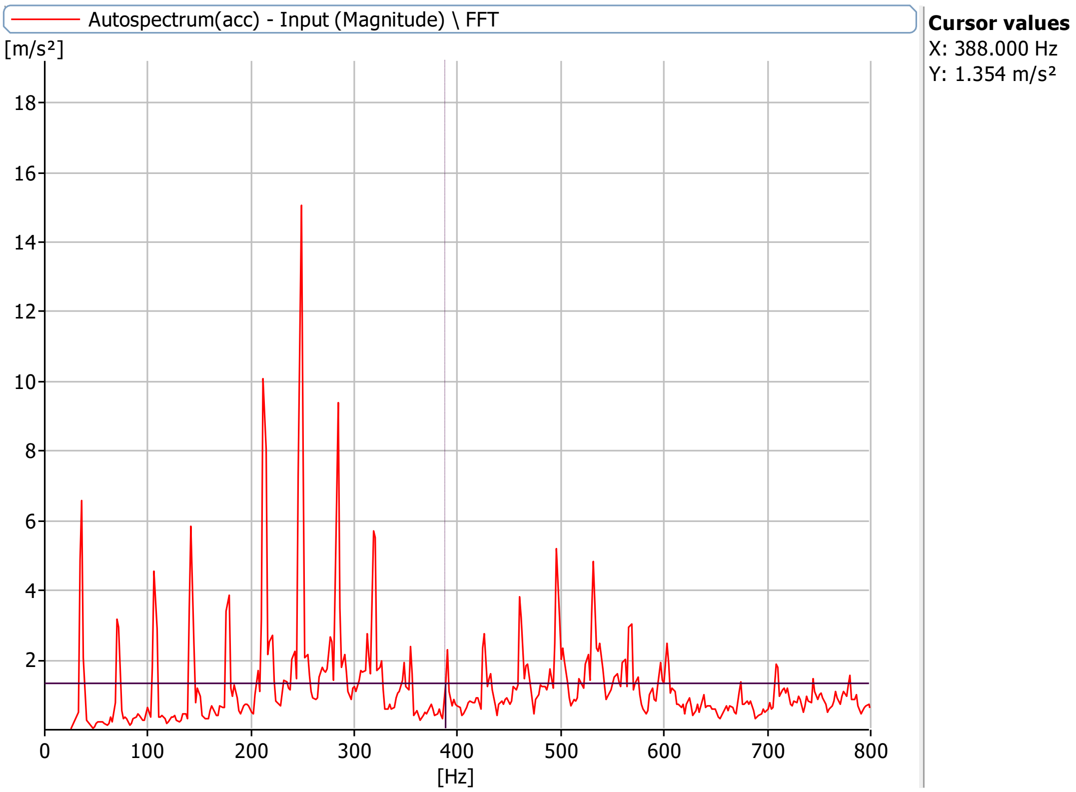


Figure 6.1.3: Vibration response of healthy bearing at a shaft speed of 2160 rpm with 0 misalignment at X axis

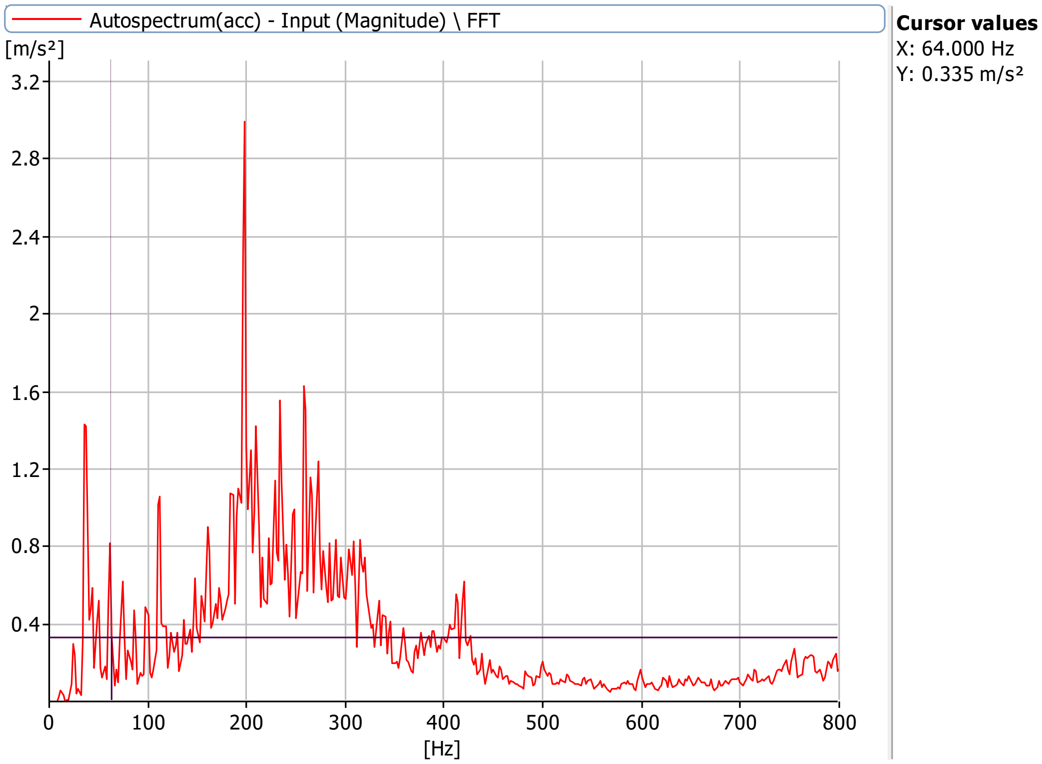


Figure 6.1.4: Vibration response of healthy bearing at a shaft speed of 720 rpm with 0 misalignment at X axis

The bearing defects results in harmonic multiples of frequency (1x, 2x, 1xBPFI ,2xBPFI), of the defect frequencies in the vibration spectrum. The repetition of vibration signals of defects can be observedas peaks in the graph of the frequency spectrum. The vibration spectrum data used for the analysis isfor four different defects state of the rolling element bearing and one representing the normal state ofthe bearing. The frequencies at different points are taken as,

1. 1x shaft frequency.
2. 2x shaft frequency.
3. 1xBPFI shaft frequency.
4. 2xBPFI shaft frequency.

The characteristics frequencies and amplitude of healthy bearing at 720 rpm,1000 rpm,1440 rpm and 2160 rpm atdifferent loading conditions.

Effect of Vibration Response of Defective Bearing

The frequency spectrums in Figs. 6.1.1 to 6.1.12 show the presence of the outer race defect BPFI. Other peaks at FTF and side-bands at BPFI reveal the presence of the distributed defect. The amplitude of vibration is significant at a different speed for distributed outer race defect. Predicted frequencies of vibration using the theoretical model (Equation 6.3) closely match the experimental results. Table 6.6 shows the comparison of the frequencies of vibrations obtained from experiments and the theoretical model. It can be observed that there is a very close match between the experimental values and theoretical values of frequencies of vibrations under different rotational speeds of the shaft.

The frequency spectrums for various defects are shown below:

Frequencies at 720 rpm, 1000 rpm,1440 rpm and 2160 rpm for defect of 0.35,5.1 mm bearing and 0 misalignment

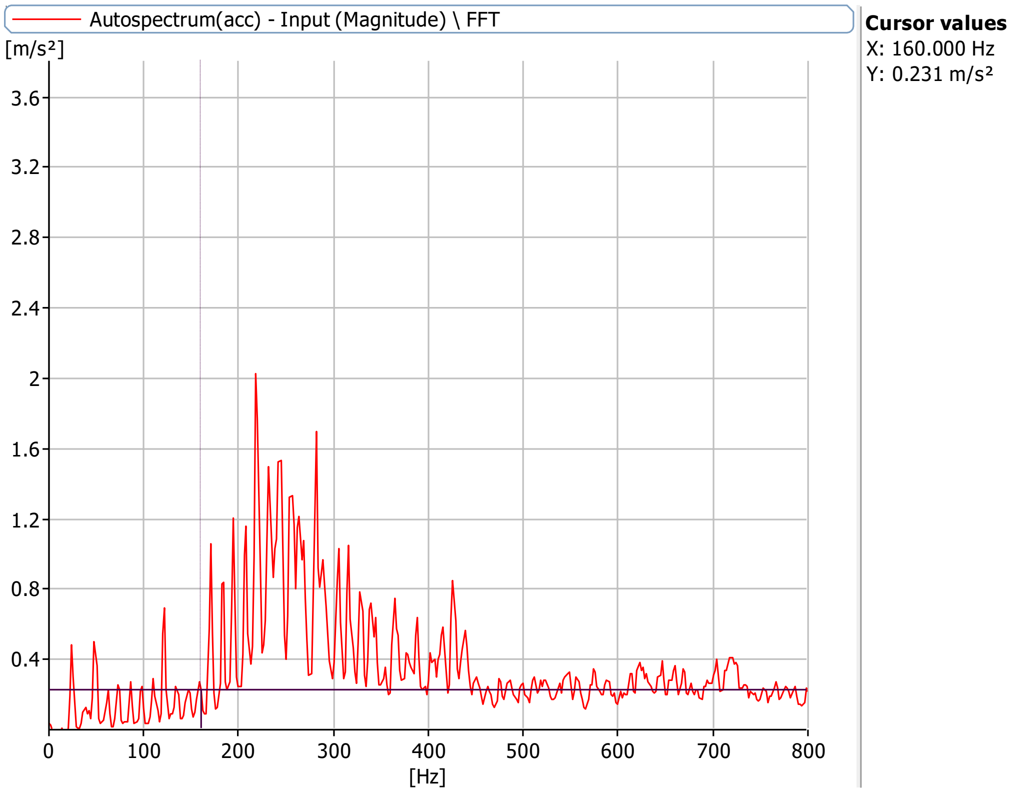


Figure 6.2.1: Frequency Spectrum at 1440 rpm with a defect size of 0.35 mm,5.1 mm with no load with 0 misalignment at X axis

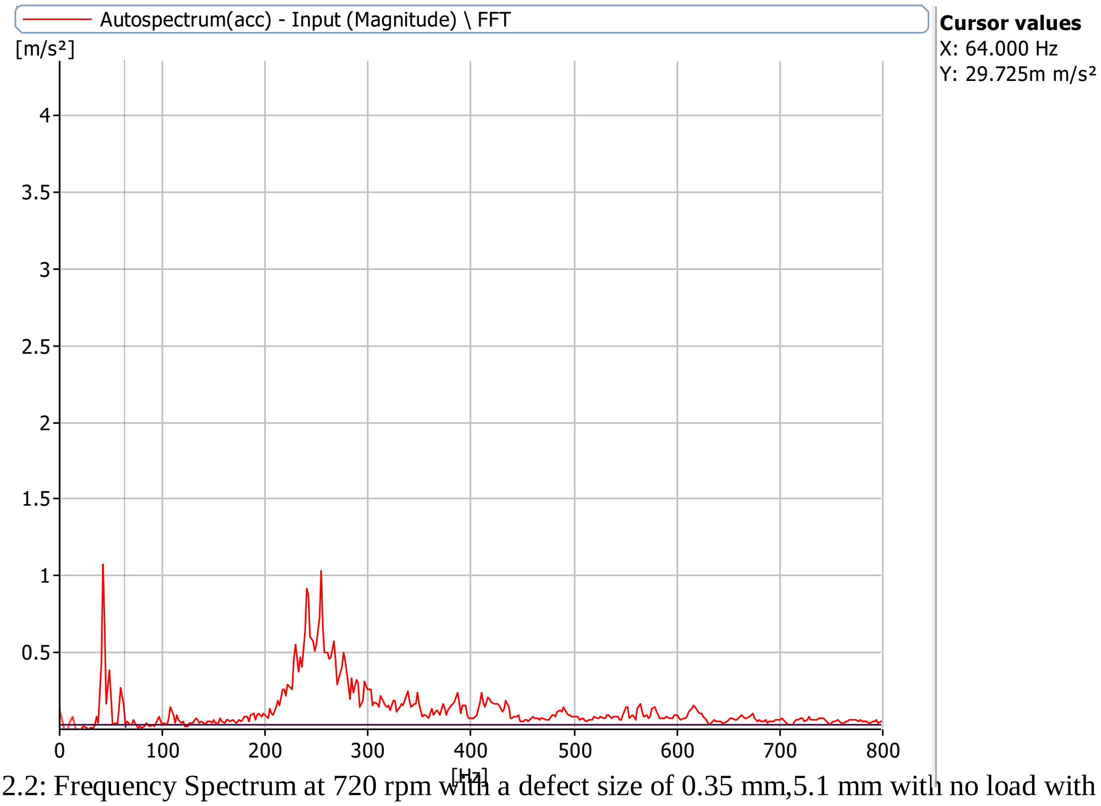


Figure 6.2.2: Frequency Spectrum at 720 rpm with a defect size of 0.35 mm,5.1 mm with no load with 0 misalignment at X axis

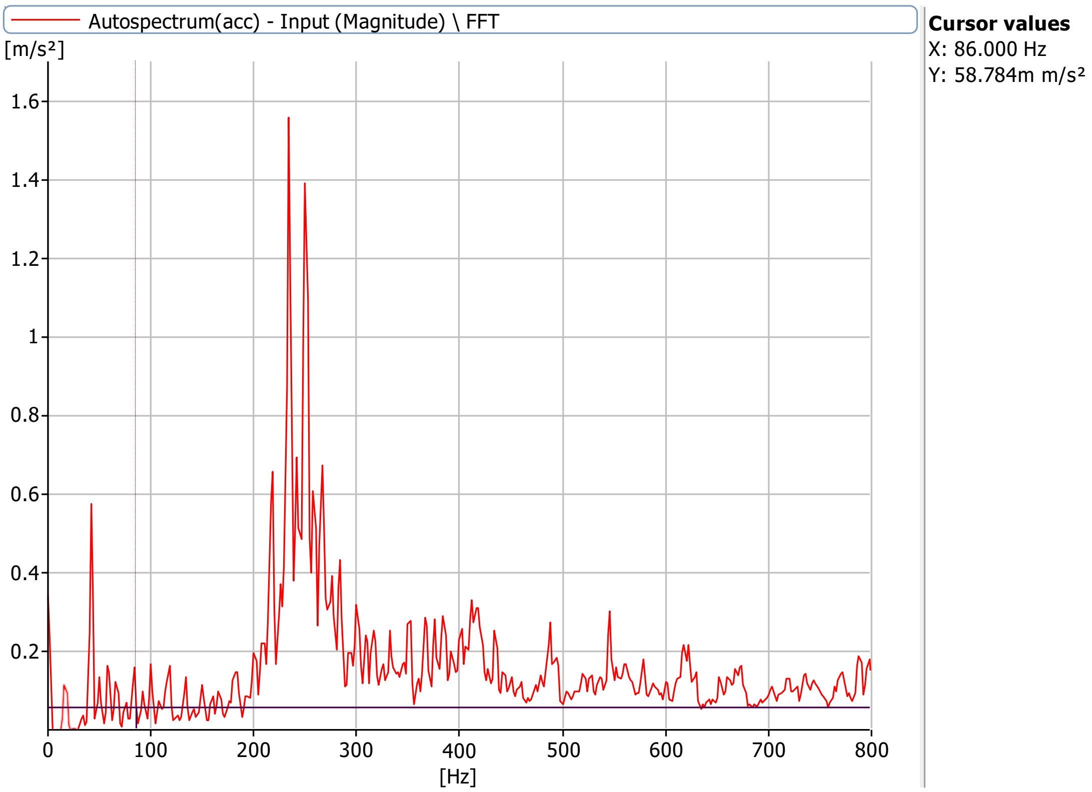


Figure 6.2.3: Frequency Spectrum at 1000 rpm with a defect size of 0.35 mm,5.1 mm with no load with 0 misalignment at X Axis

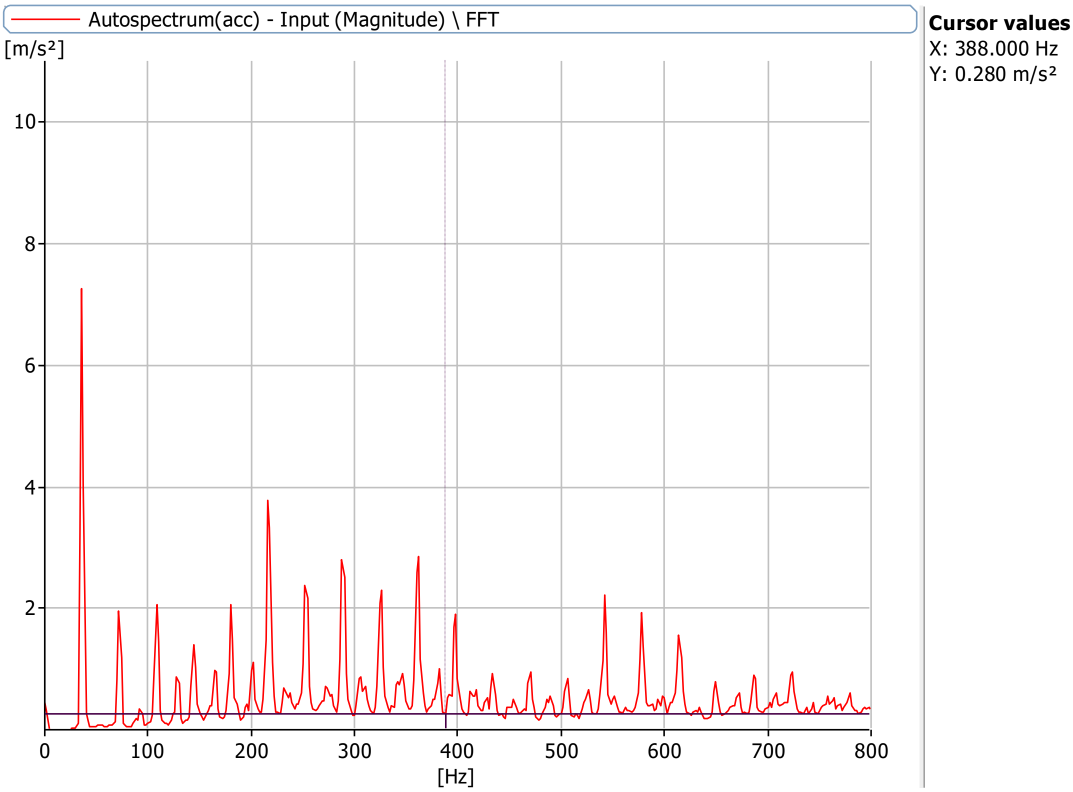


Figure 6.2.4: Frequency Spectrum at 2160 rpm with a defect size of 0.35 mm, ,5.1 mm with no load with 0 misalignment at X Axis

Figure 6.3:Frequencies at 720 rpm, 1000 rpm,1440 rpm and 2160 rpm for defect of 0.32,8.2 mm bearing and 0 misalignment

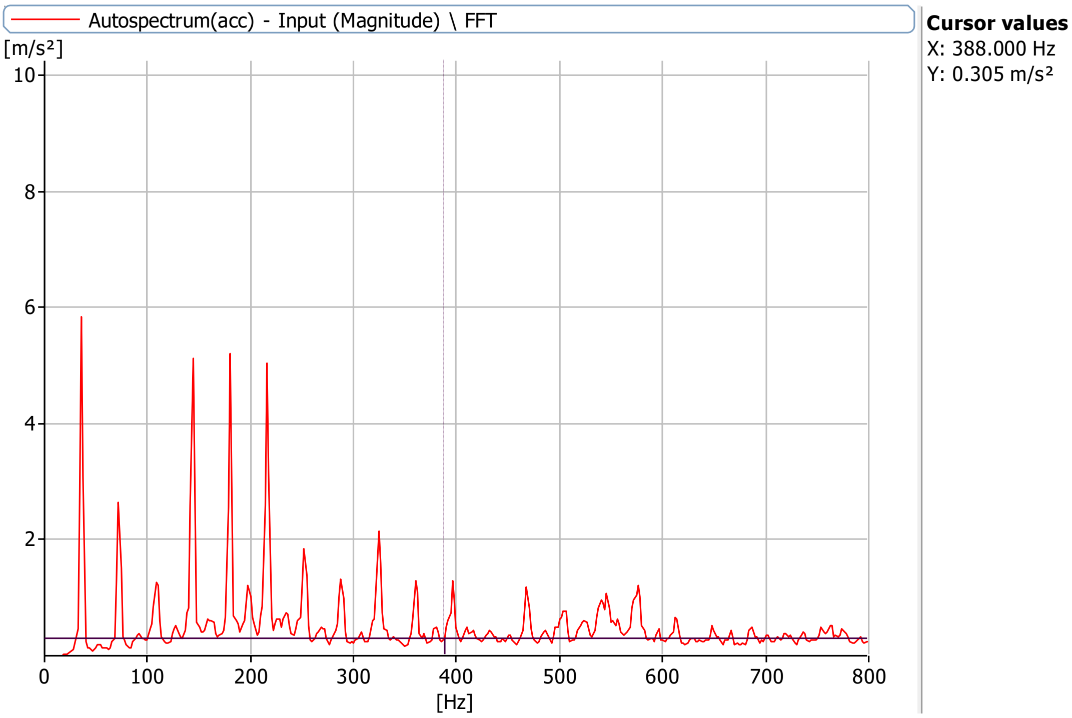


Figure 6.3.1: Frequency Spectrum at 2160 rpm with a defect size of 0.32 mm,8.2 mm with no load with 0 misalignment at X Axis

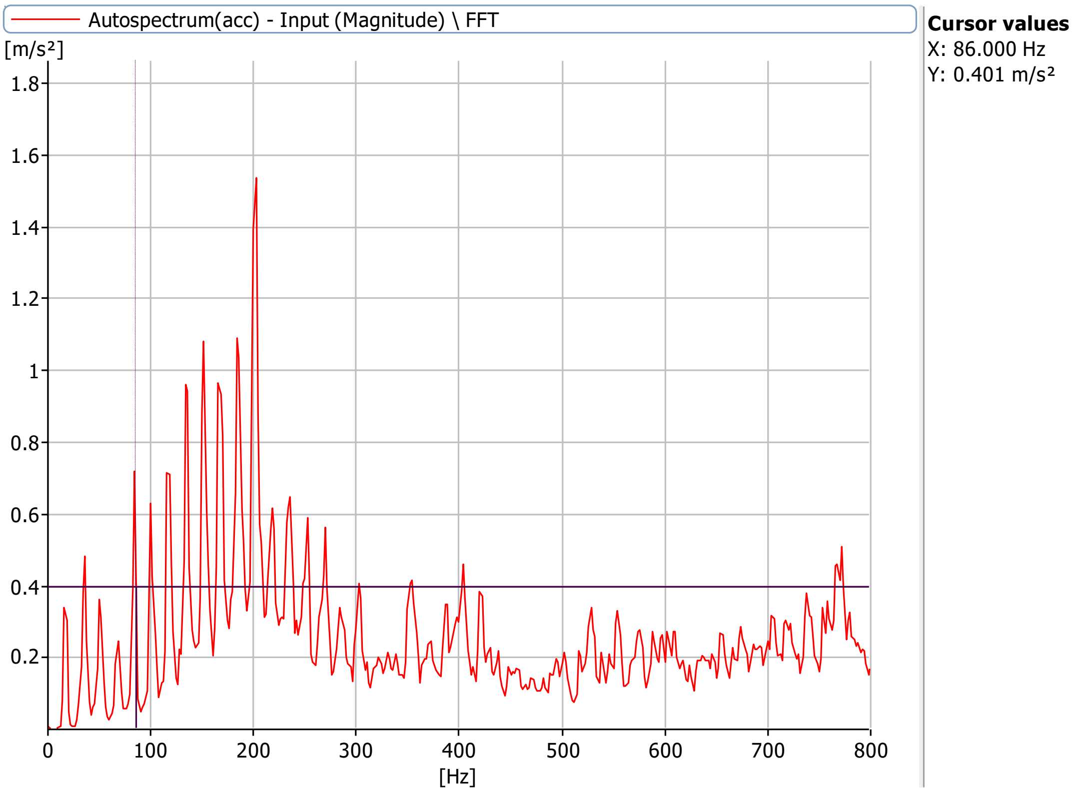


Figure 6.3.2: Frequency Spectrum at 1000 rpm with a defect size of 0.32 mm,8.2 mm with no load with 0 misalignment at X Axis

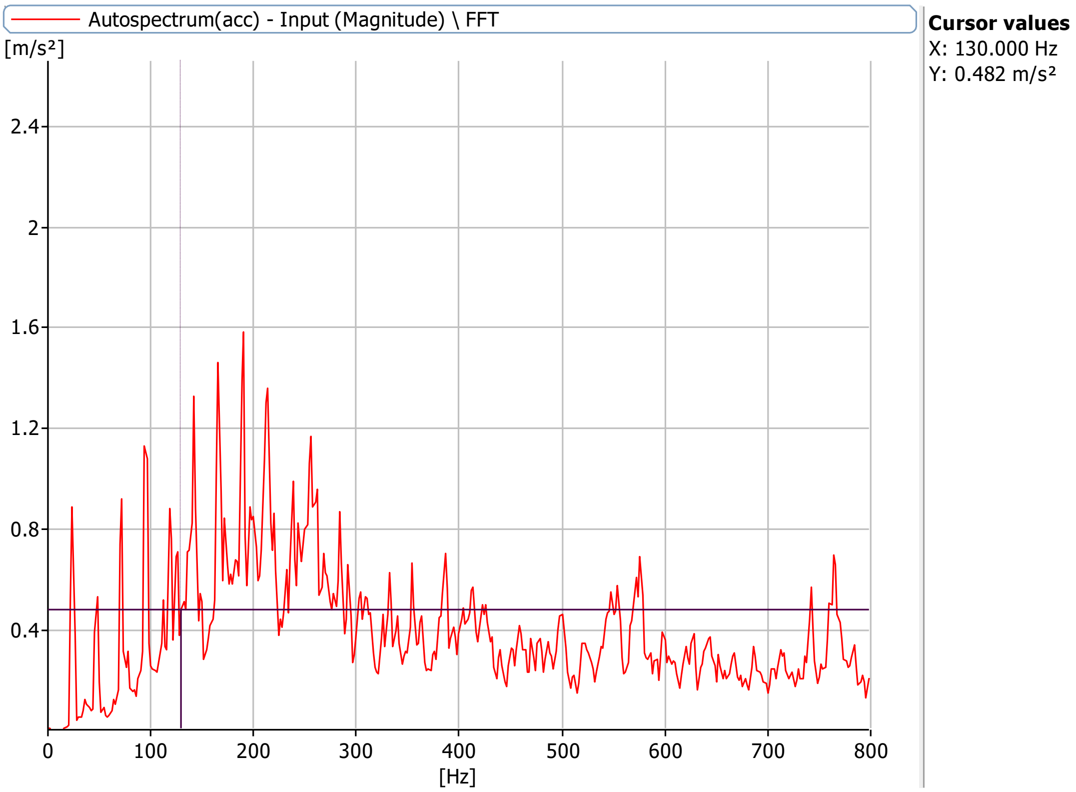


Figure 6.3.3: Frequency Spectrum at 1440 rpm with a defect size of 0.32 mm,8.2 mm with no load with 0 misalignment at X Axis

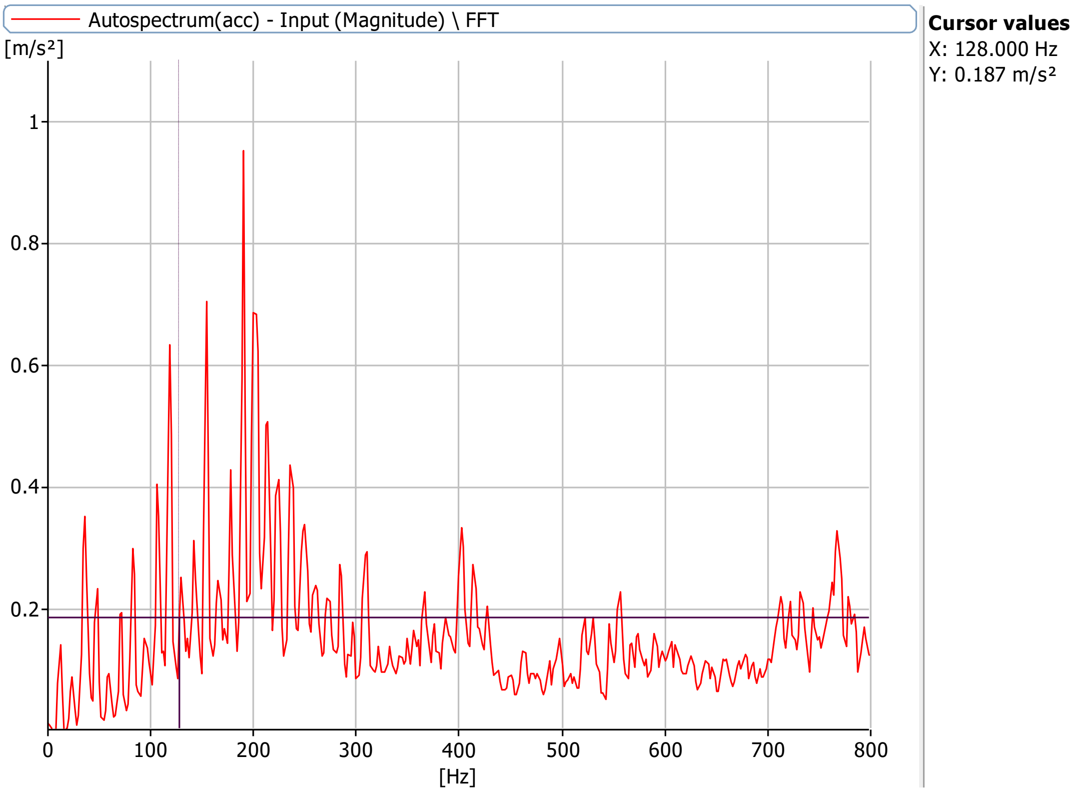


Figure 6.3.4: Frequency Spectrum at 720 rpm with a defect size of 0.32 mm,8.2 mm with no load with 0 misalignment at X Axis

Frequencies at 720 rpm, 1000 rpm,1440 rpm and 2160 rpm for defect of 5.1 mm bearing and 0 misalignment

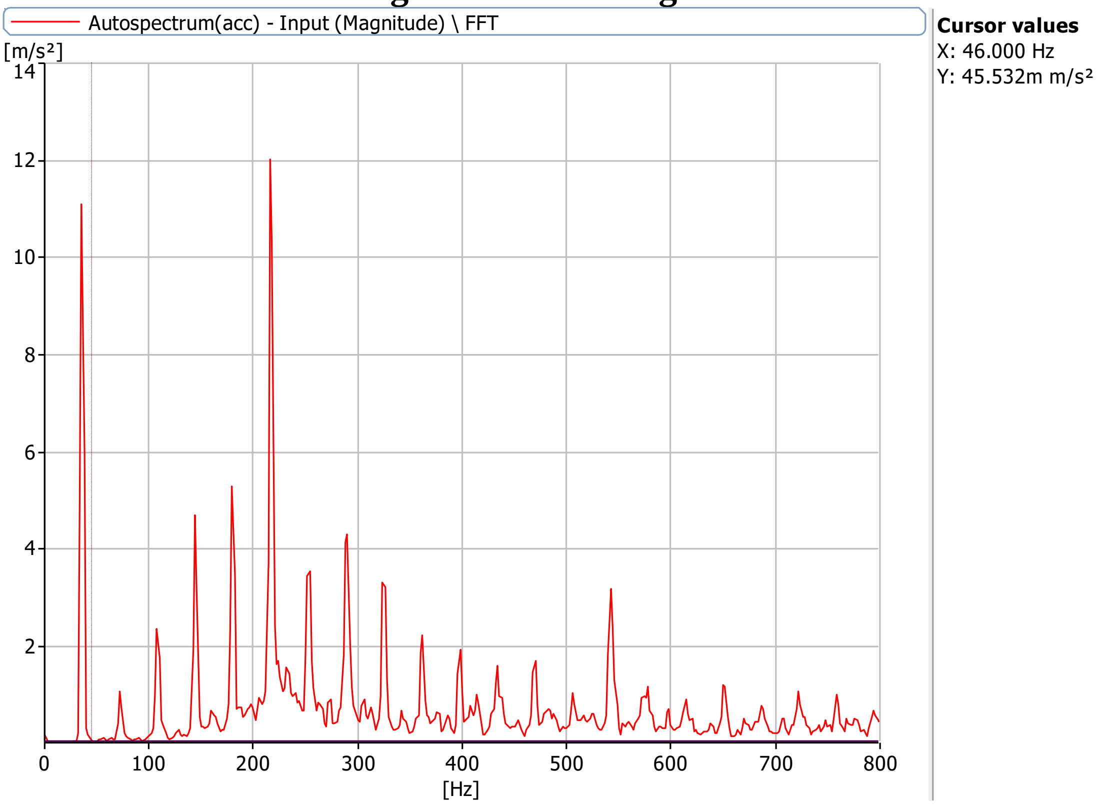


Figure 6.4.1: Frequency Spectrum at 2160 rpm with a defect size of 5.1 mm with no load with 0 misalignment at X Axis

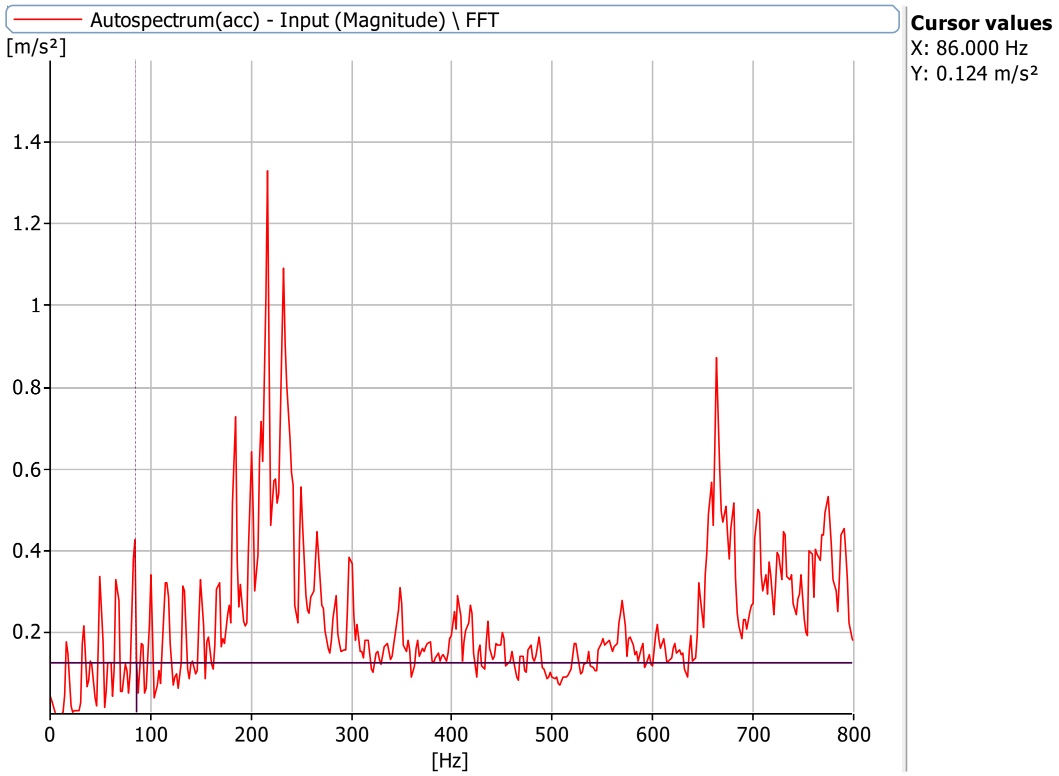


Figure 6.4.2: Frequency Spectrum at 1000 rpm with a defect size of 5.1 mm with no load with 0 misalignment at X Axis

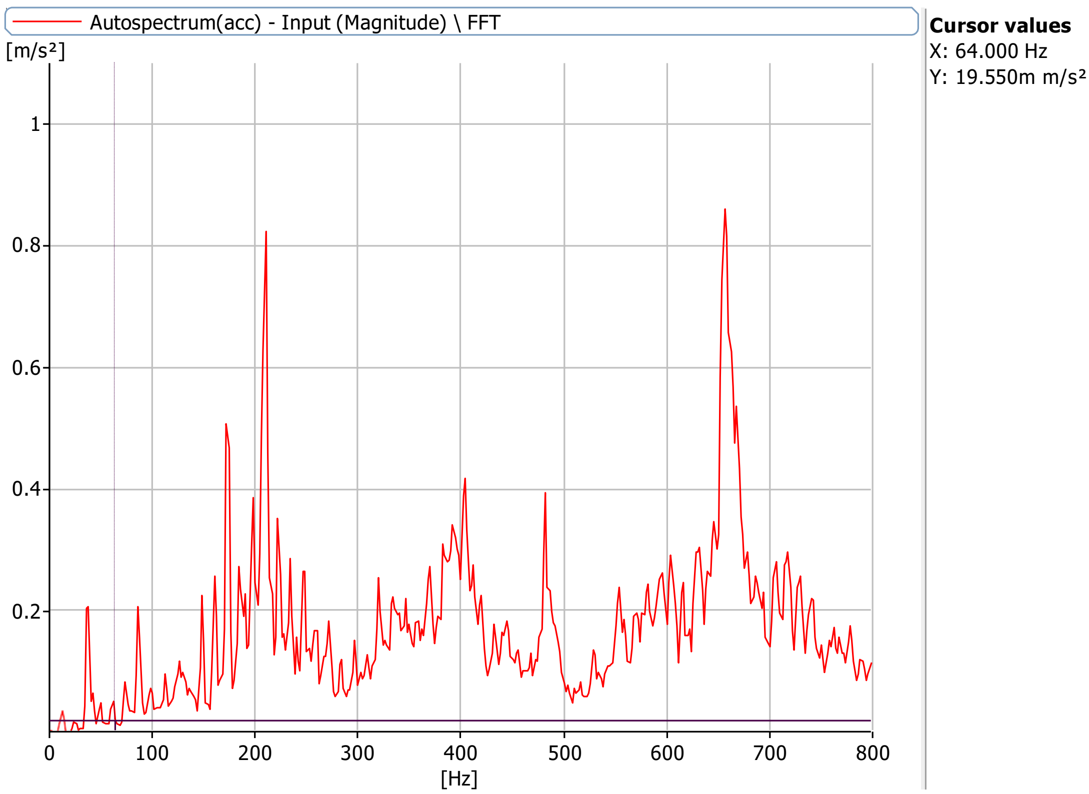


Figure 6.4.3: Frequency Spectrum at 720 rpm with a defect size of 5.1 mm with no load with 0 misalignment at X Axis

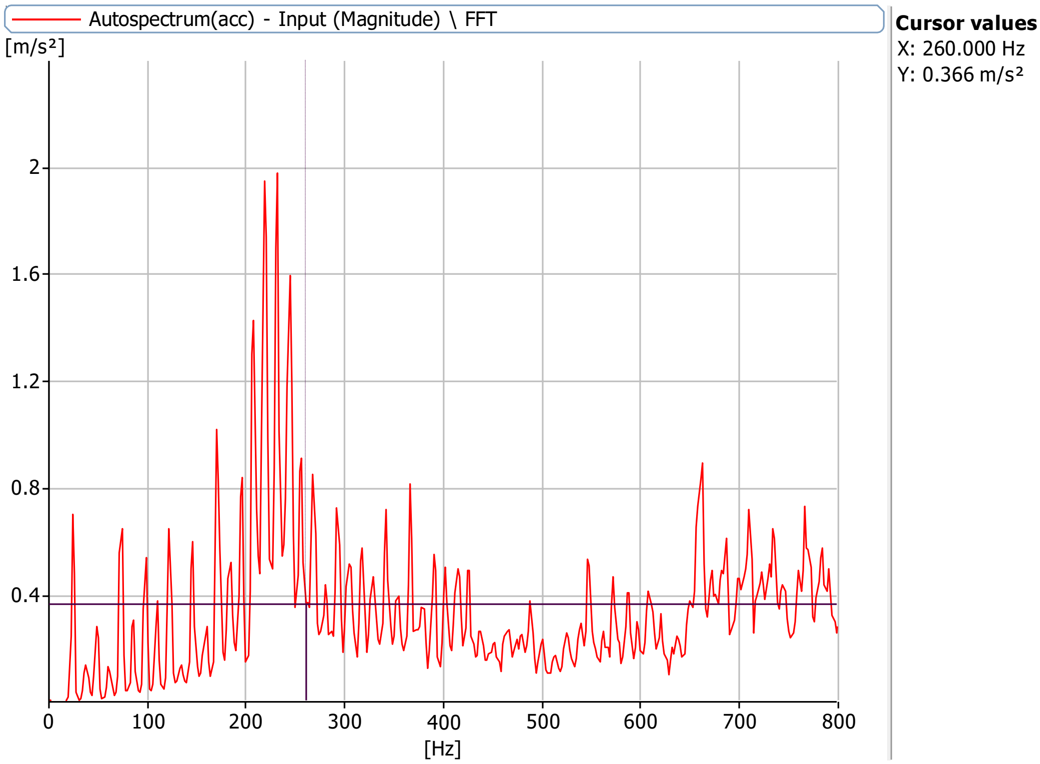


Figure 6.4.3: Frequency Spectrum at 1440 rpm with a defect size of 5.1 mm with no load with 0 misalignment at X Axis

Frequencies at 720 rpm, 1000 rpm,1440 rpm and 2160 rpm for defect of 5.1 mm bearing and 0 misalignment

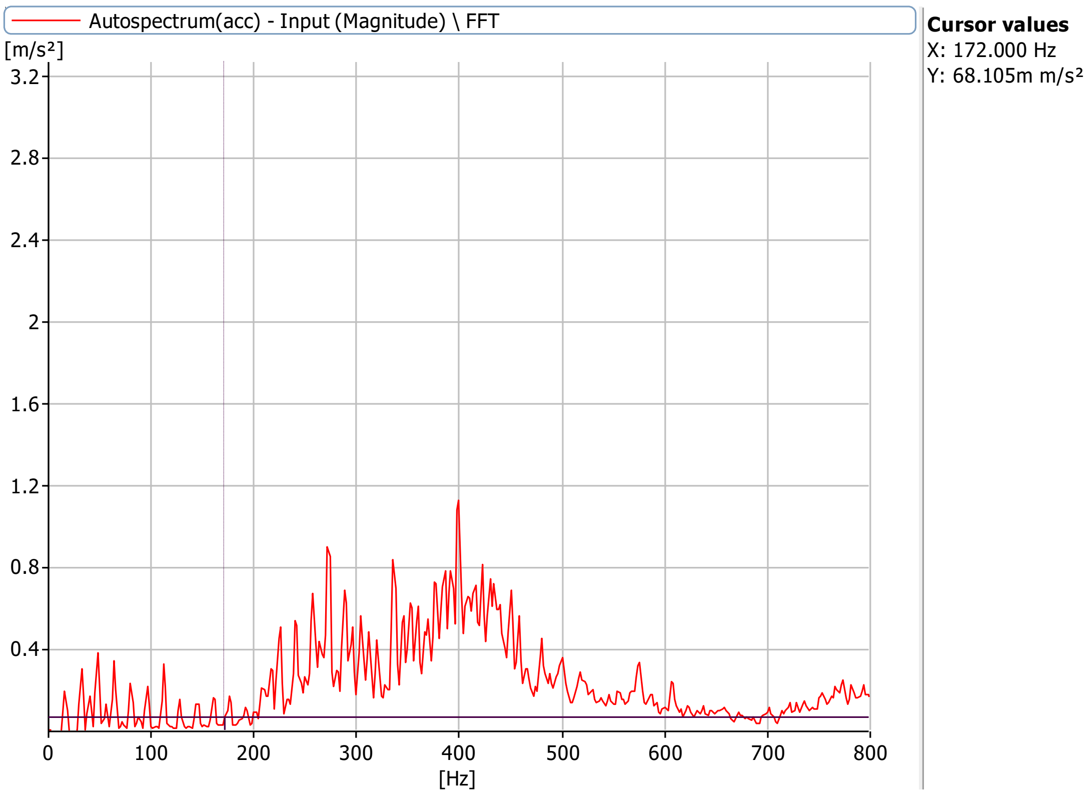


Figure 6.5.1: Frequency Spectrum at 1000 rpm with a defect size of 8 mm with no load with 0

misalignment at X Axis

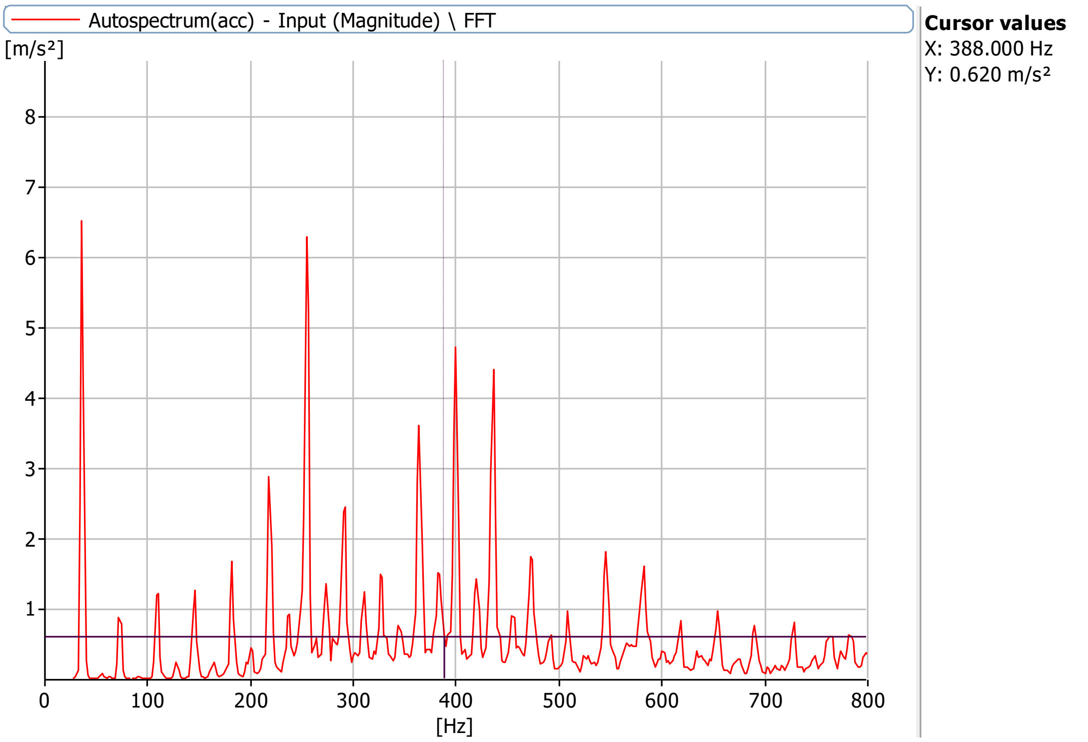


Figure 6.5.2: Frequency Spectrum at 2160 rpm with a defect size of 8 mm with no load with 0 misalignment at X Axis

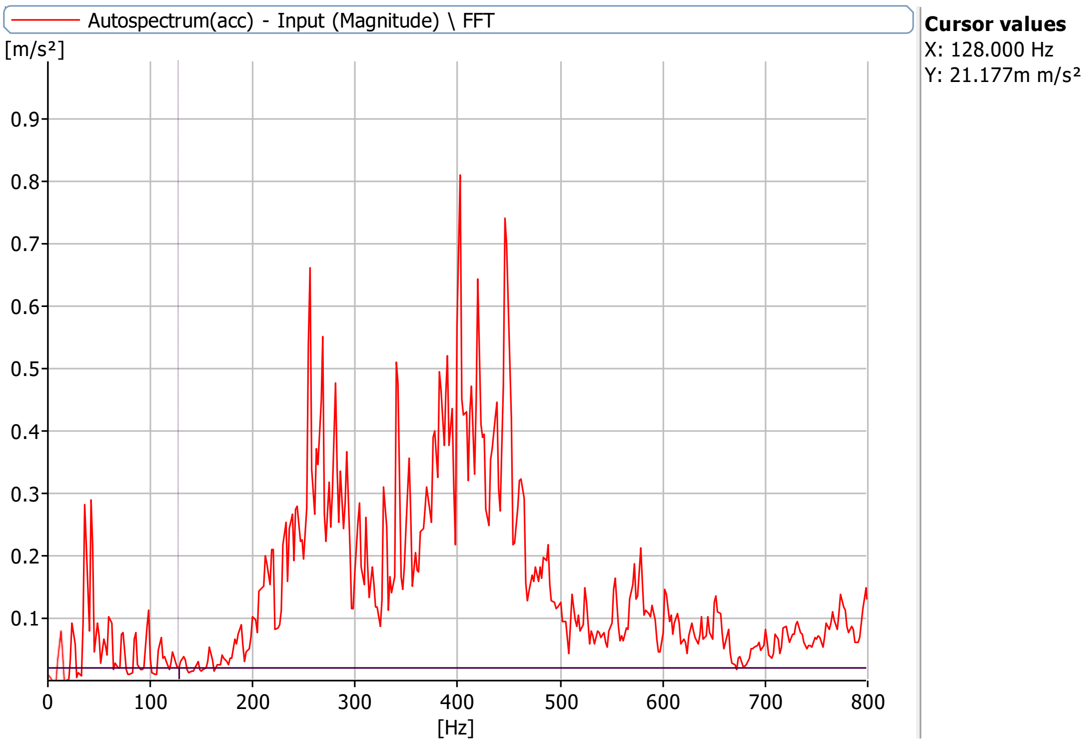


Figure 6.5.3: Frequency Spectrum at 720 rpm with a defect size of 8 mm with no load with 0 misalignment at X Axis

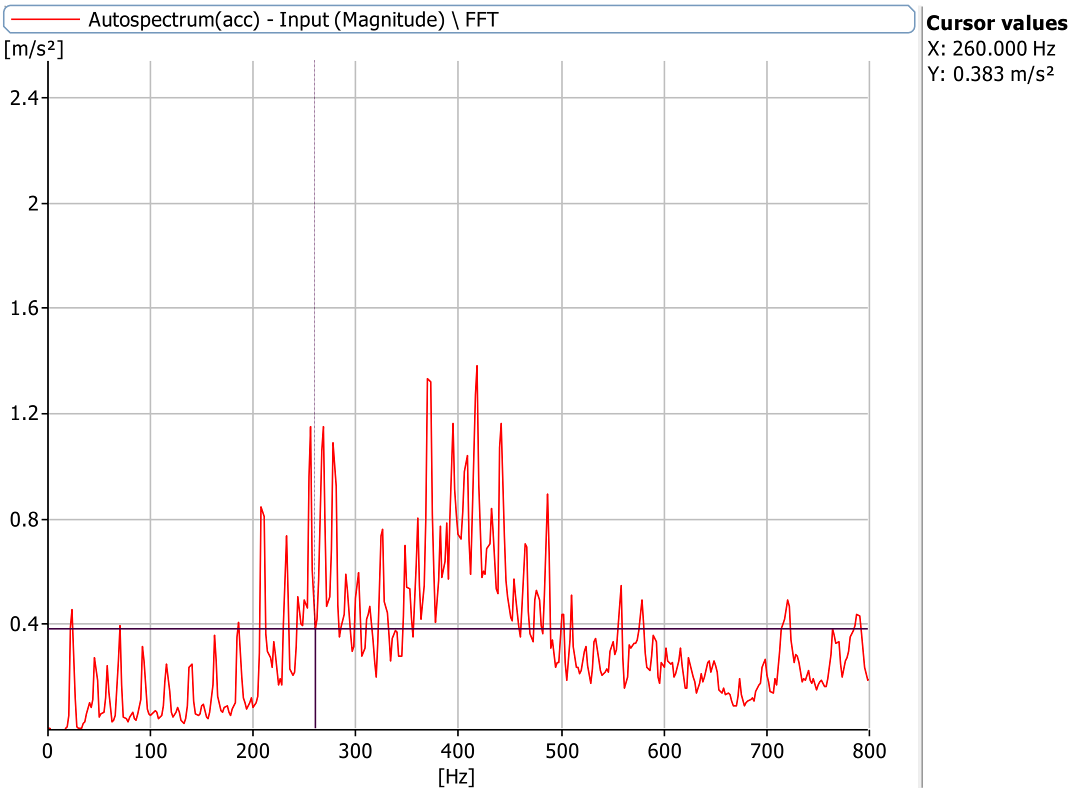


Figure 6.5.4: Frequency Spectrum at 1440 rpm with a defect size of 8 mm with no load with 0 misalignment at X Axis

Frequencies at 720 rpm, 1000 rpm,1440 rpm and 2160 rpm for defect of 0 mm bearing and 0.25 mm misalignment

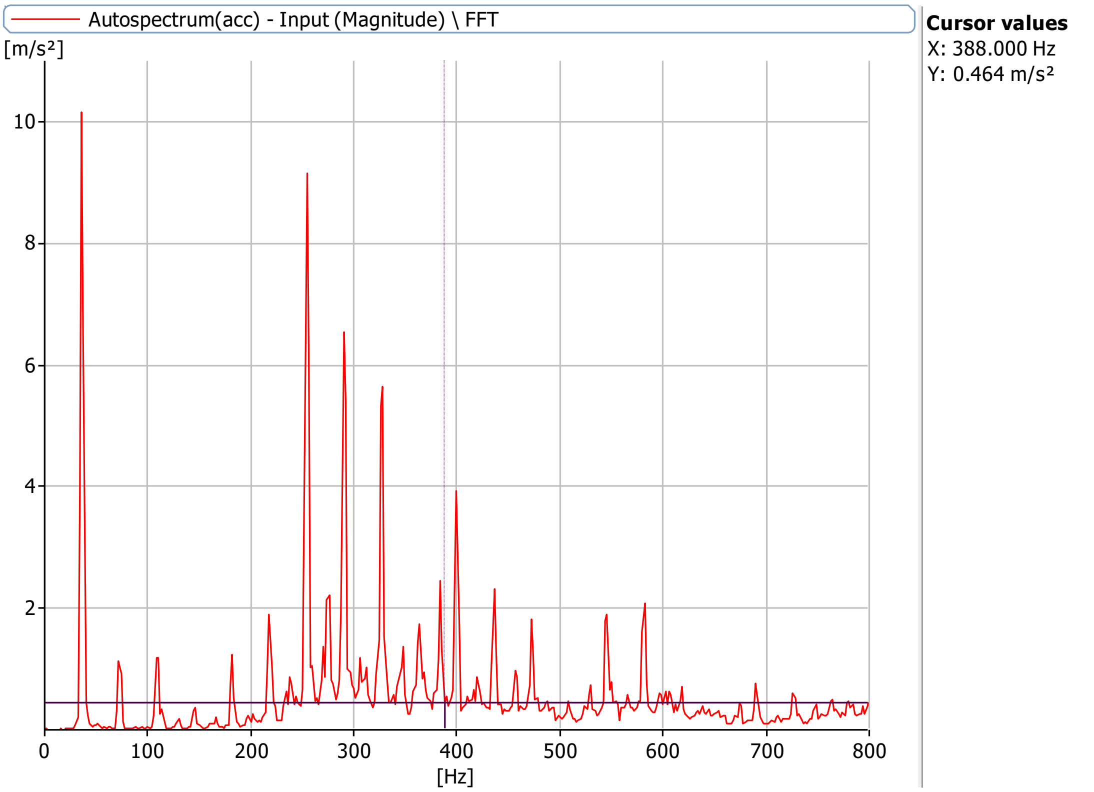


Figure 6.6.1: Frequency Spectrum at 2160 rpm with a defect size of 0 mm with no load with 0.25 misalignment at X Axis

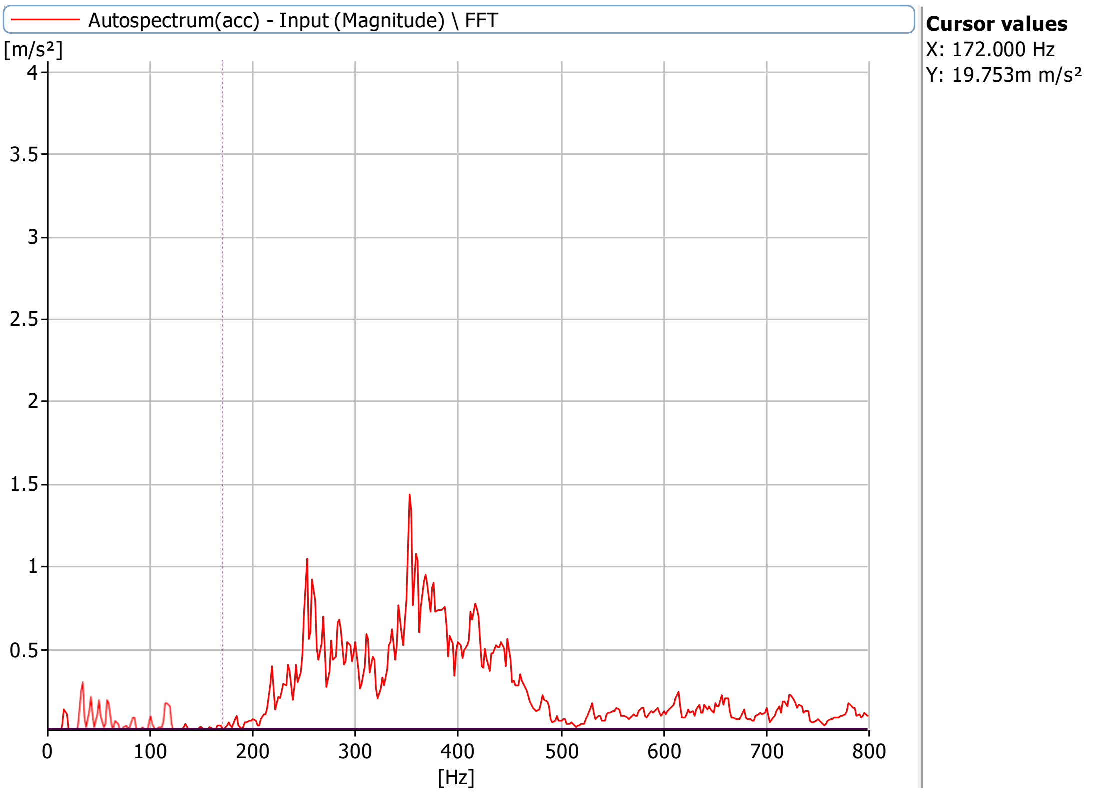


Figure 6.6.2: Frequency Spectrum at 1000 rpm with a defect size of 0 mm with no load with 0.25

misalignment at X Axis

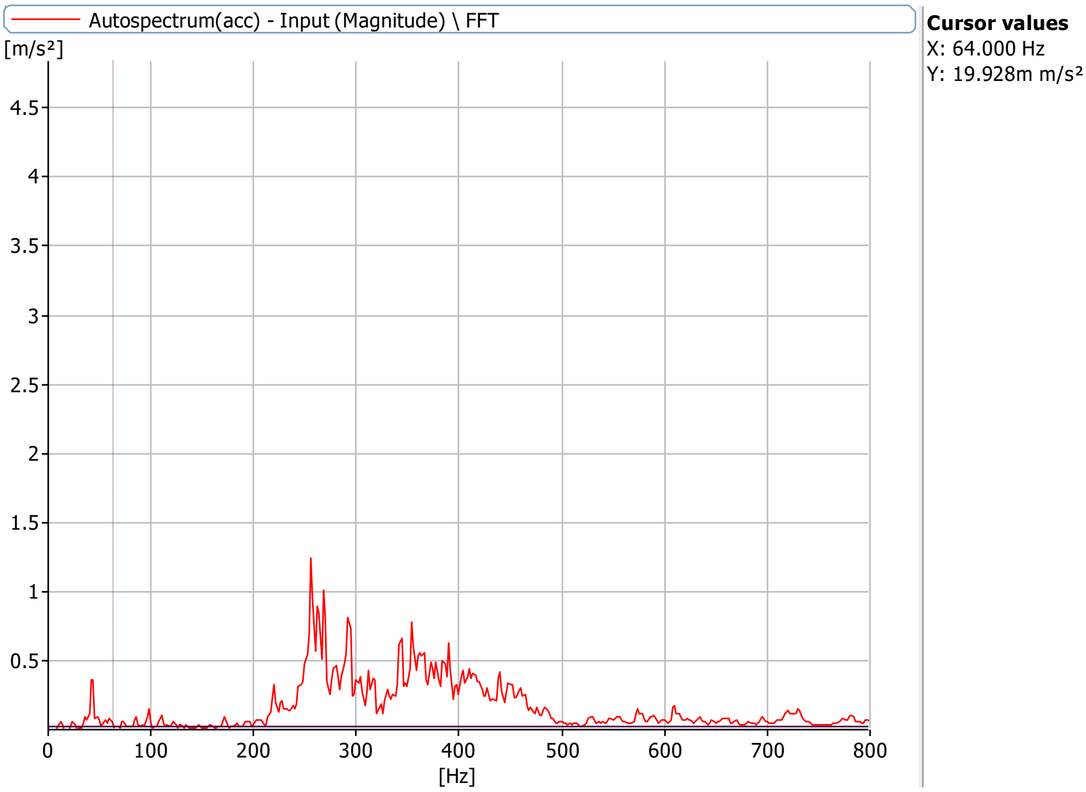


Figure 6.6.3: Frequency Spectrum at 720 rpm with a defect size of 0 mm with no load with 0.25 misalignment at X Axis

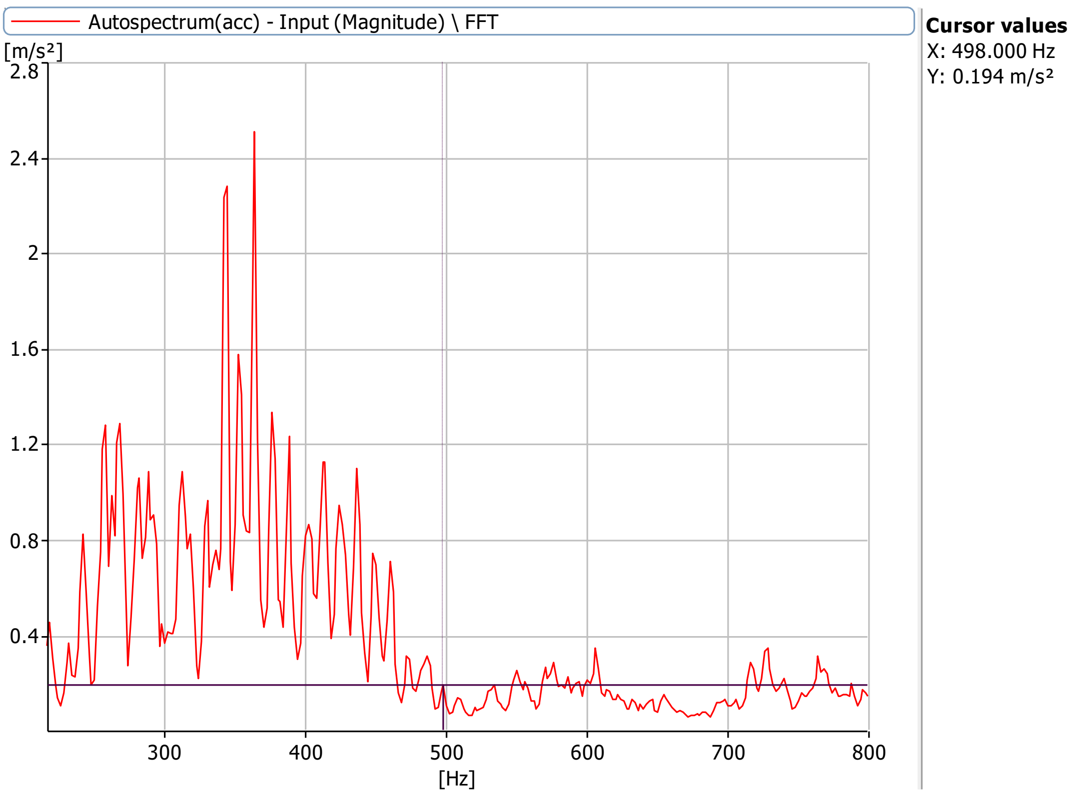


Figure 6.6.4: Frequency Spectrum at 1440 rpm with a defect size of 0 mm with no load with 0.25 misalignment at X Axis

Frequencies at 720 rpm, 1000 rpm,1440 rpm and 2160 rpm for defect of 3.2 mm,5.2mm bearing and 0.25 misalignment

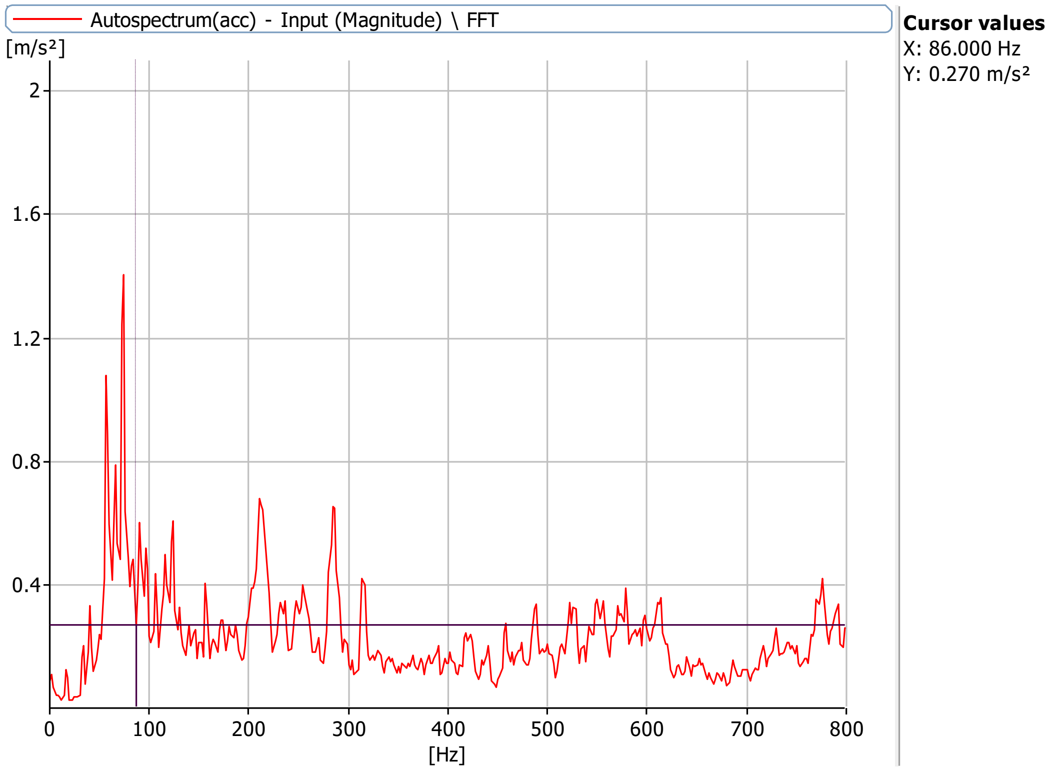


Figure 6.7.1: Frequency Spectrum at 2160 rpm with a defect size of 3.2 mm,5.1 mm with no load with 0.25 misalignment at X Axis

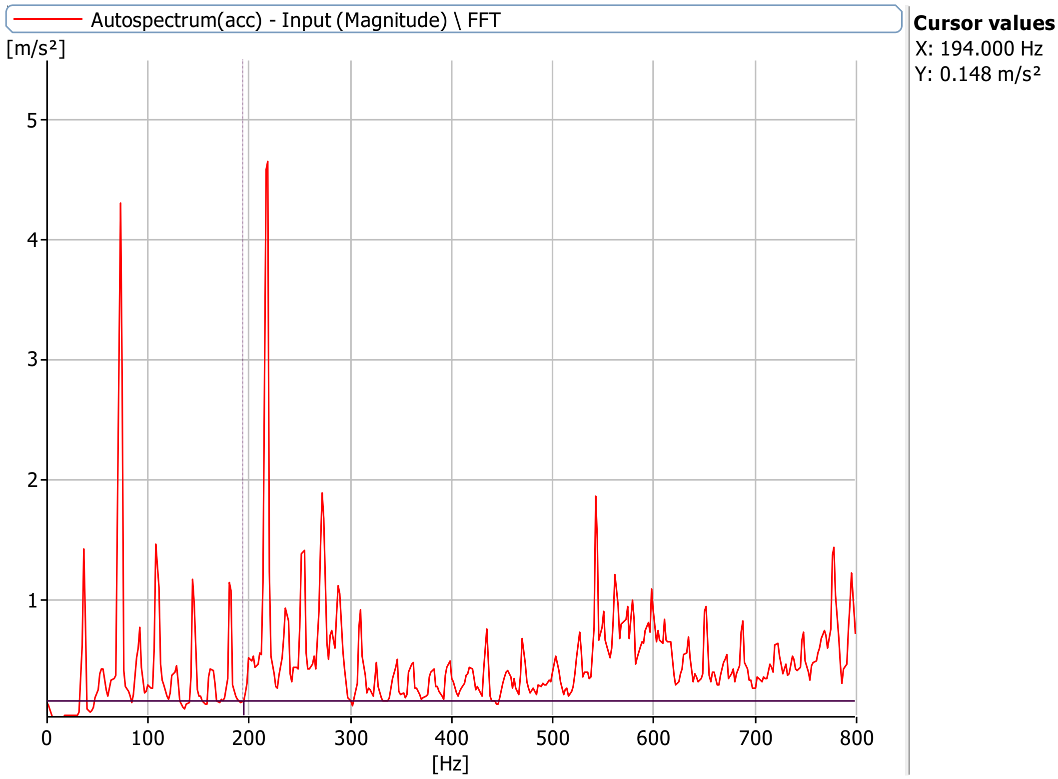


Figure 6.7.2: Frequency Spectrum at 1000 rpm with a defect size of 3.2 mm,5.1 mm with no load with 0.25 misalignment at X Axis

Frequencies at 720 rpm, 1000 rpm,1440 rpm and 2160 rpm for defect of 3.2 mm,5.2mm bearing and 0.25

misalignment

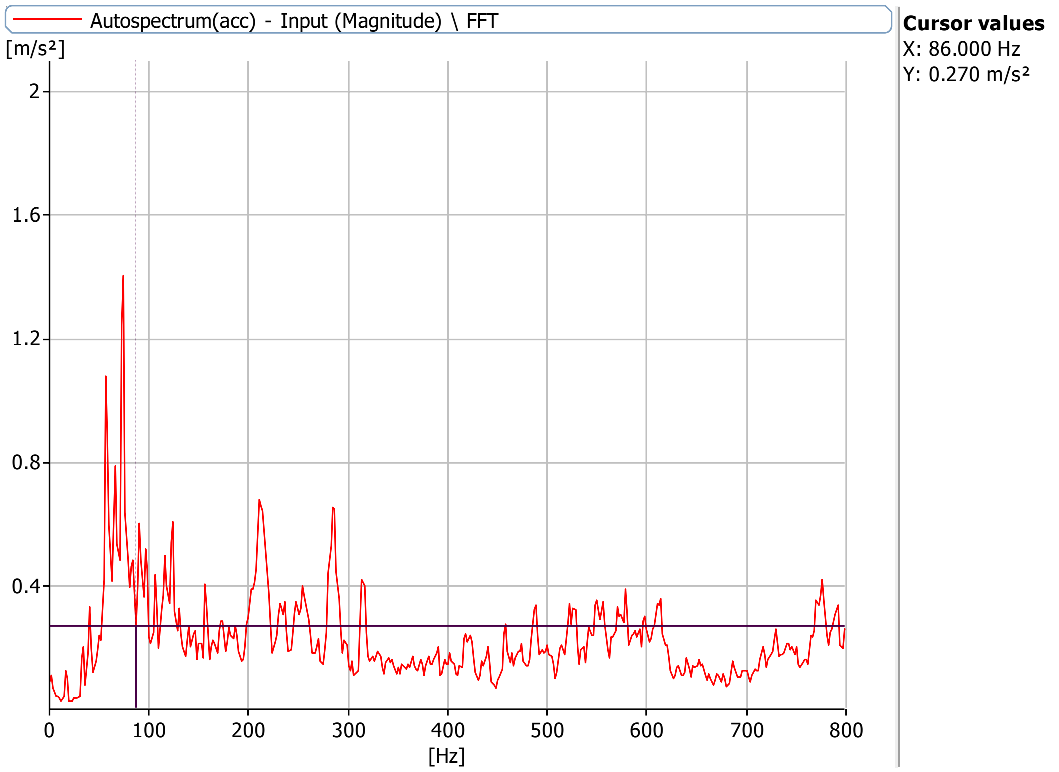


Figure 6.7.1: Frequency Spectrum at 2160 rpm with a defect size of 3.2 mm,5.1 mm with no load with 0.25 misalignment at X Axis

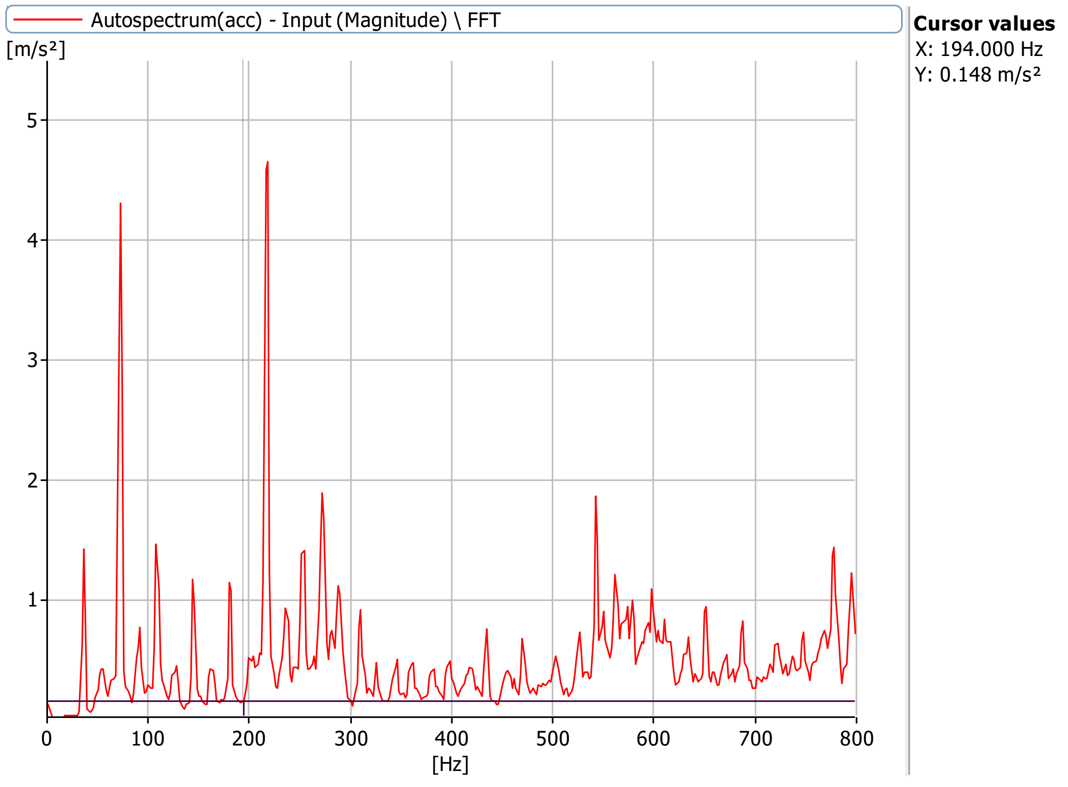


Figure 6.7.2: Frequency Spectrum at 1000 rpm with a defect size of 3.2 mm,5.1 mm with no load with 0.25 misalignment at X Axis

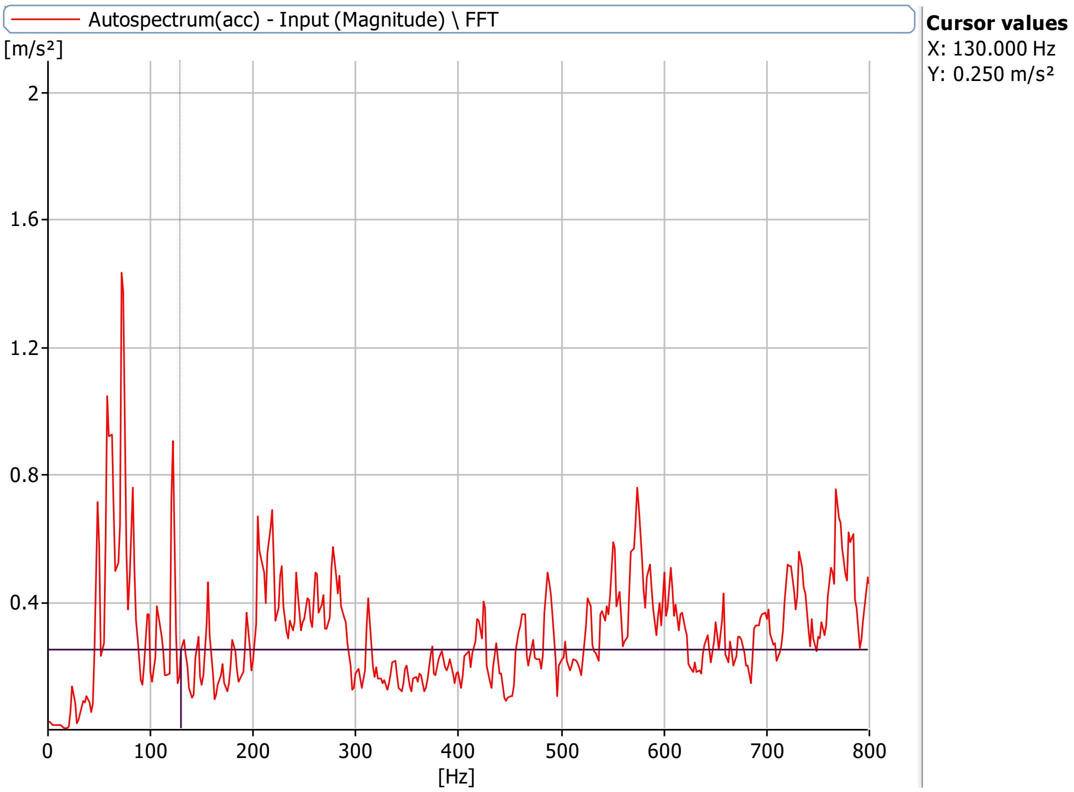


Figure 6.7.3: Frequency Spectrum at 1440 rpm with a defect size of 3.2 mm,5.1 mm with no load with 0.25 misalignment at X Axis

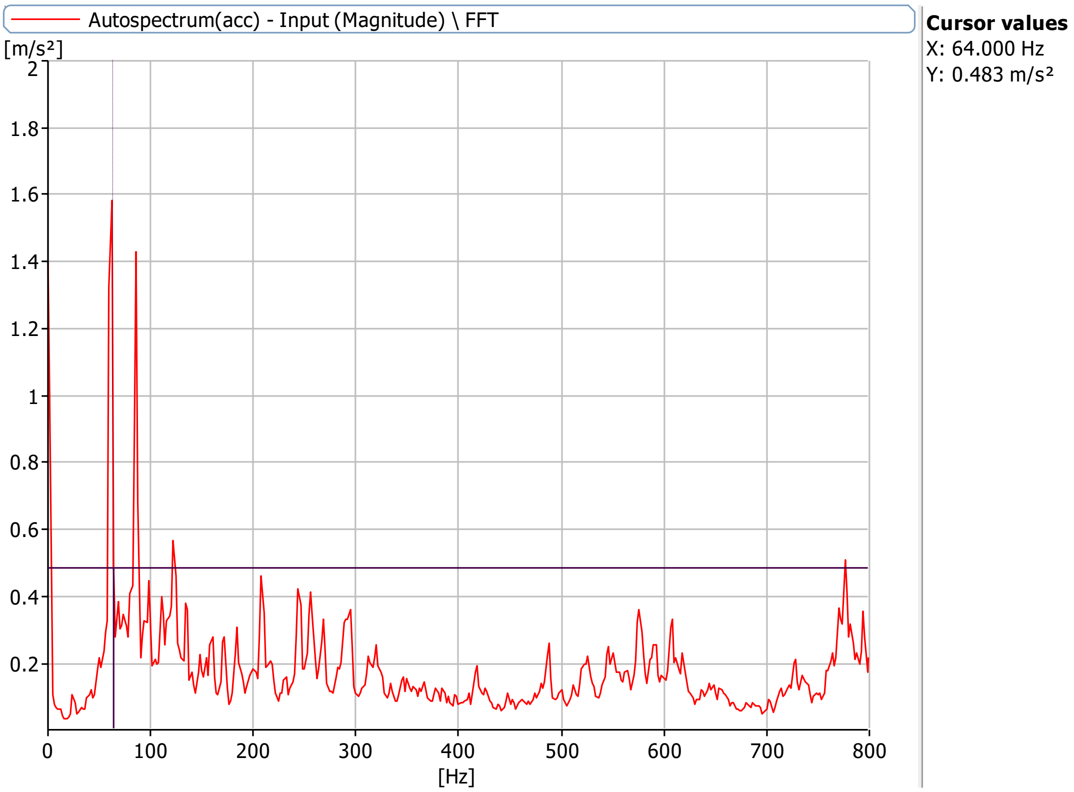


Figure 6.7.4: Frequency Spectrum at 720 rpm with a defect size of 3.2 mm,5.1 mm with no load with 0.25 misalignment at X Axis

Frequencies at 720 rpm, 1000 rpm,1440 rpm and 2160 rpm for defect of 3.2mm,8.2 mm bearing and 0.25

misalignment

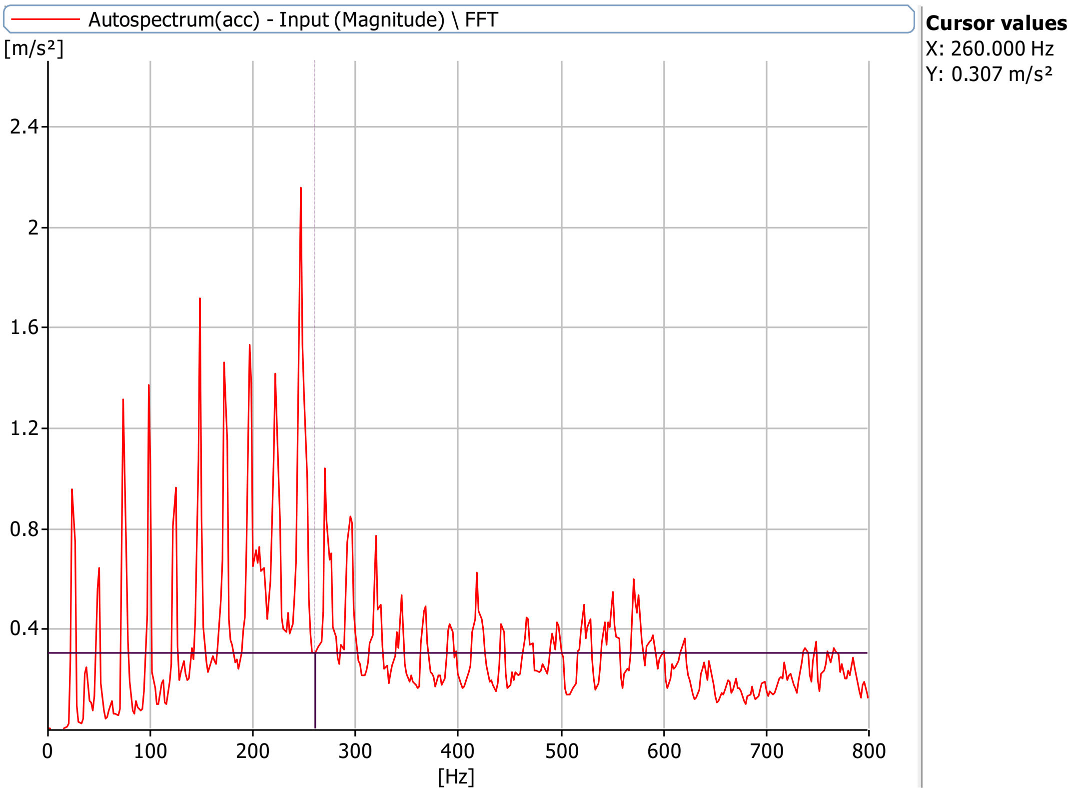


Figure 6.8.1: Frequency Spectrum at 1440 rpm with a defect size of 3.2 mm,8.2mm with no load with 0.25 misalignment at X Axis

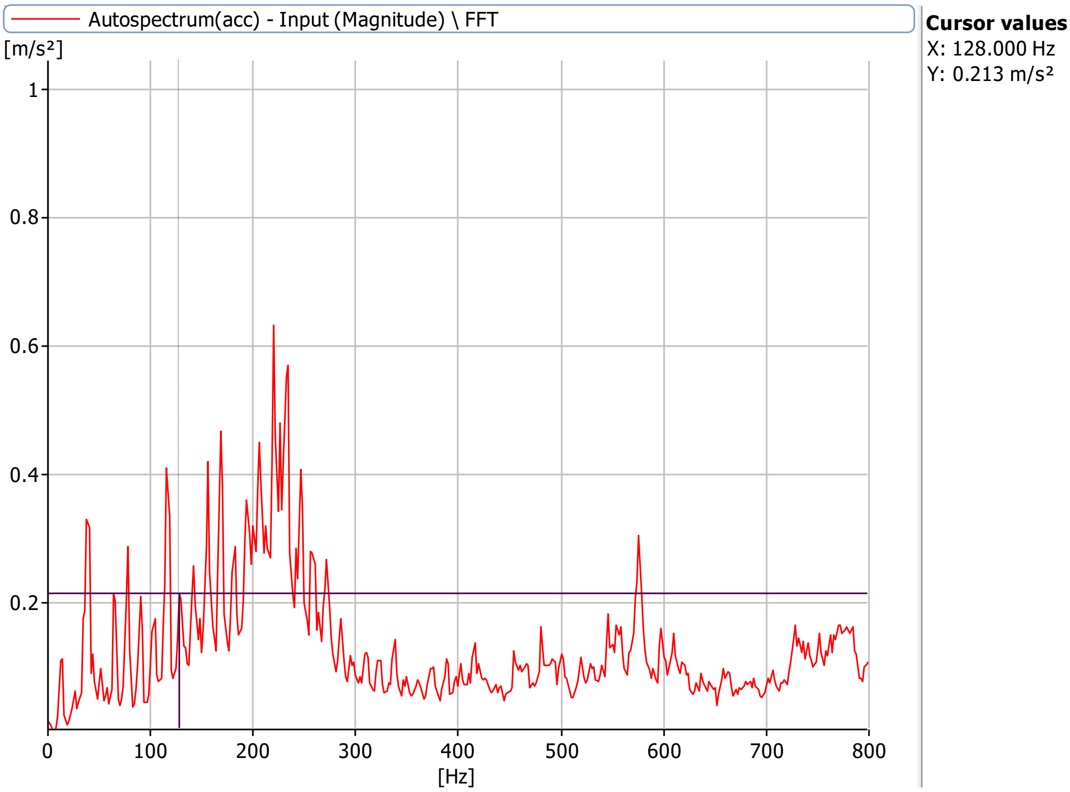


Figure 6.8.2: Frequency Spectrum at 720 rpm with a defect size of 3.2 mm,8.2mm with no load with 0.25 misalignment at X Axis

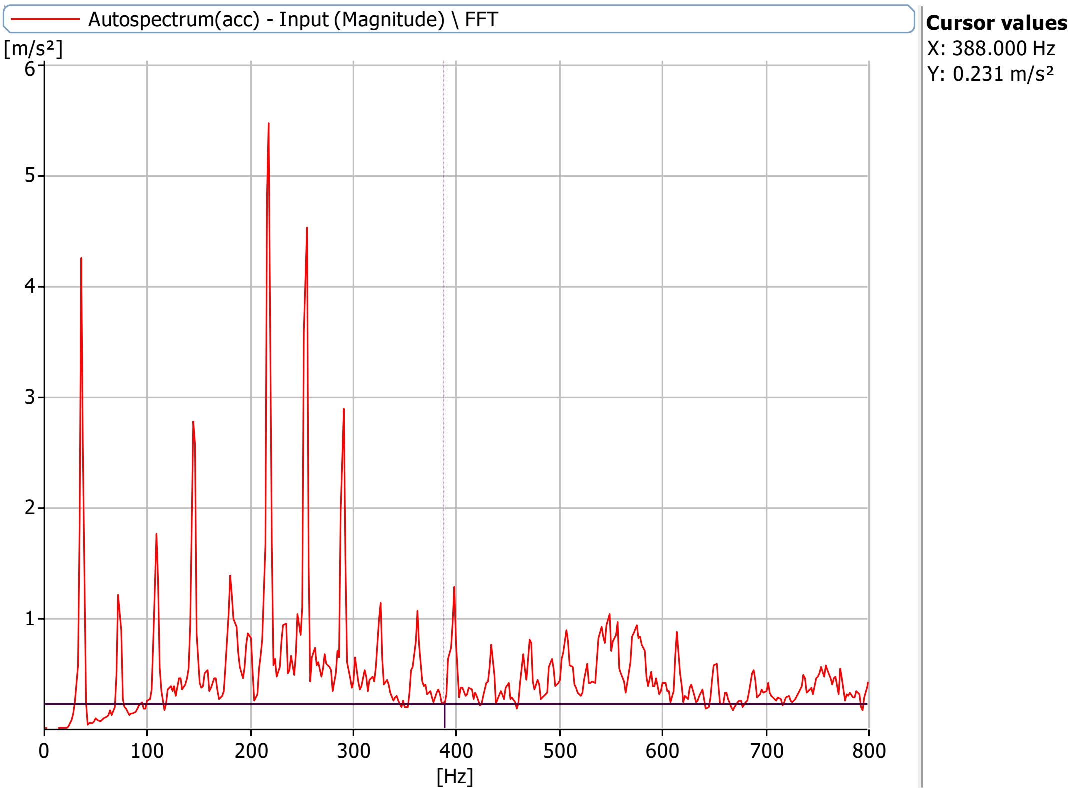


Figure 6.8.3: Frequency Spectrum at 2160 rpm with a defect size of 3.2 mm,8.2mm with no load with 0.25 misalignment at X Axis

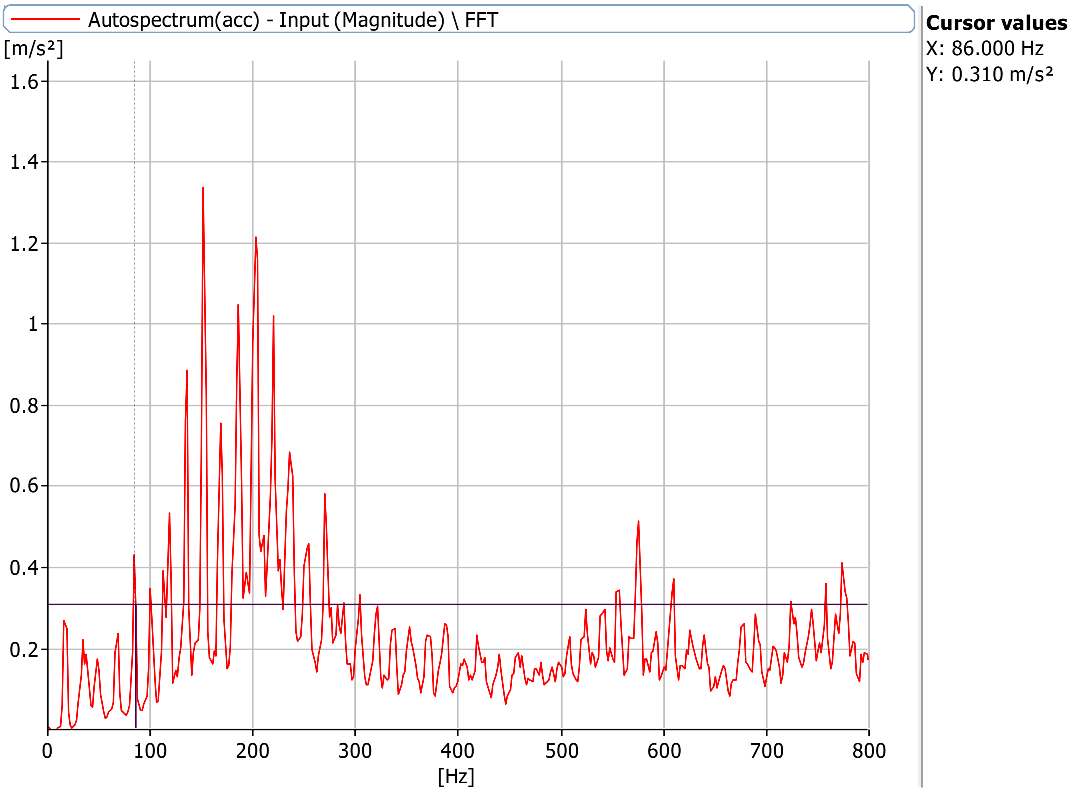


Figure 6.8.4: Frequency Spectrum at 1000 rpm with a defect size of 3.2 mm,8.2mm with no load with 0.25 misalignment at X Axis

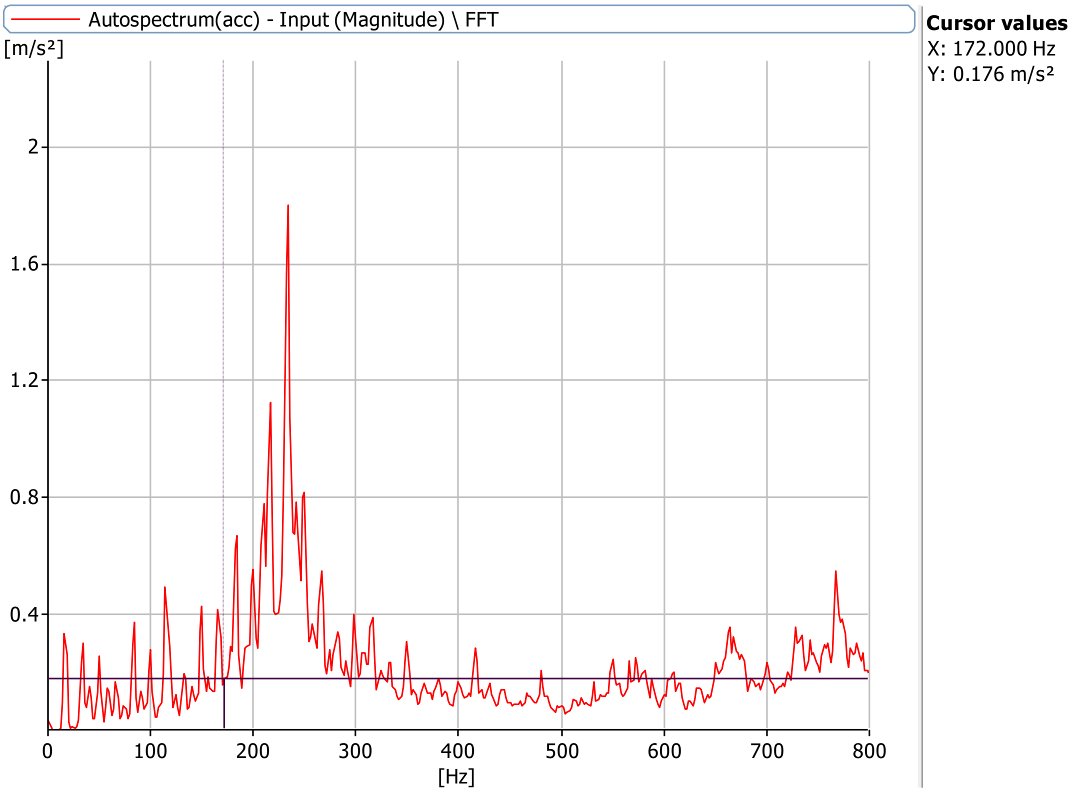


Figure 6.9.1: Frequency Spectrum at 1000 rpm with a defect size of 5.1mm with no load with 0.25 misalignment at X Axis

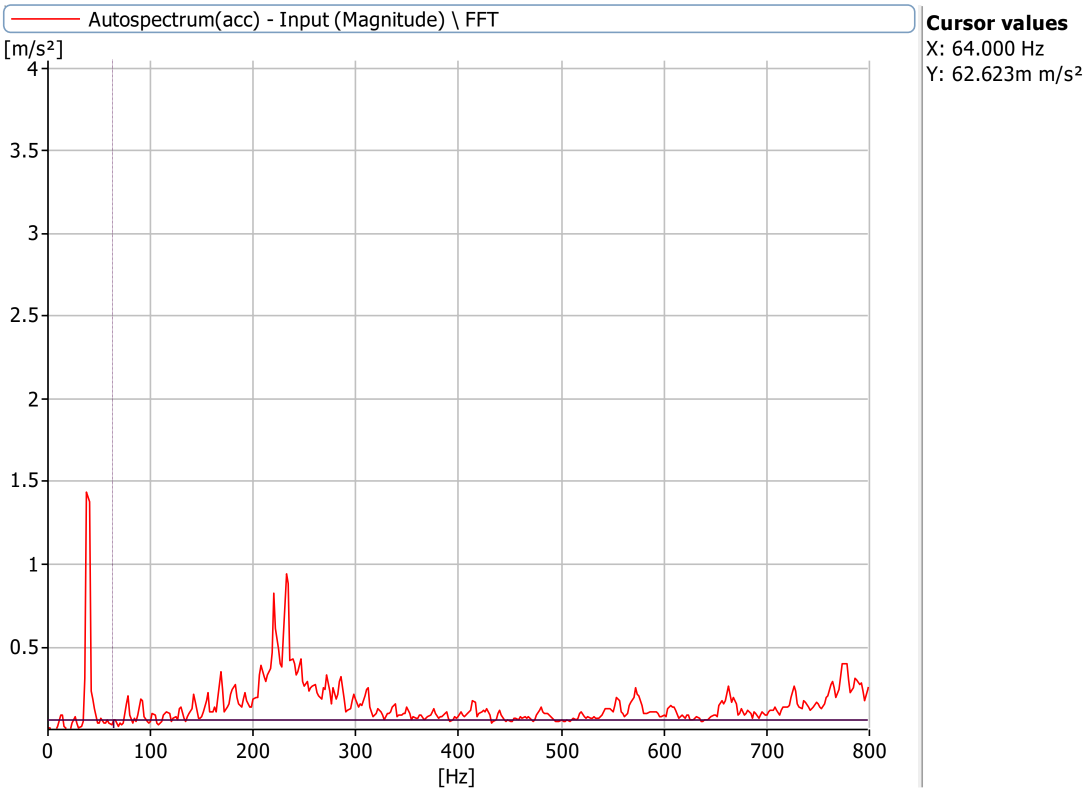


Figure 6.9.2: Frequency Spectrum at 720 rpm with a defect size of 5.1 mm with no load with 0.25 misalignment at X Axis

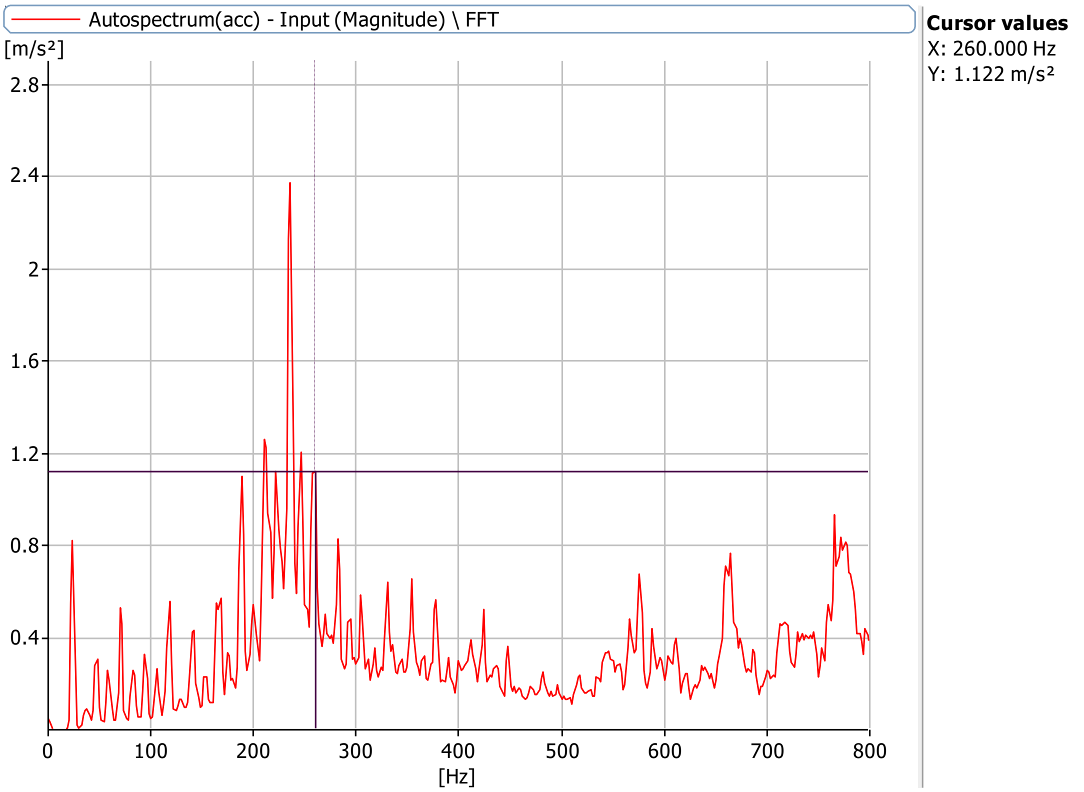


Figure 6.9.3: Frequency Spectrum at 1440 rpm with a defect size of 5.1 mm with no load with 0.25 misalignment at X Axis

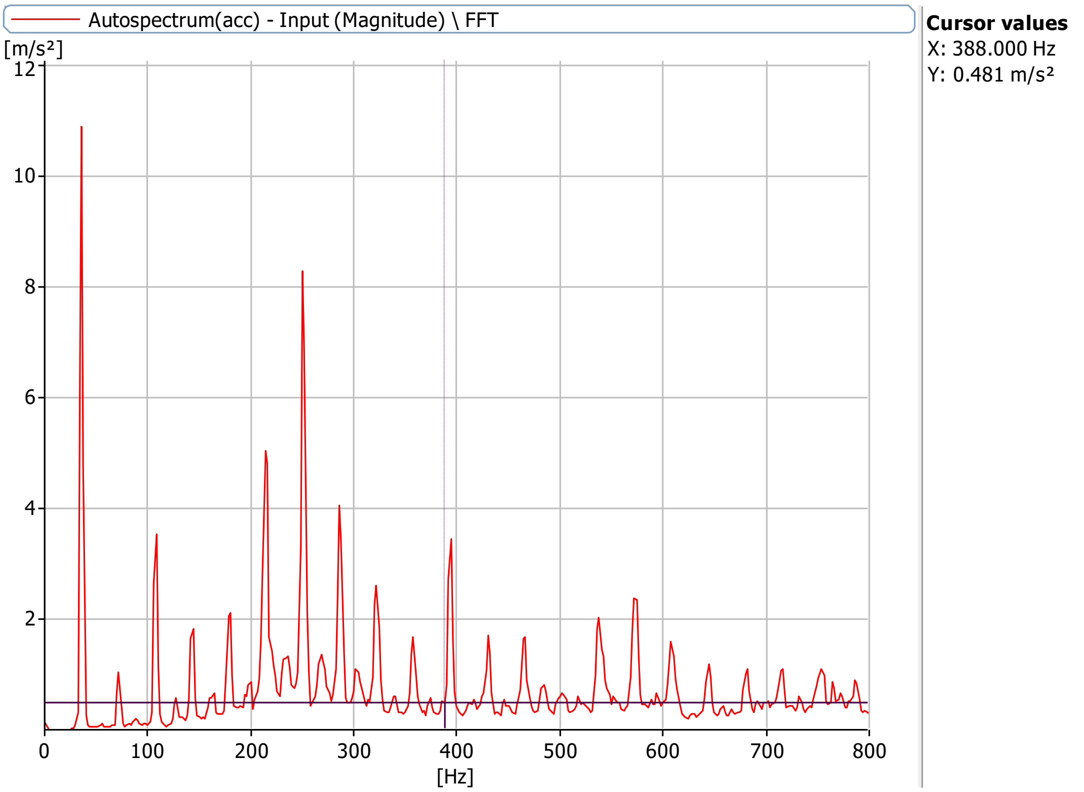


Figure 6.9.4: Frequency Spectrum at 2160 rpm with a defect size of 5.1 mm with no load with 0.25 misalignment at X Axis

Frequencies at 720 rpm, 1000 rpm,1440 rpm and 2160 rpm for defect of 8 mm bearing and 0.25 misalignmen

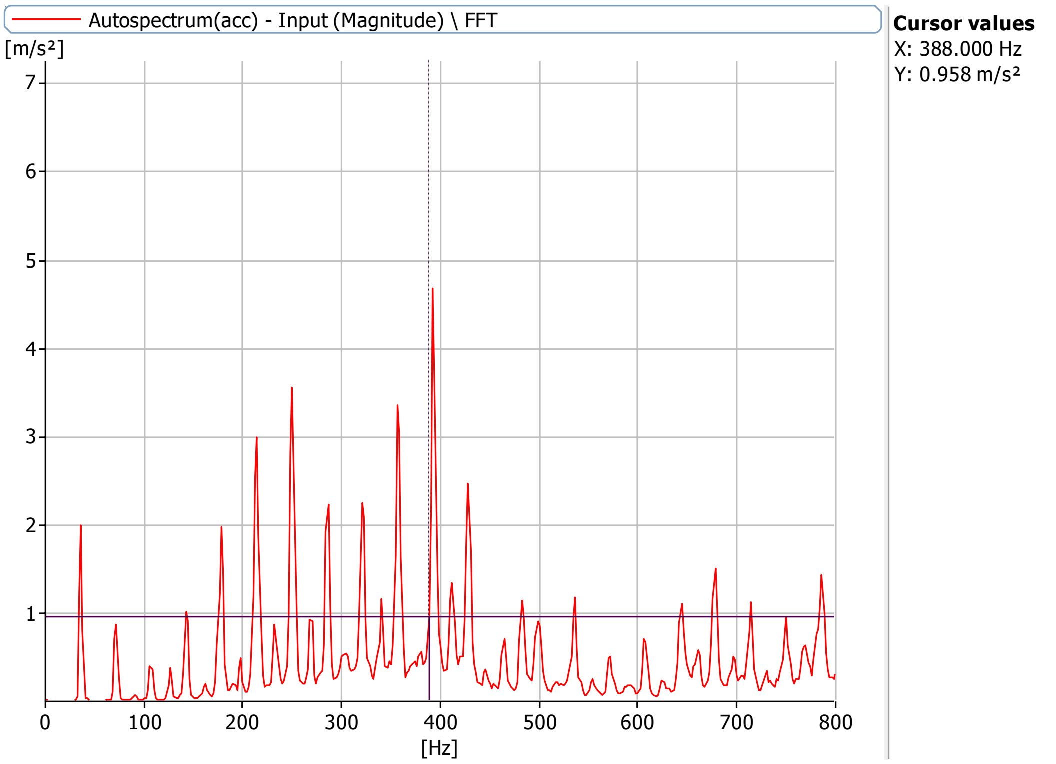


Figure 6.10.1: Frequency Spectrum at 2160 rpm with a defect size of 8 mm with no load with 0.25 misalignment at X Axis

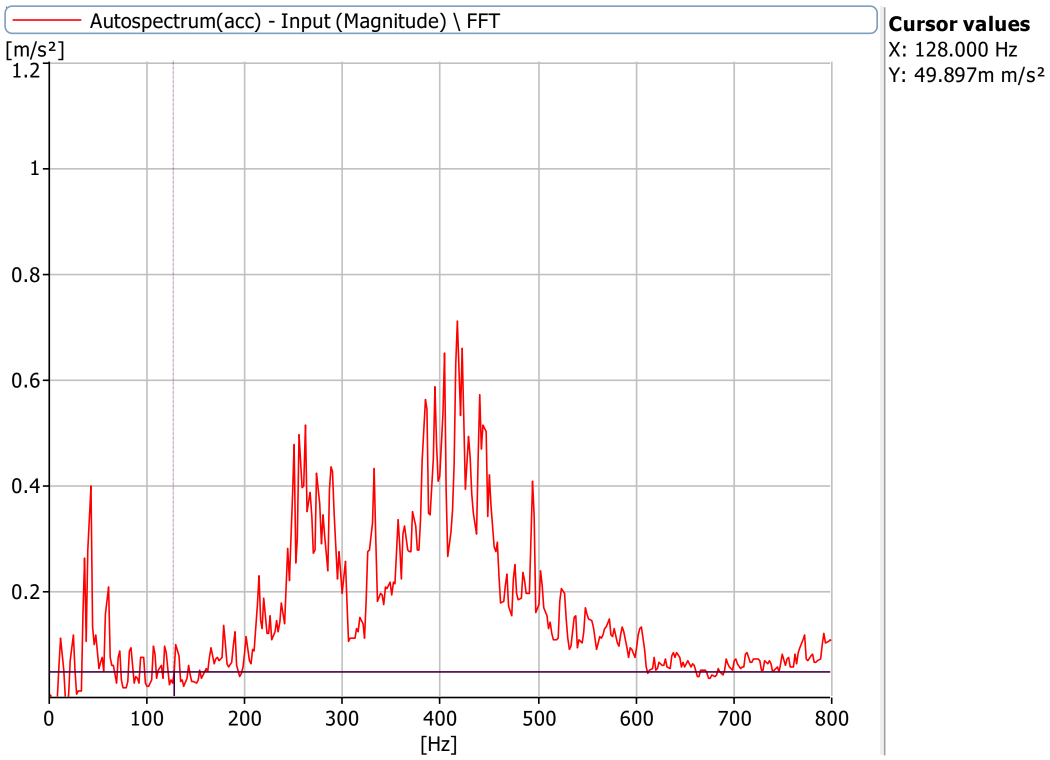


Figure 6.10.2: Frequency Spectrum at 720 rpm with a defect size of 8 mm with no load with 0.25 misalignment at X Axis

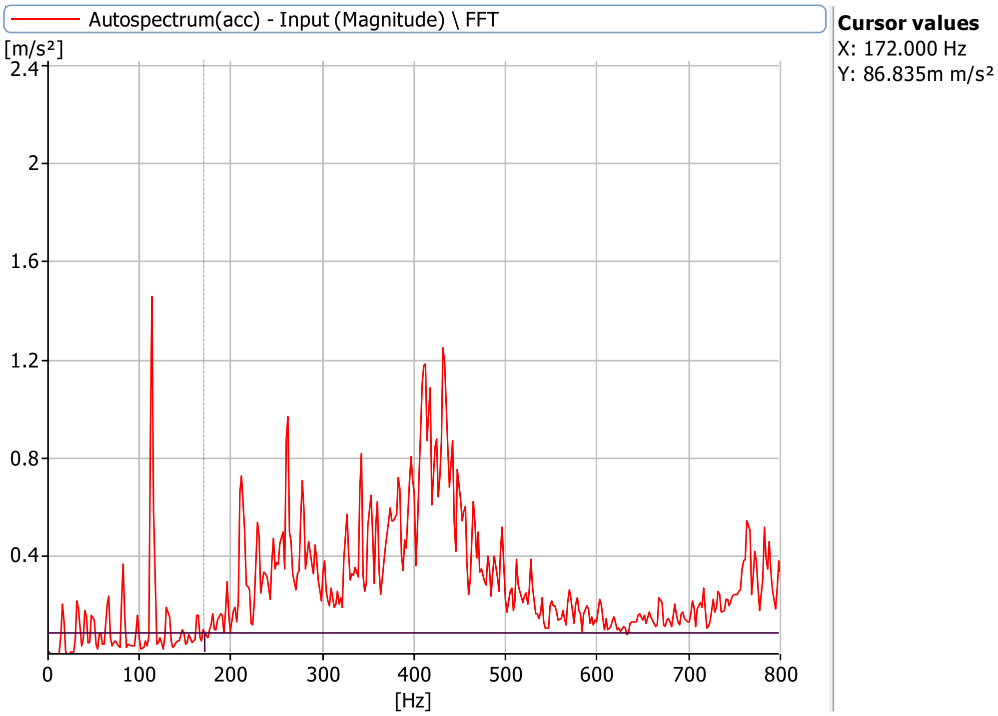


Figure 6.10.3: Frequency Spectrum at 1000 rpm with a defect size of 8 mm with no load with 0.25 misalignment at X Axis

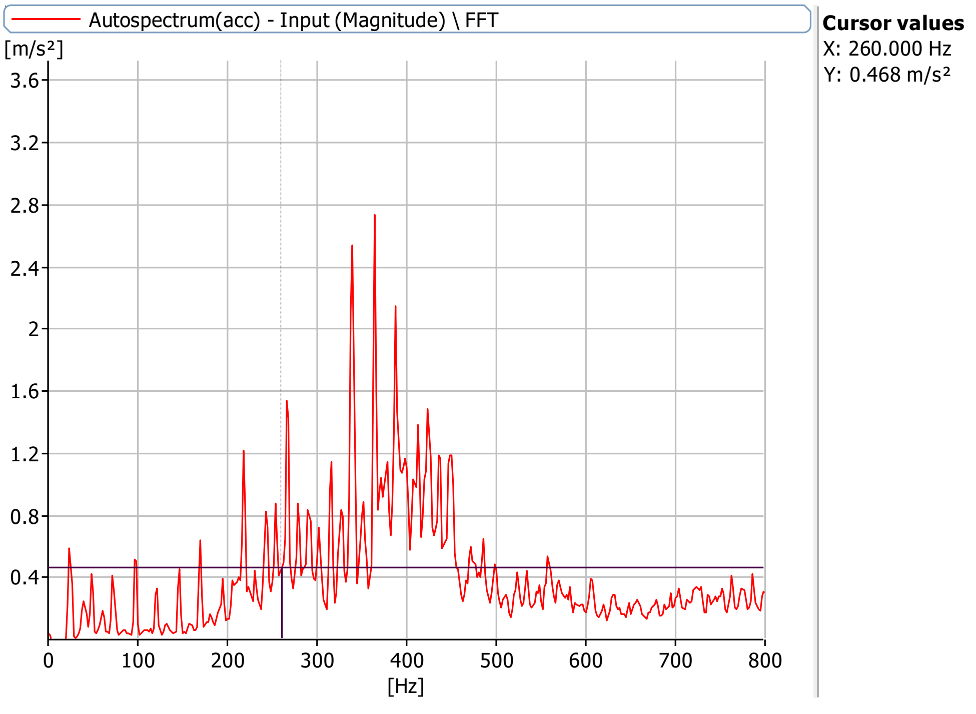


Figure 6.10.4: Frequency Spectrum at 1440 rpm with a defect size of 8 mm with no load with 0.25 misalignment at X Axis

Frequencies at 720 rpm, 1000 rpm,1440 rpm and 2160 rpm for defect of 0 mm bearing and 0.5 misalignment

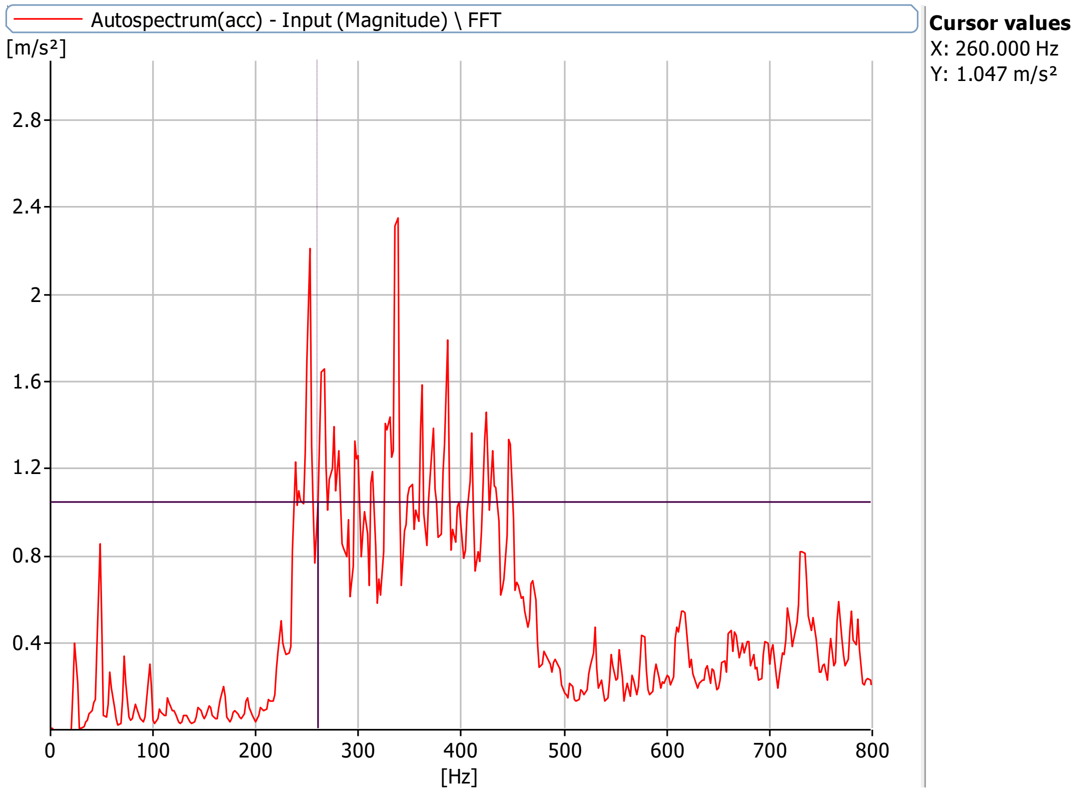


Figure 6.11.1: Frequency Spectrum at 1440 rpm with a defect size of 8 mm with no load with 0.5 misalignment at X Axis

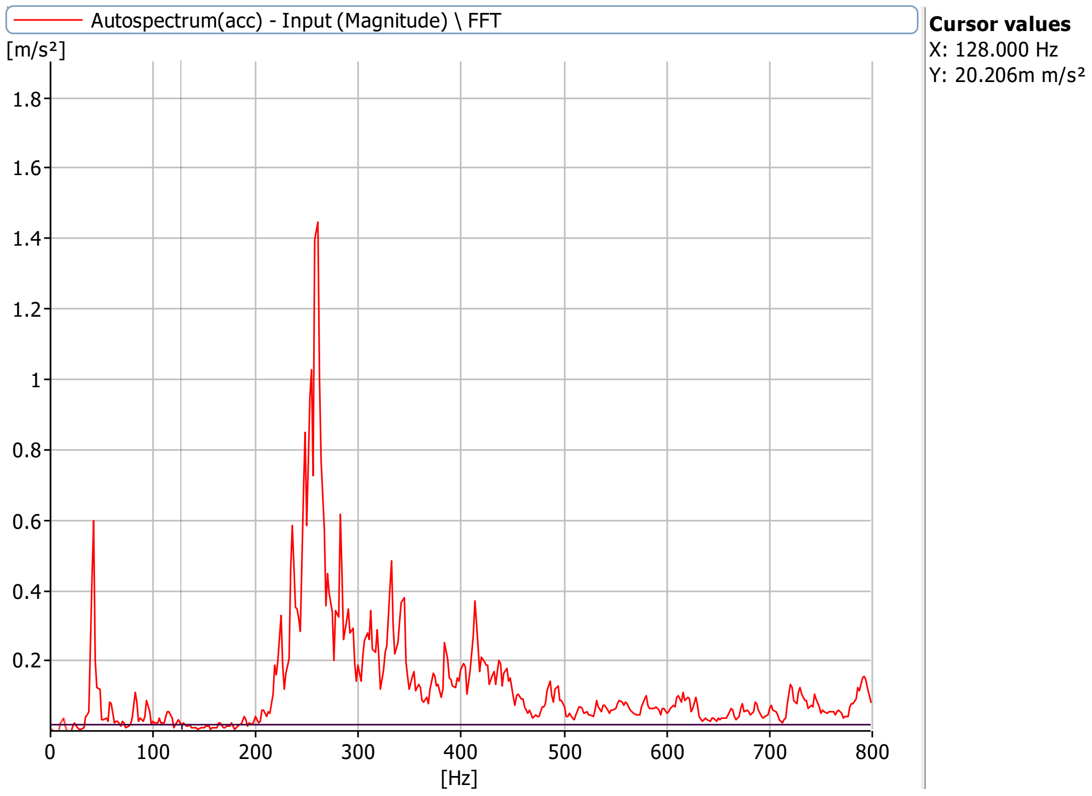


Figure 6.11.2: Frequency Spectrum at 720 rpm with a defect size of 8 mm with no load with 0.5 misalignment at X Axis

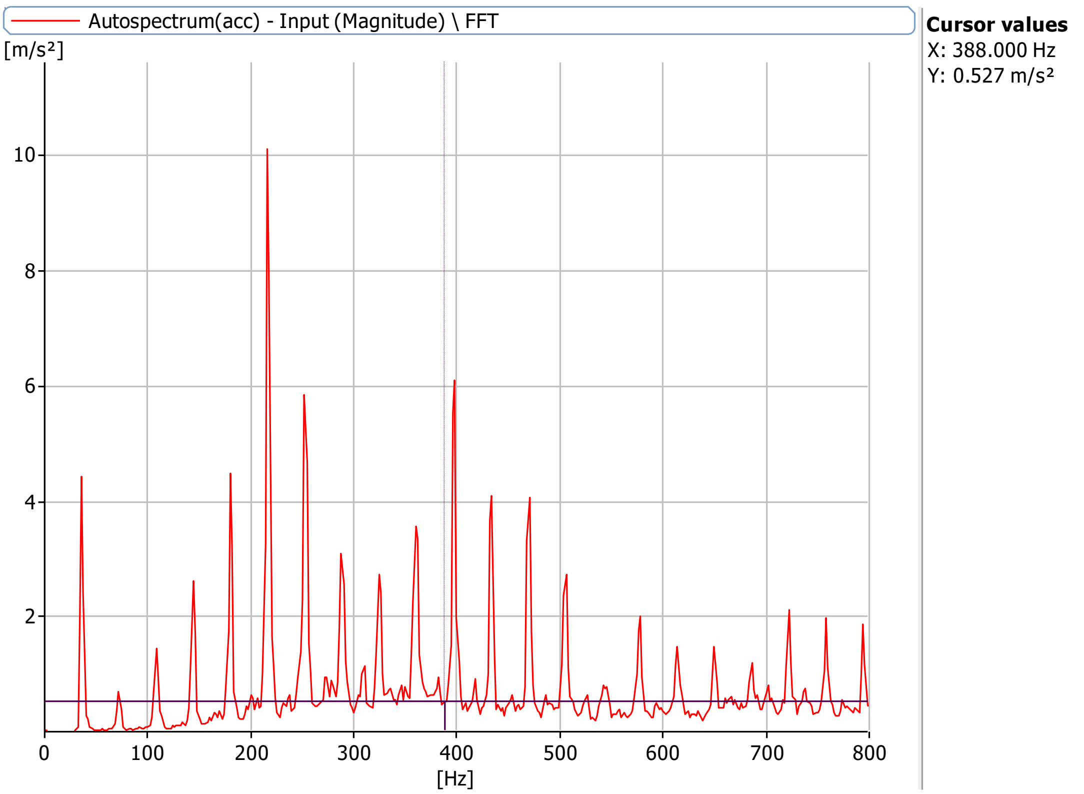


Figure 6.11.3: Frequency Spectrum at 2160 rpm with a defect size of 0 mm with no load with 0.5 misalignment at X Axis

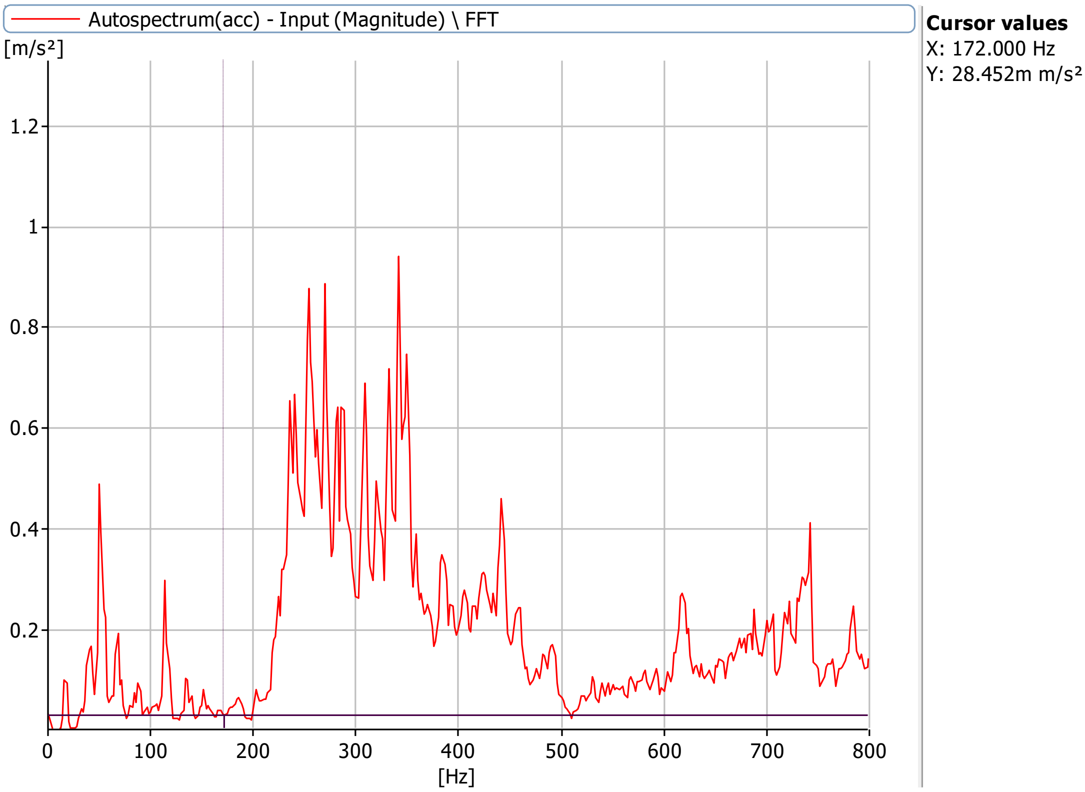


Figure 6.11.4: Frequency Spectrum at 1000 rpm with a defect size of 0 mm with no load with 0.5 misalignment at X Axis

Frequencies at 720 rpm, 1000 rpm,1440 rpm and 2160 rpm for defect of 3.2 mm,5.1mm bearing and 0.5mm misalignment

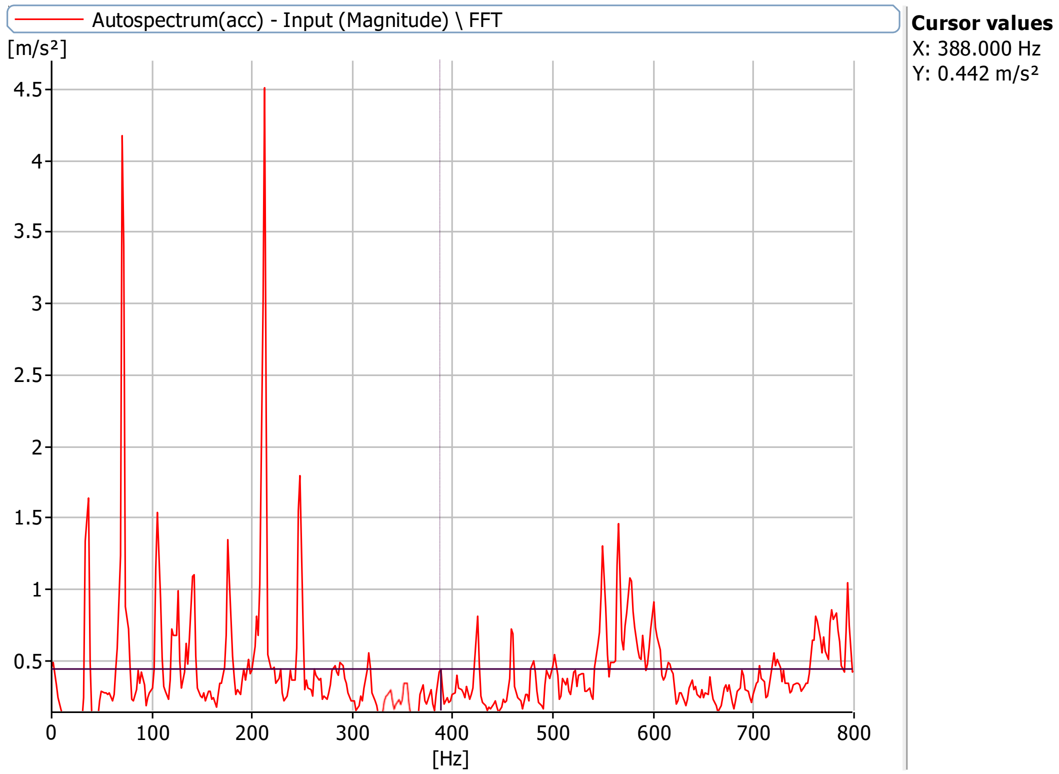


Figure 6.12.1: Frequency Spectrum at 2160 rpm with a defect size of 3.2 mm,5.1mm with no load with 0.5 misalignment at X Axis

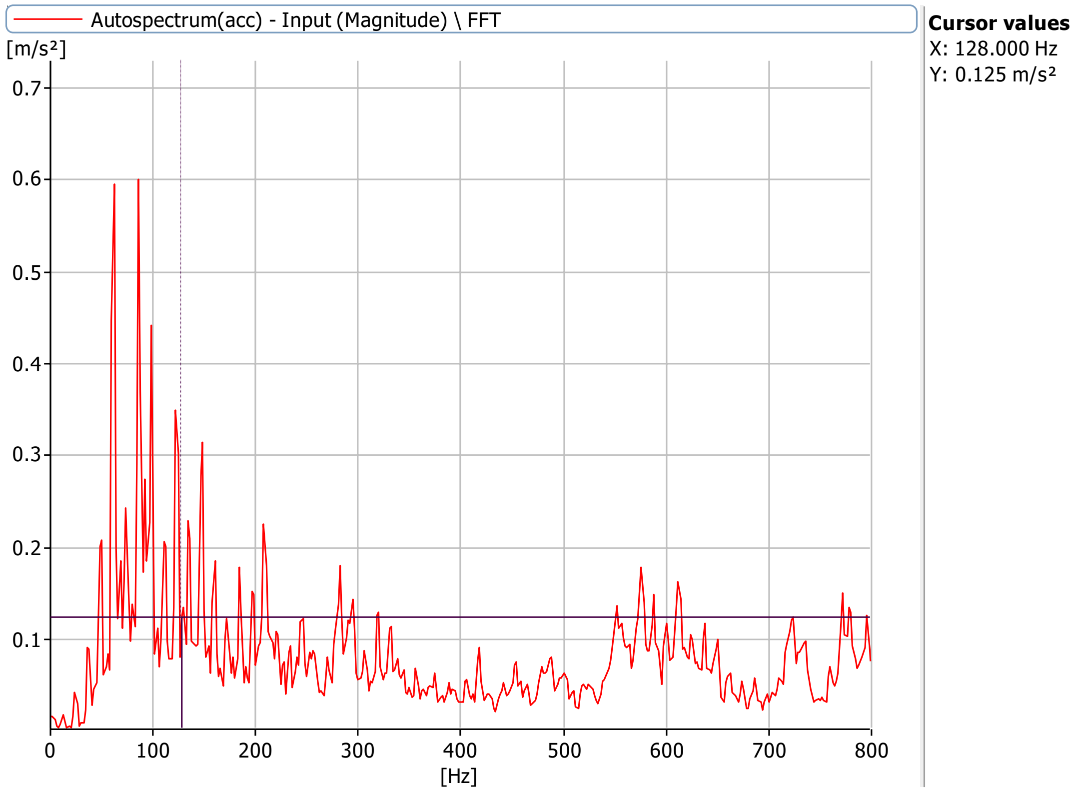


Figure 6.12.2: Frequency Spectrum at 720 rpm with a defect size of 3.2 mm,5.1mm with no load with 0.5 misalignment at X Axis

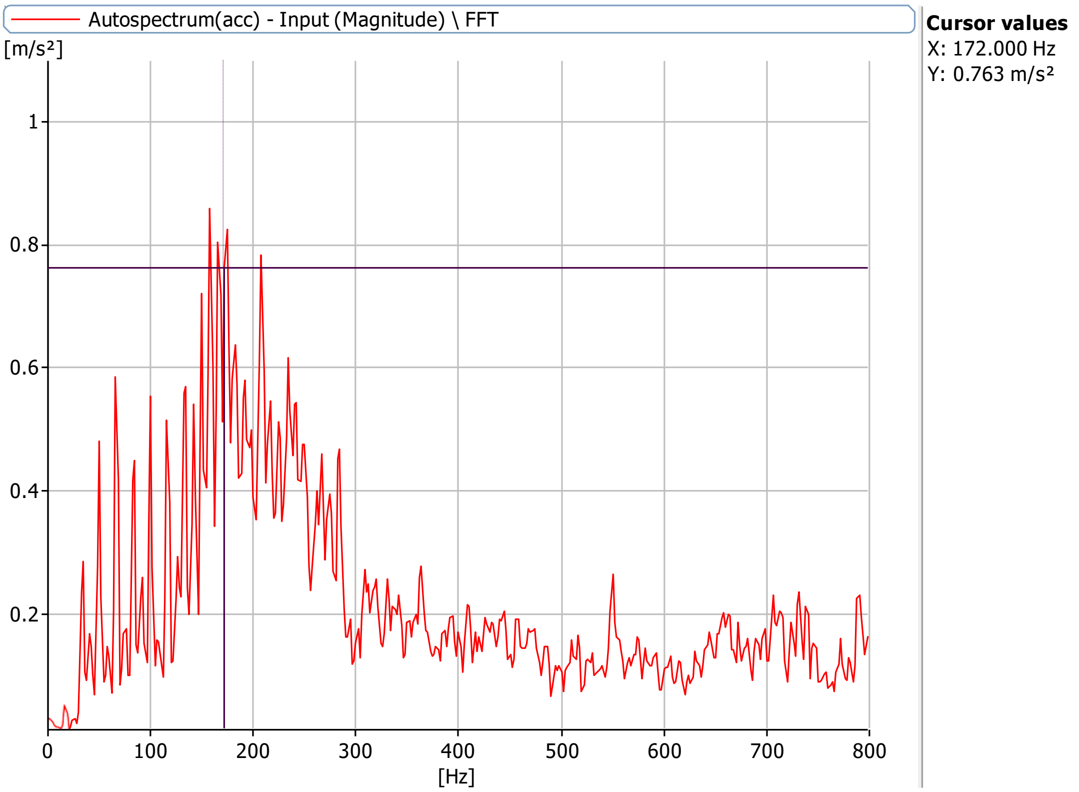


Figure 6.12.3: Frequency Spectrum at 1000 rpm with a defect size of 3.2 mm,5.1mm with no load with 0.5 misalignment at X Axis

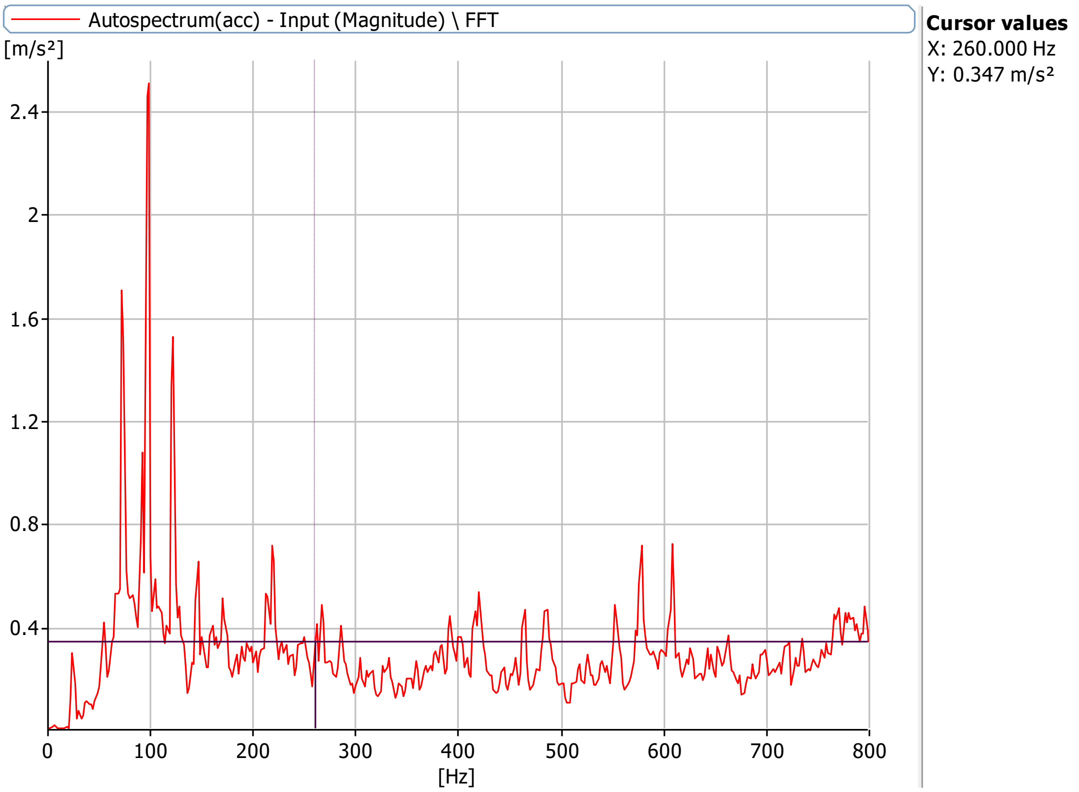


Figure 6.12.4: Frequency Spectrum at 1440 rpm with a defect size of 3.2 mm,5.1mm with no load with 0.5 misalignment at X Axis

Frequencies at 720 rpm, 1000 rpm,1440 rpm and 2160 rpm for defect of 3.2 mm,8.2mm bearing and 0.5mm misalignment

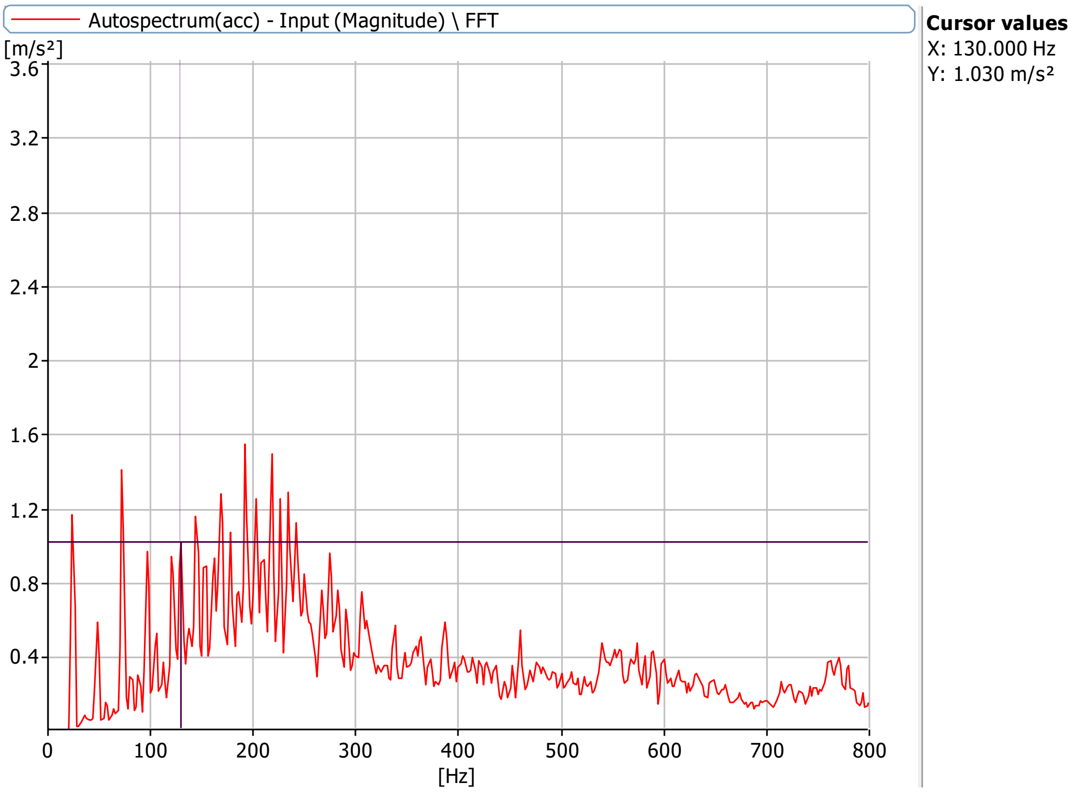


Figure 6.13.1: Frequency Spectrum at 1440 rpm with a defect size of 3.2 mm,8.2mm with no load with 0.5 misalignment at X Axis

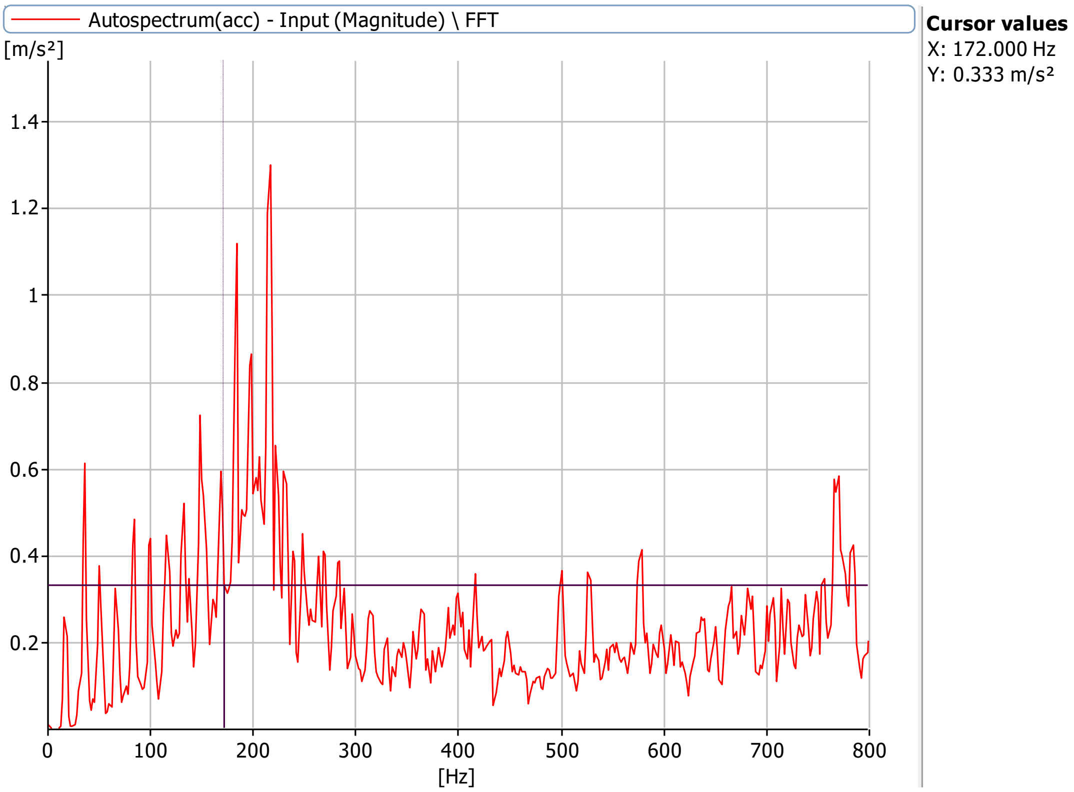


Figure 6.13.2: Frequency Spectrum at 1000 rpm with a defect size of 3.2 mm,8.2mm with no load with 0.5 misalignment at X Axis

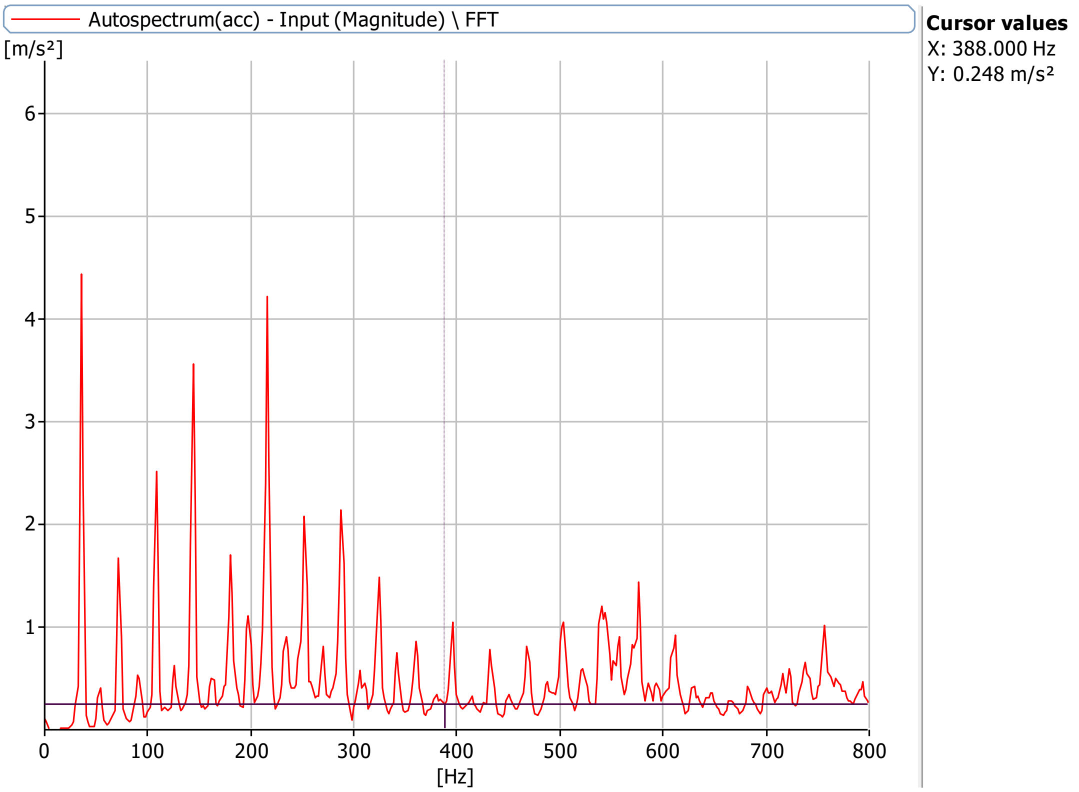


Figure 6.13.3: Frequency Spectrum at 2160 rpm with a defect size of 3.2 mm,8.2mm with no load with 0.5 misalignment at X Axis

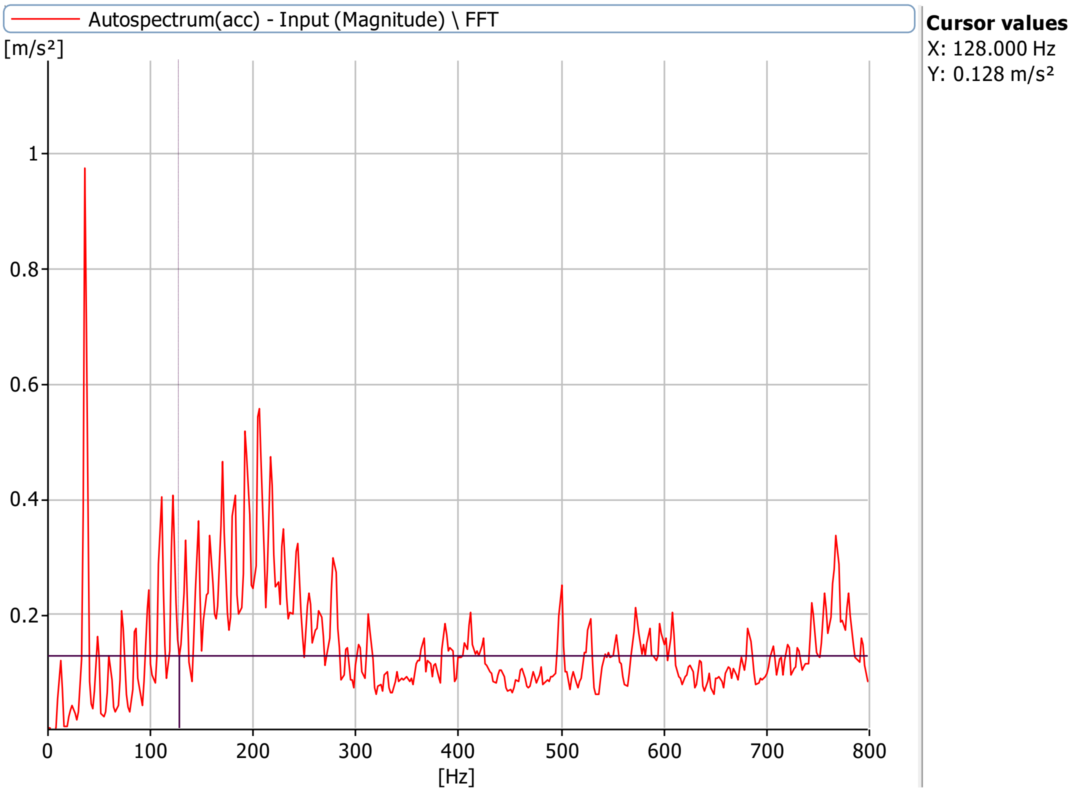


Figure 6.13.4: Frequency Spectrum at 720 rpm with a defect size of 3.2 mm,8.2mm with no load with 0.5 misalignment at X Axis

Frequencies at 720 rpm, 1000 rpm,1440 rpm and 2160 rpm for defect of 5.1mm bearing and 0.5 misalignment

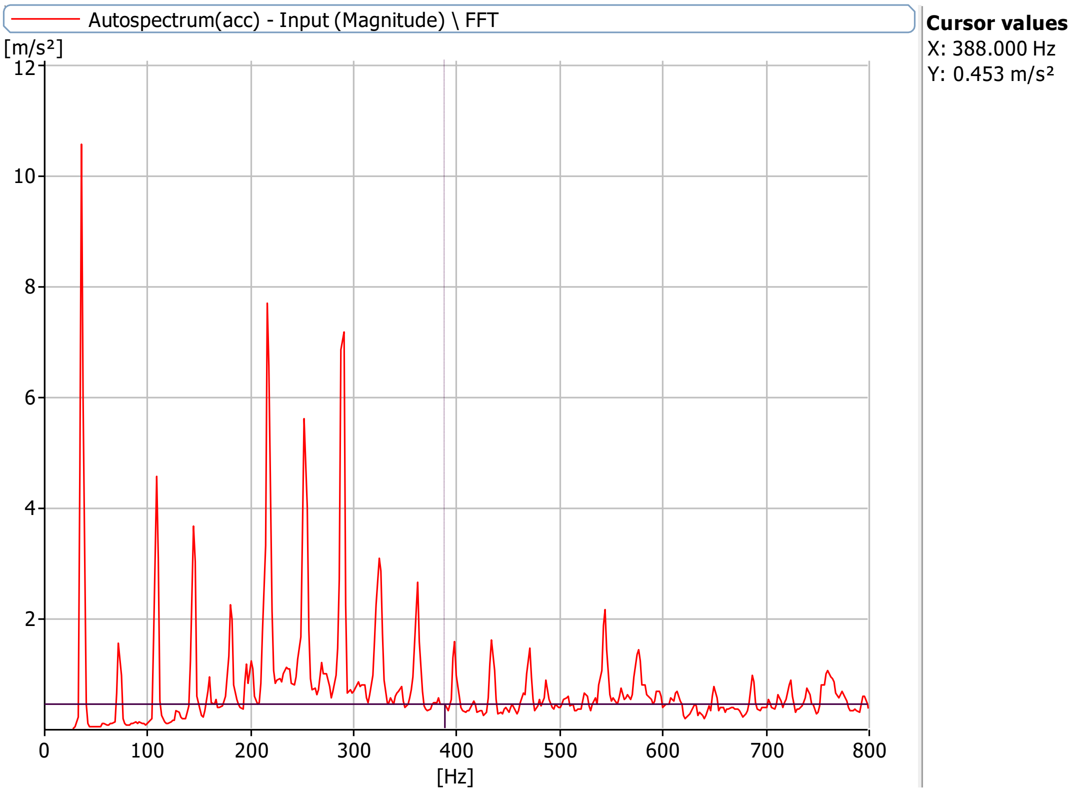


Figure 6.14.1: Frequency Spectrum at 2160 rpm with a defect size of 5.1mm with no load with 0.5 misalignment at X Axis

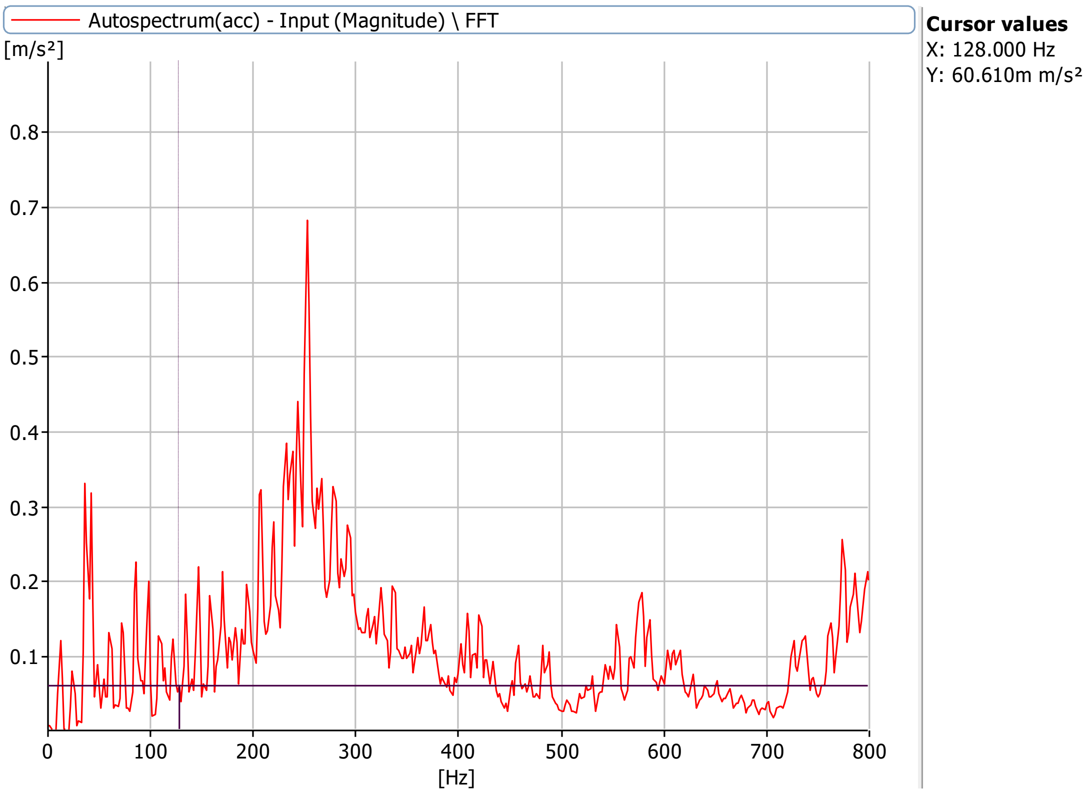


Figure 6.14.2: Frequency Spectrum at 1000 rpm with a defect size of 5.1mm with no load with 0.5 misalignment at X Axis

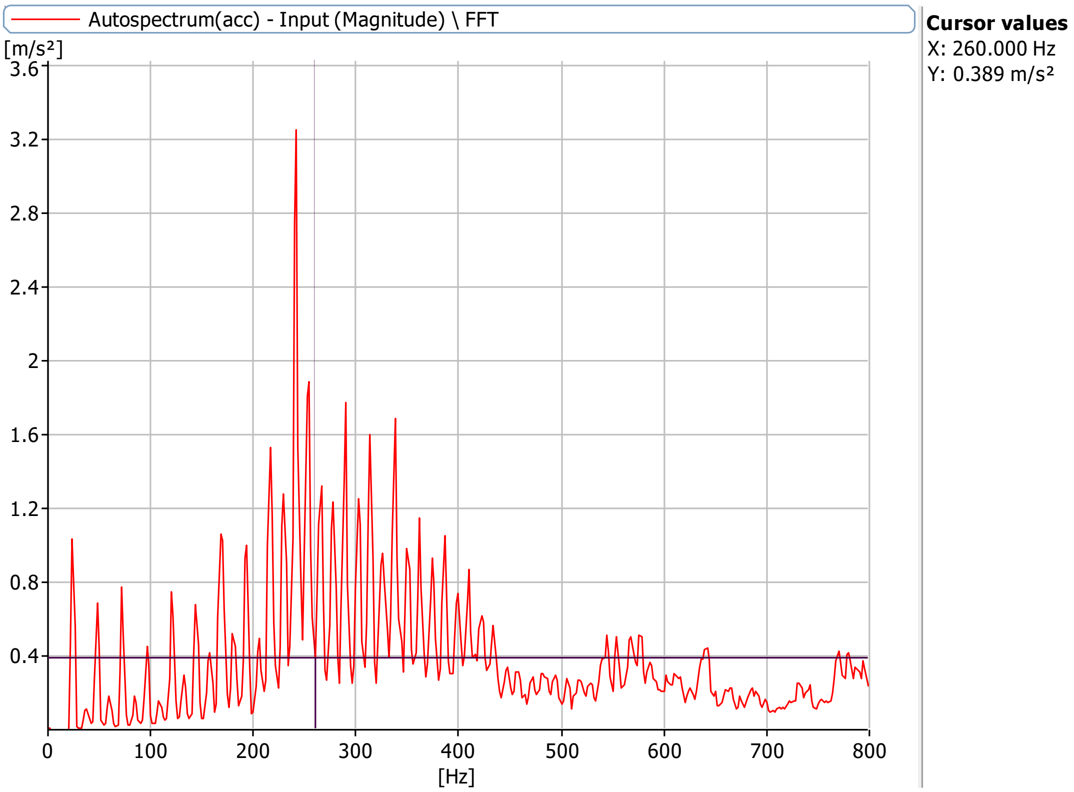


Figure 6.14.3: Frequency Spectrum at 720 rpm with a defect size of 5.1mm with no load with 0.5 misalignment at X Axis

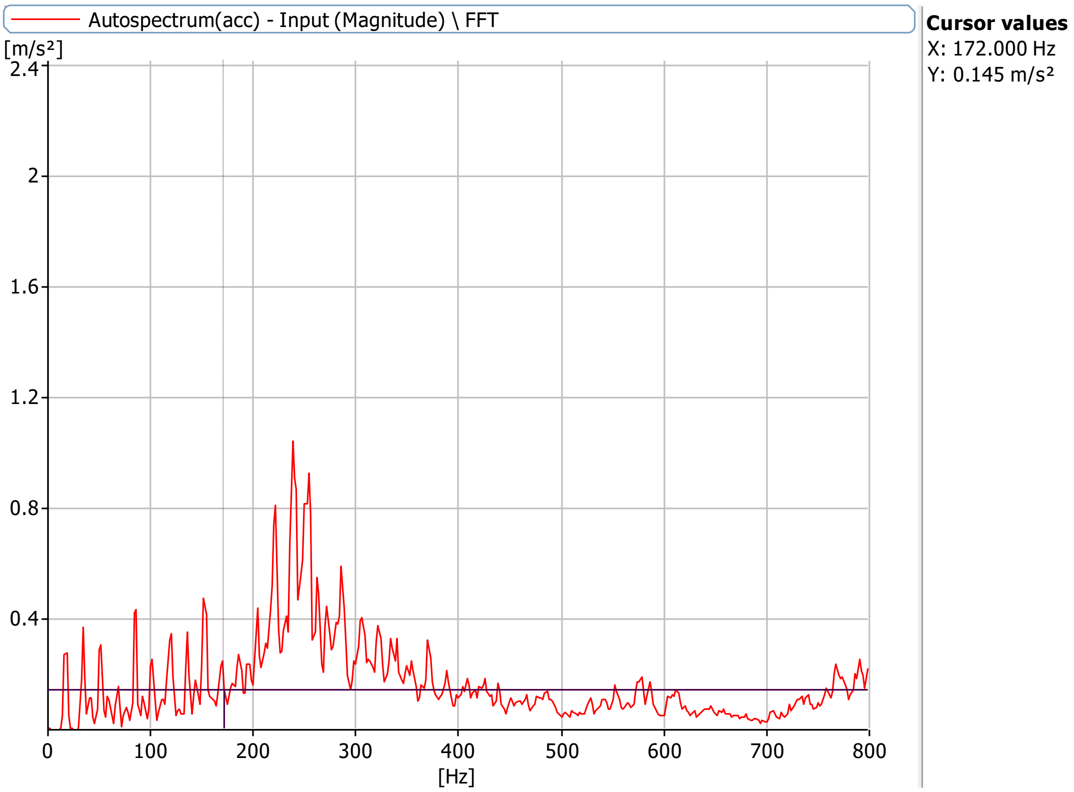


Figure 6.14.4: Frequency Spectrum at 1440 rpm with a defect size of 5.1mm with no load with 0.5 misalignment at X Axis

Frequencies at 720 rpm, 1000 rpm,1440 rpm and 2160 rpm for defect of 8mm bearing and 0.5 misalignment

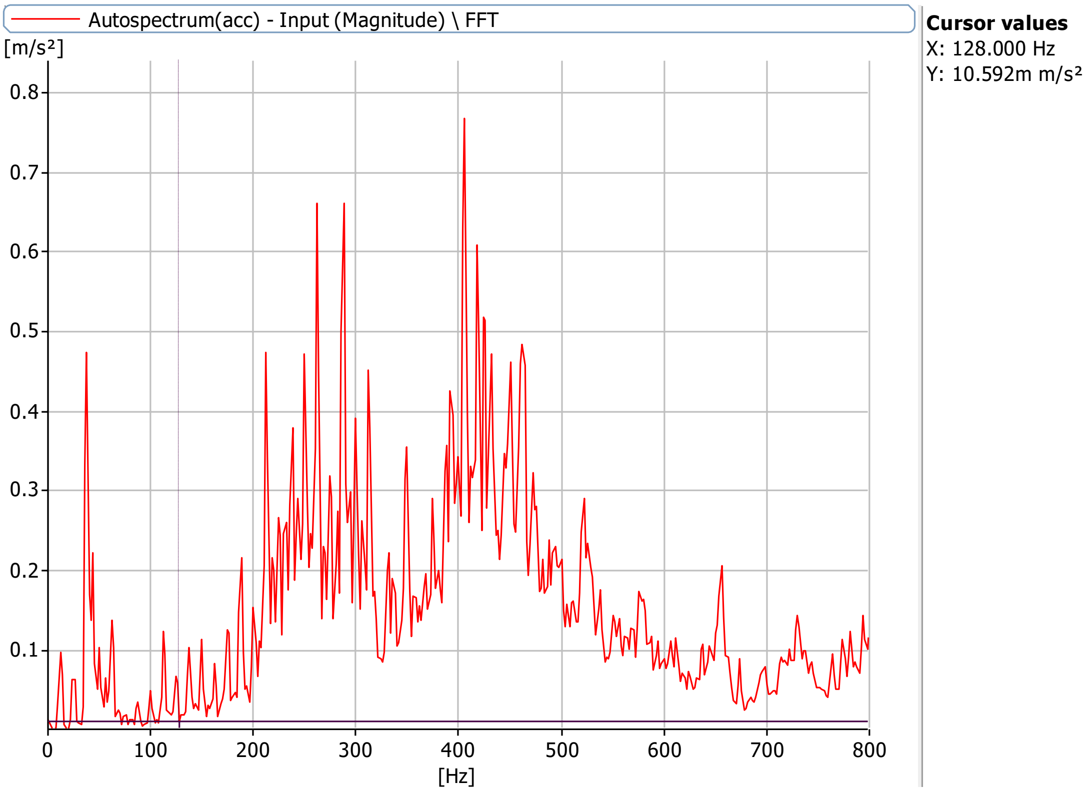


Figure 6.15.1: Frequency Spectrum at 1000 rpm with a defect size of 8mm with no load with 0.5 misalignment at X Axis

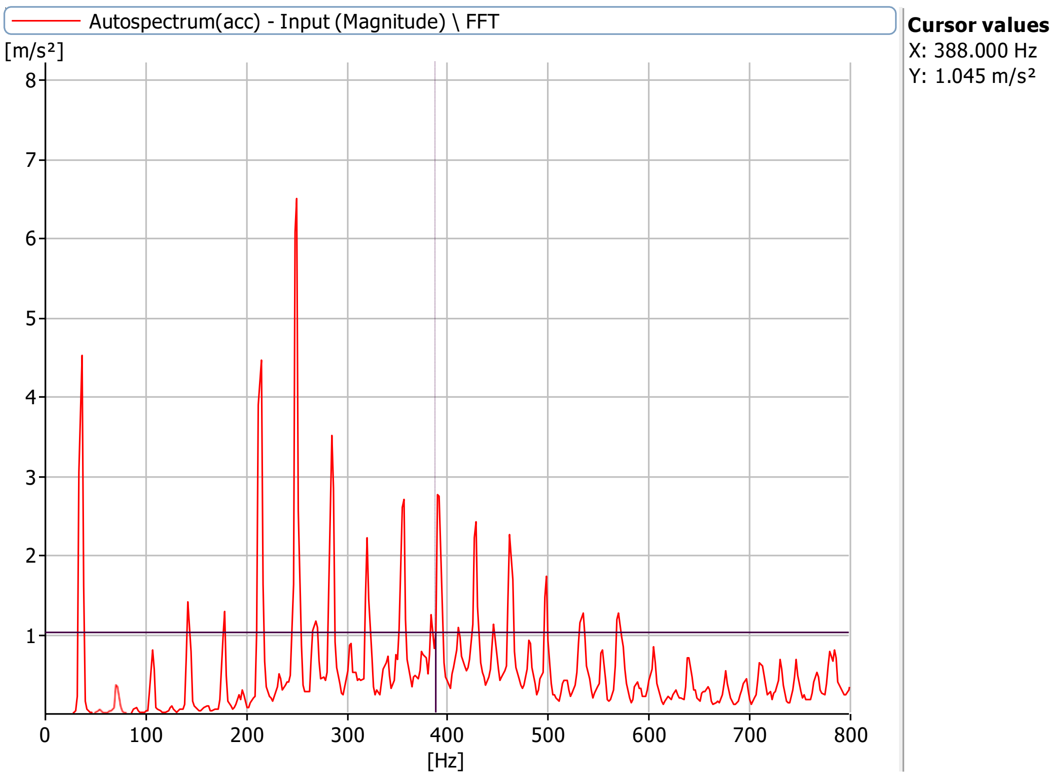


Figure 6.15.2: Frequency Spectrum at 2160 rpm with a defect size of 8mm with no load with 0.5 misalignment at X Axis

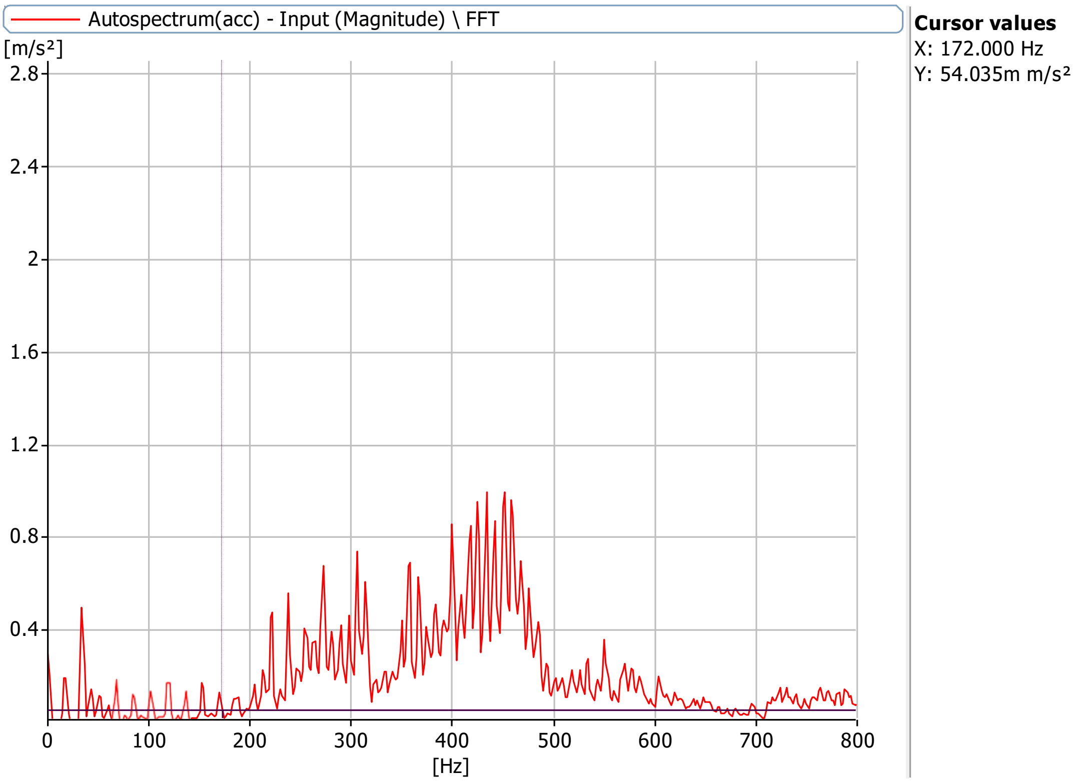


Figure 6.15.3: Frequency Spectrum at 720 rpm with a defect size of 8mm with no load with 0.5 misalignment at X Axis

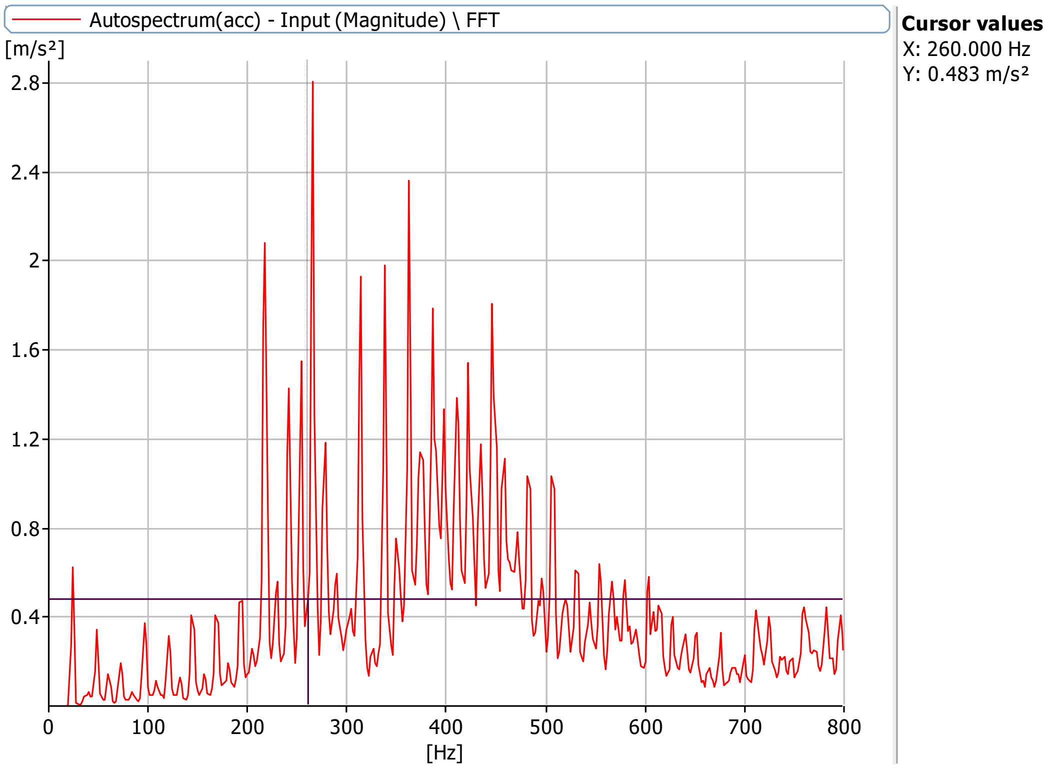


Figure 6.15.4: Frequency Spectrum at 1440 rpm with a defect size of 8mm with no load with 0.5 misalignment at X Axis

Frequencies at 720 rpm, 1000 rpm,1440 rpm and 2160 rpm for defect of 0mm bearing and 0.75 misalignment

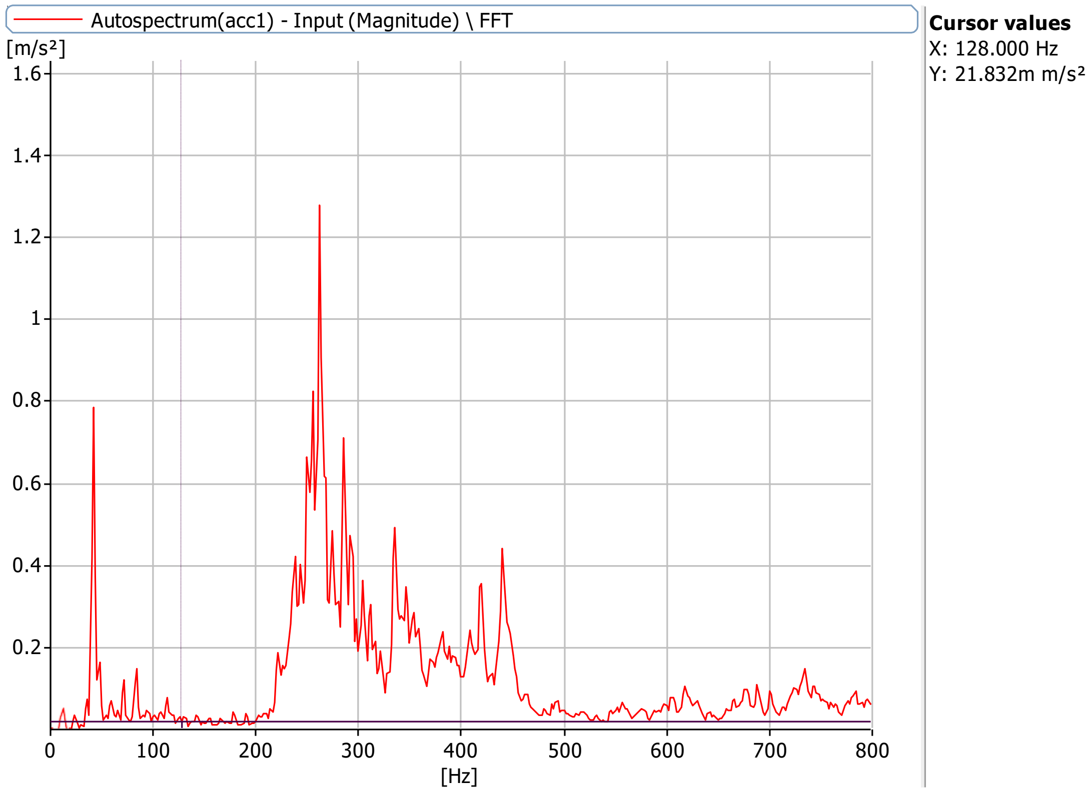


Figure 6.16.1: Frequency Spectrum at 720 rpm with a defect size of 0 mm with no load with 0.75 misalignment at X Axis

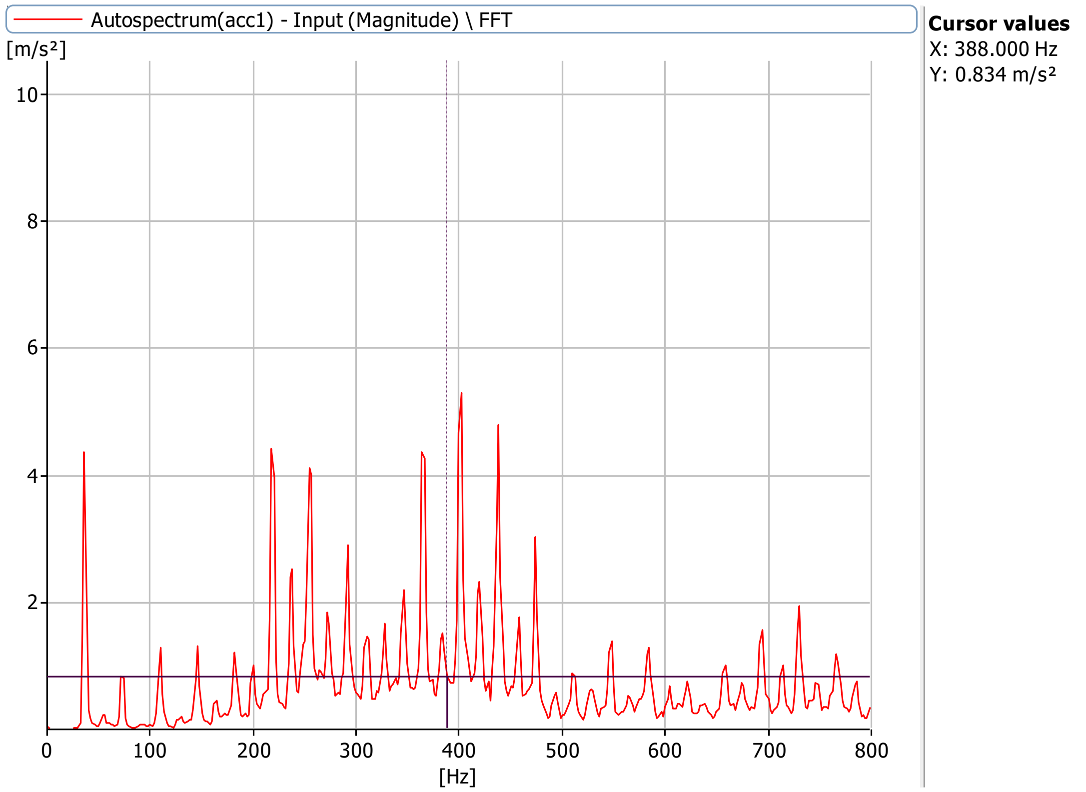


Figure 6.16.2: Frequency Spectrum at 2160 rpm with a defect size of 0 mm with no load with 0.75 misalignment at X Axis

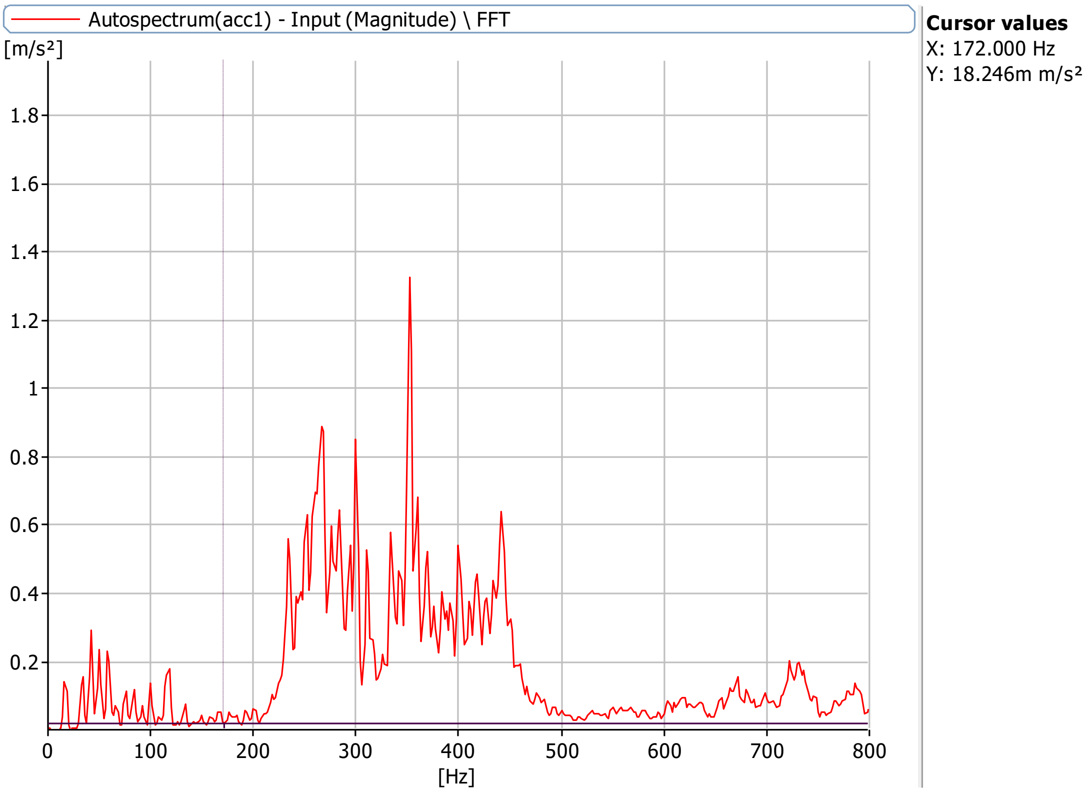


Figure 6.16.3: Frequency Spectrum at 1000 rpm with a defect size of 0 mm with no load with 0.75 misalignment at X Axis

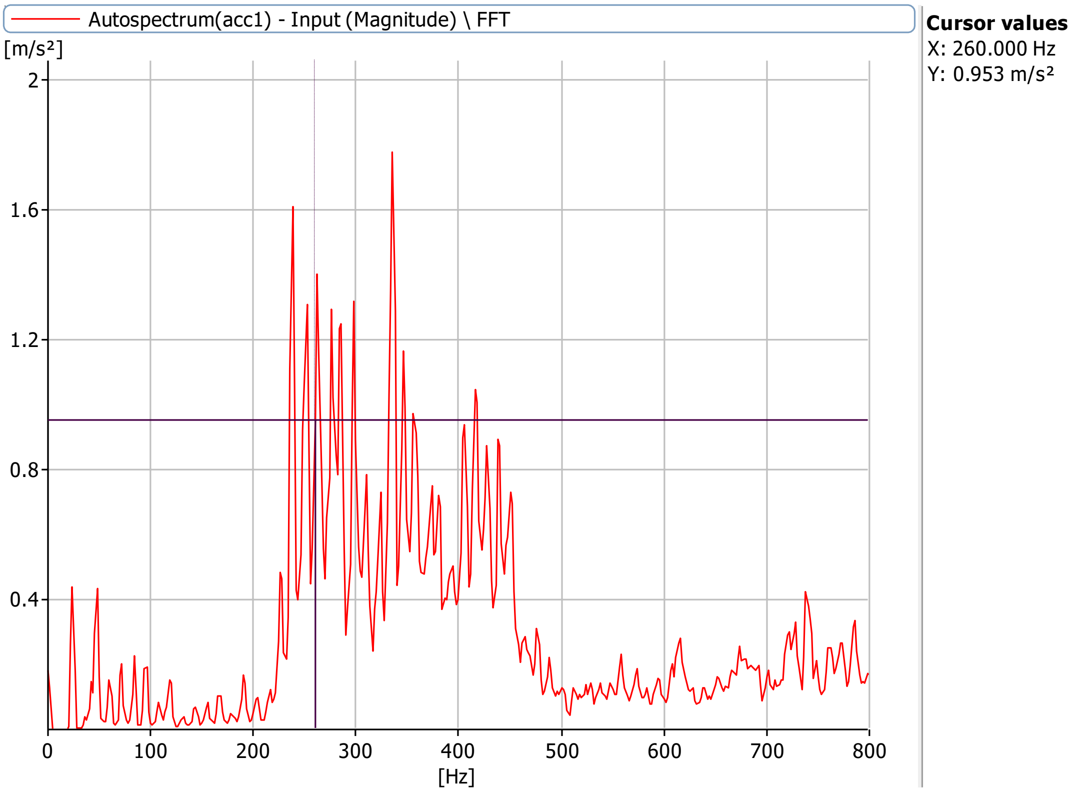


Figure 6.16.4: Frequency Spectrum at 1440 rpm with a defect size of 0 mm with no load with 0.75

misalignment at X Axis

Frequencies at 720 rpm, 1000 rpm,1440 rpm and 2160 rpm for defect of 3.2 mm,5.1 mm bearing and 0.75mm misalignment

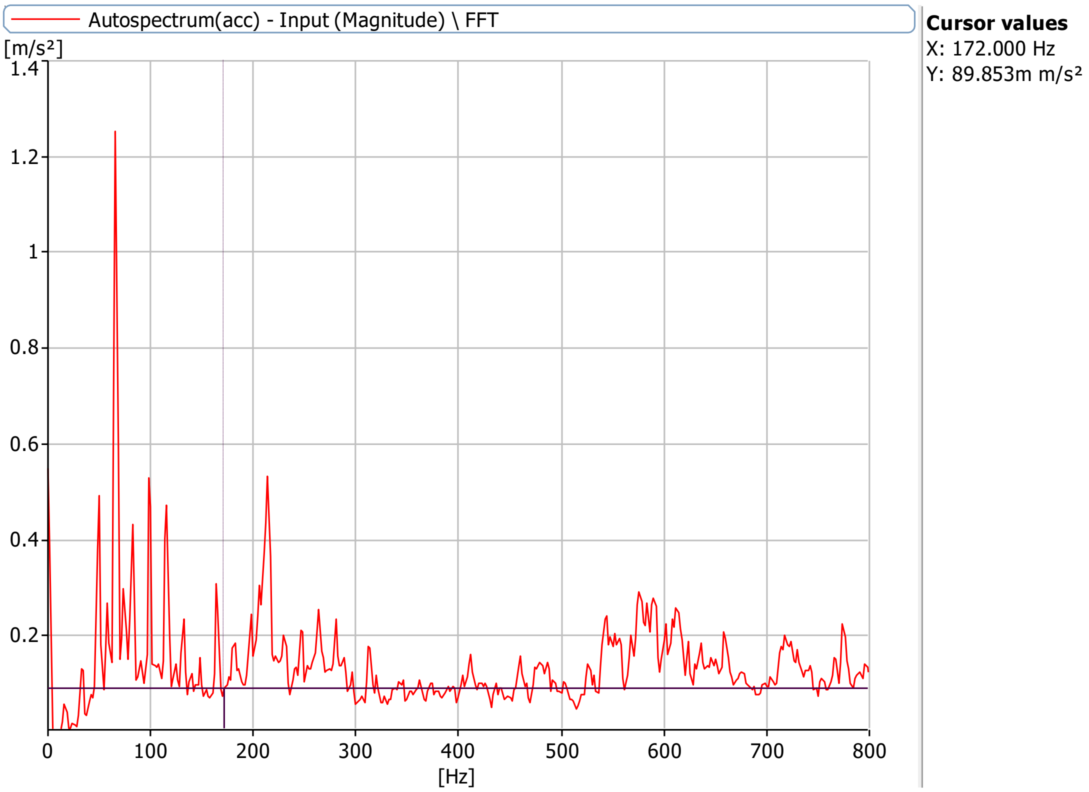


Figure 6.17.1: Frequency Spectrum at 1000 rpm with a defect size of 3.2 mm,5.1mm with no load with 0.75mm misalignment at X Axis

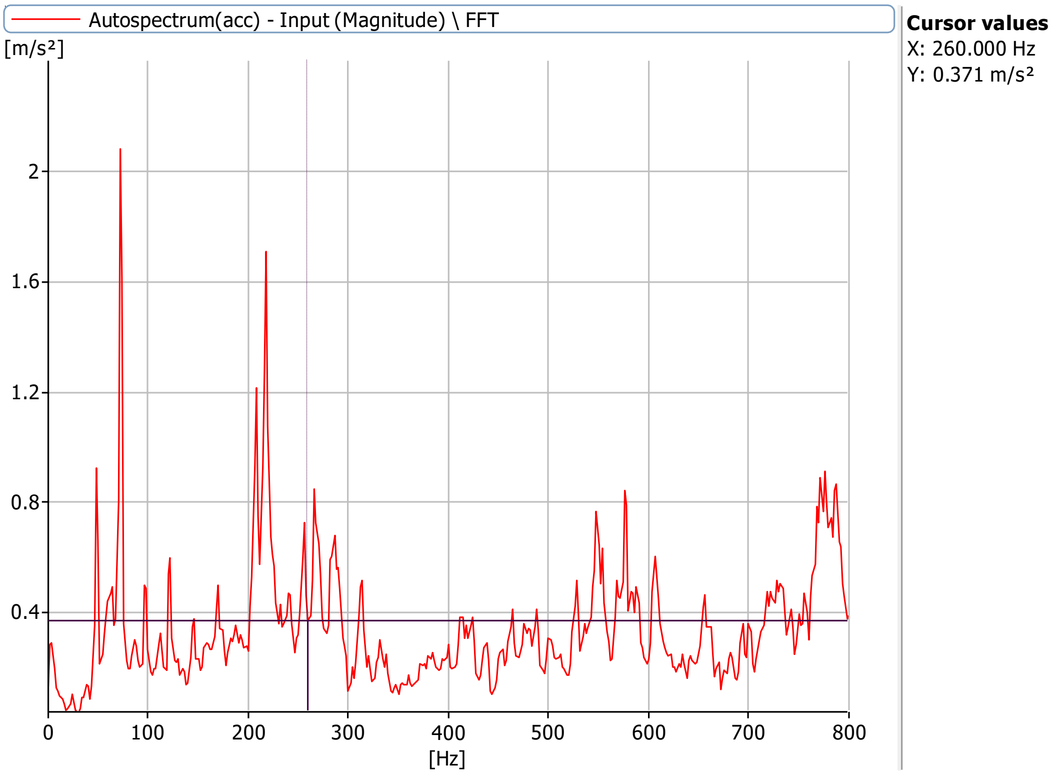


Figure 6.17.2: Frequency Spectrum at 1440 rpm with a defect size of 3.2 mm,5.1mm with no load with 0.75mm misalignment at X Axis

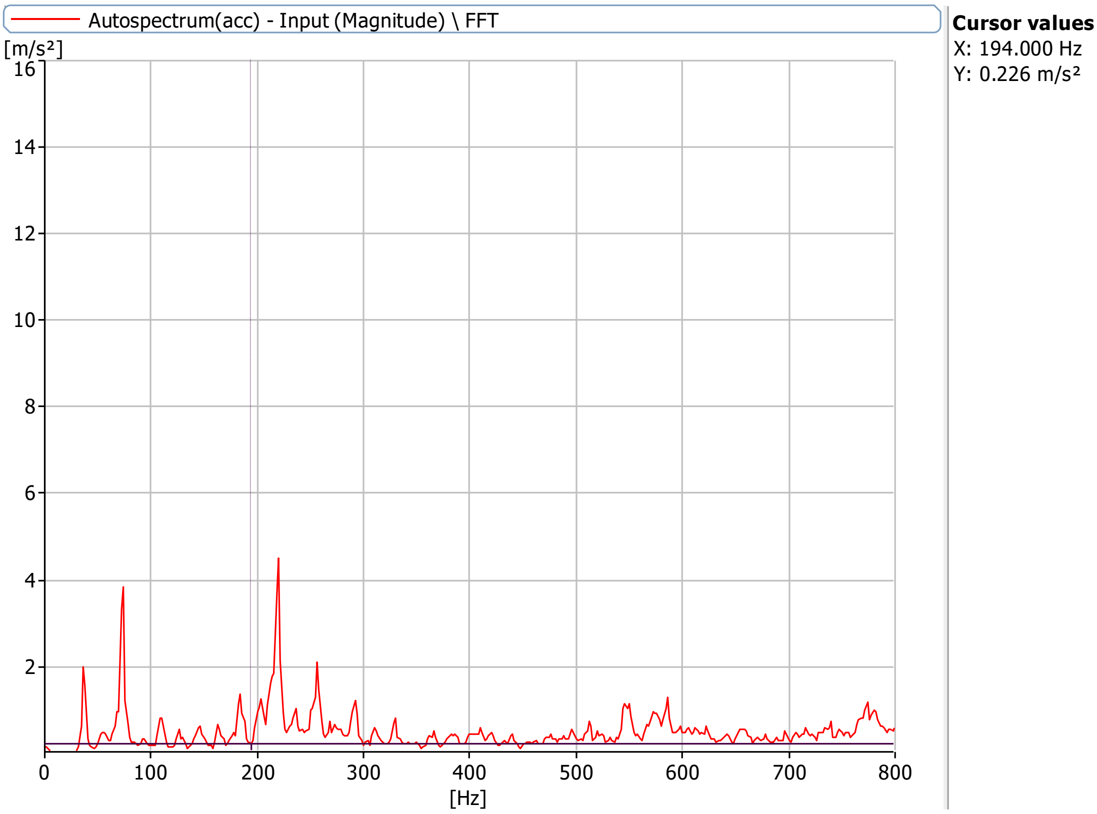


Figure 6.17.3: Frequency Spectrum at 2160 rpm with a defect size of 3.2 mm,5.1mm with no load with 0.75mm misalignment at X Axis

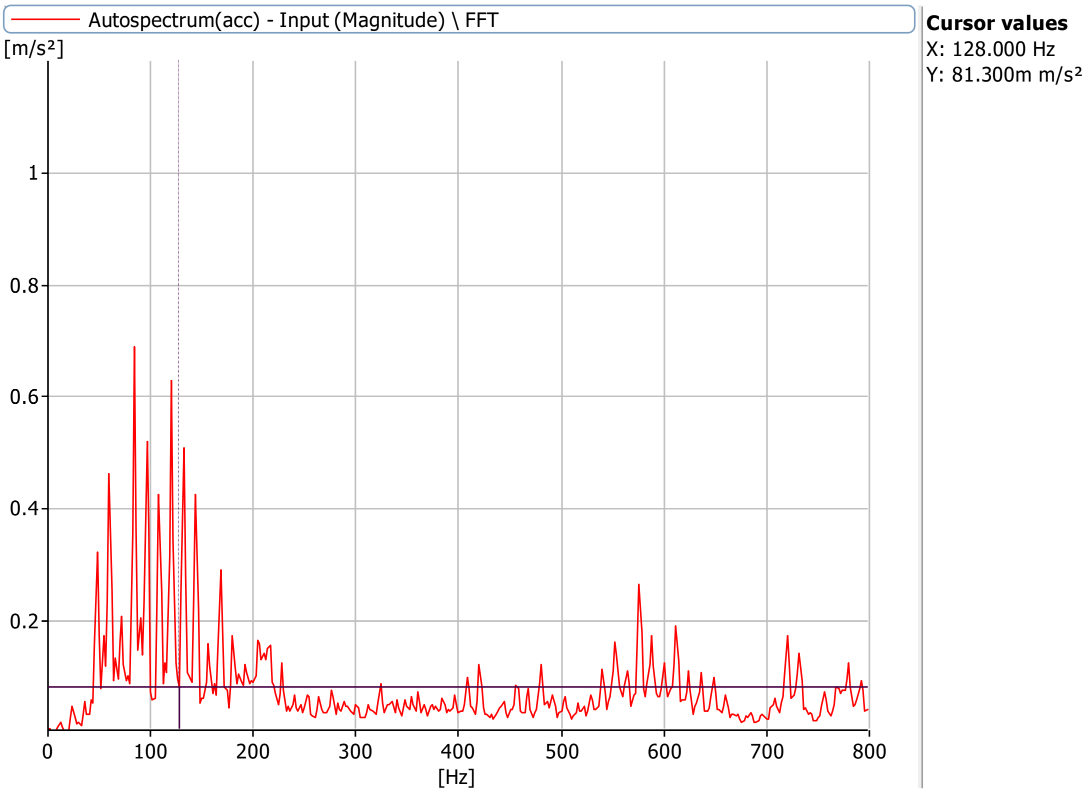


Figure 6.17.4: Frequency Spectrum at 720 rpm with a defect size of 3.2 mm,5.1mm with no load with 0.75mm misalignment at X Axis

Frequencies at 720 rpm, 1000 rpm,1440 rpm and 2160 rpm for defect of 3.2 mm,8.2 mm bearing and 0.75 misalignment

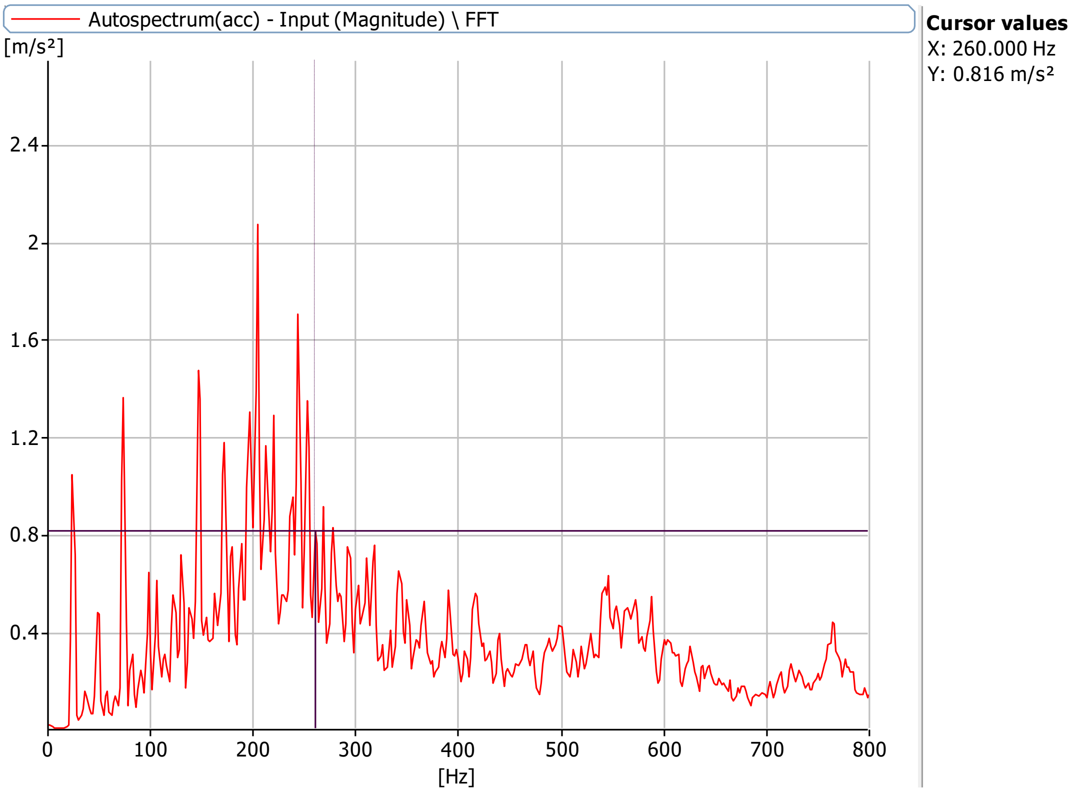


Figure 6.18.1: Frequency Spectrum at 2160 rpm with a defect size of 3.2 mm,8.2 mm with no load with 0.75mm misalignment at X Axis

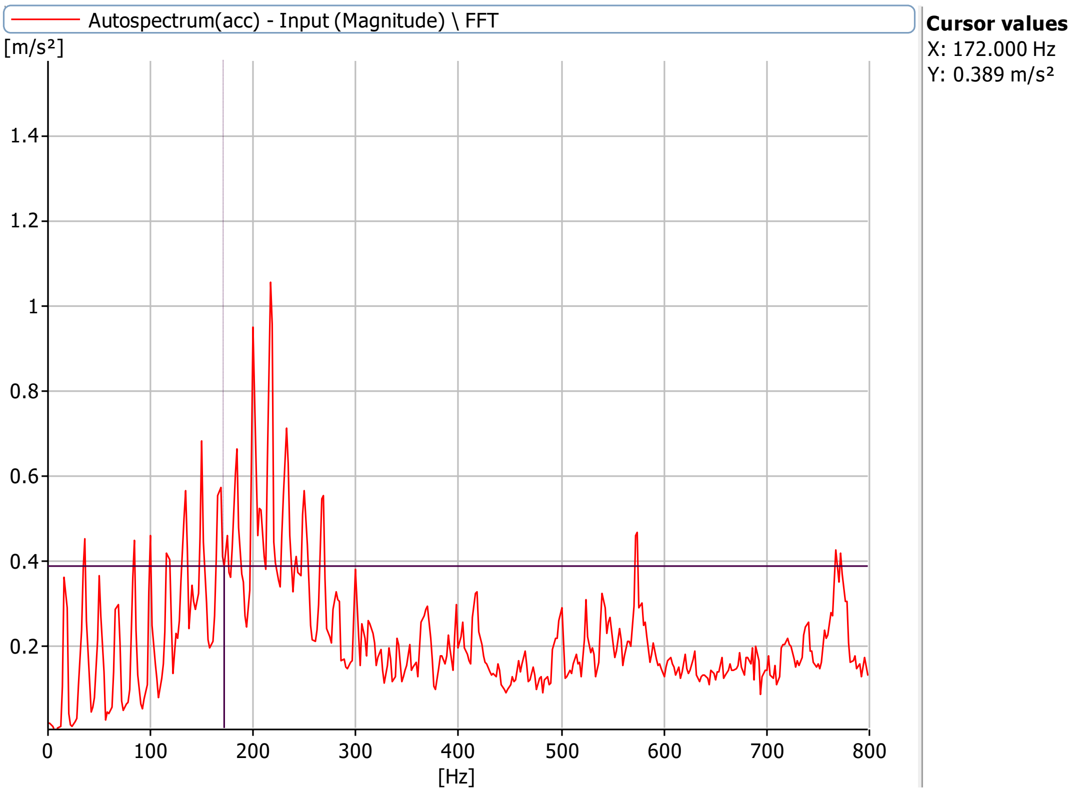


Figure 6.18.2: Frequency Spectrum at 1440 rpm with a defect size of 3.2 mm,8.2 mm with no load with 0.75mm misalignment at X Axis

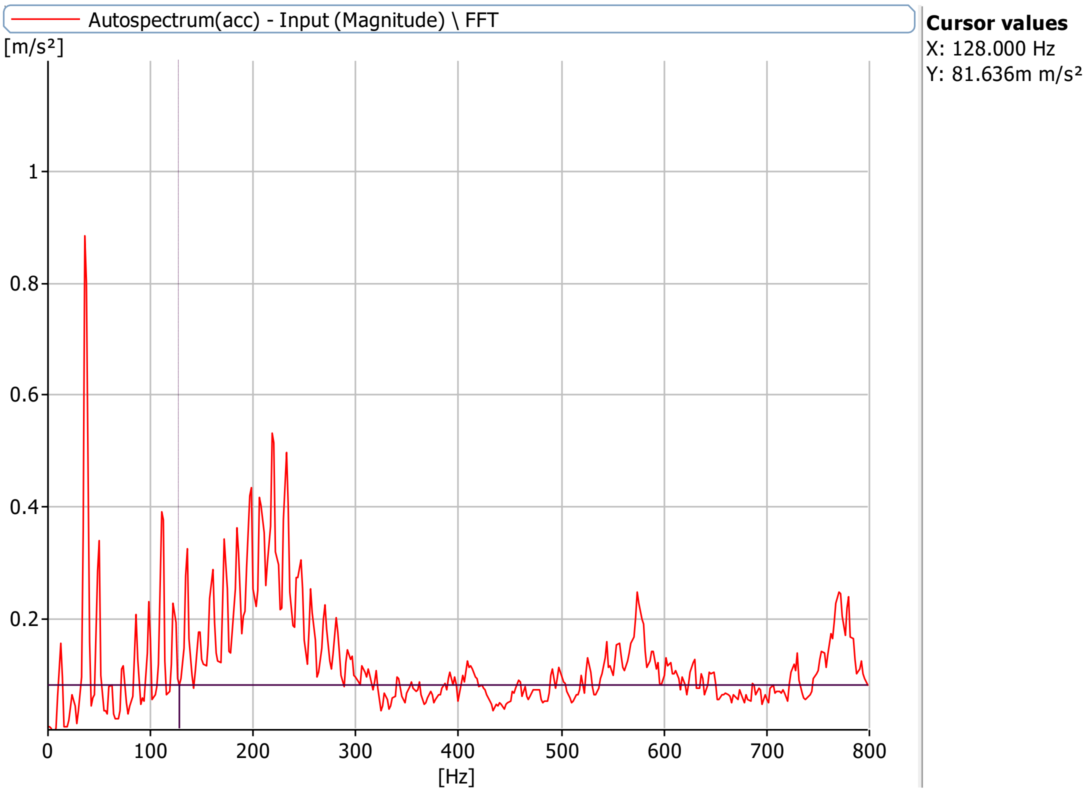


Figure 6.18.3: Frequency Spectrum at 1000 rpm with a defect size of 3.2 mm,8.2 mm with no load with 0.75mm misalignment at X Axis

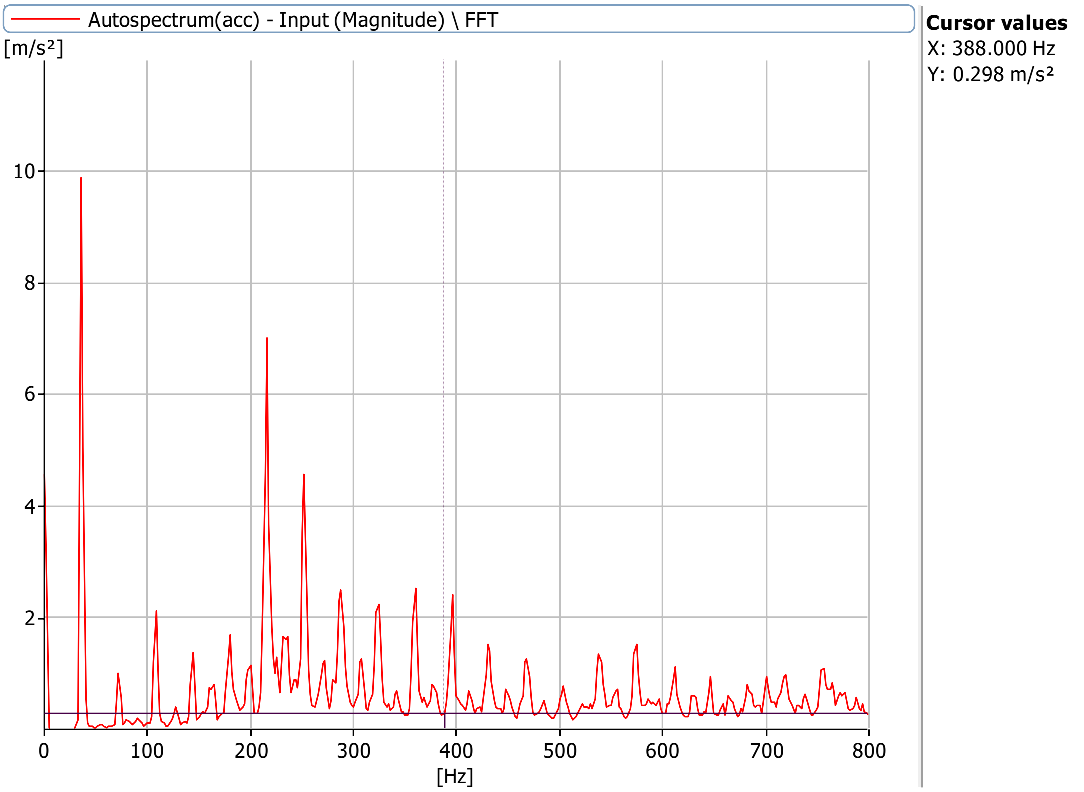


Figure 6.18.3: Frequency Spectrum at 720 rpm with a defect size of 3.2 mm,8.2 mm with no load with 0.75mm misalignment at X Axis.

Frequencies at 720 rpm, 1000 rpm,1440 rpm and 2160 rpm for defect of 5.1 mm bearing and 0.75mm misalignment

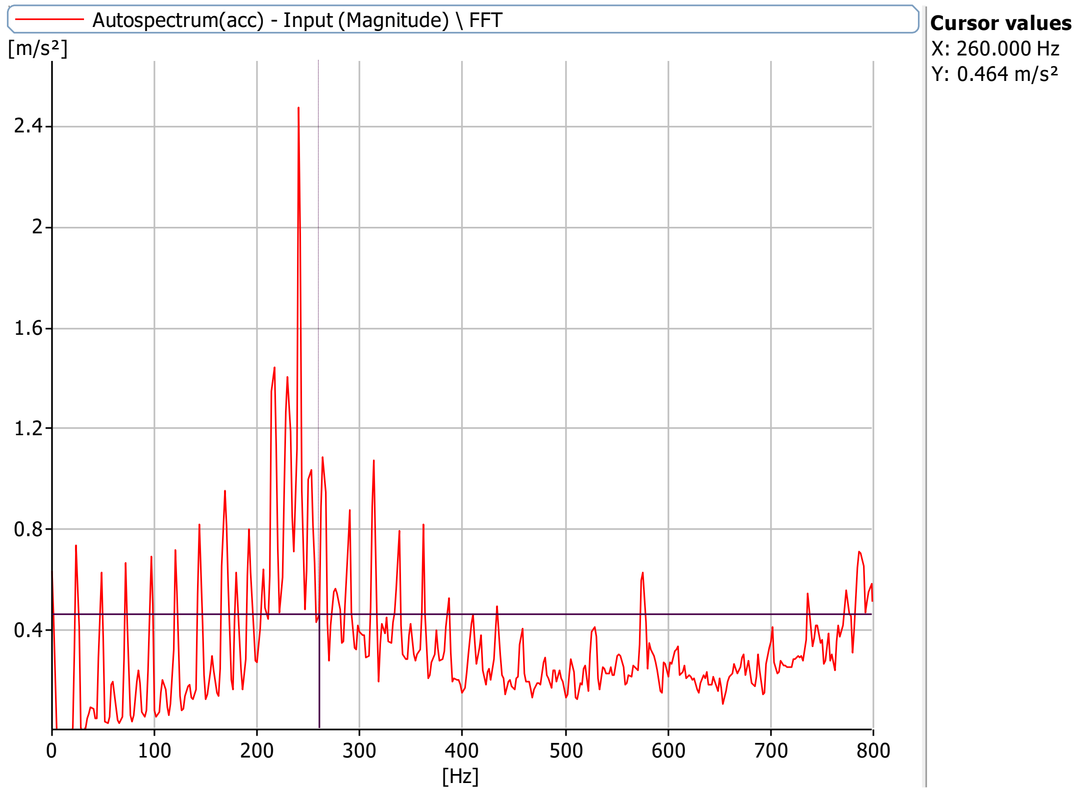


Figure 6.19.1: Frequency Spectrum at 2160 rpm with a defect size of 5.1 mm with no load with 0.75mm misalignment at X Axis

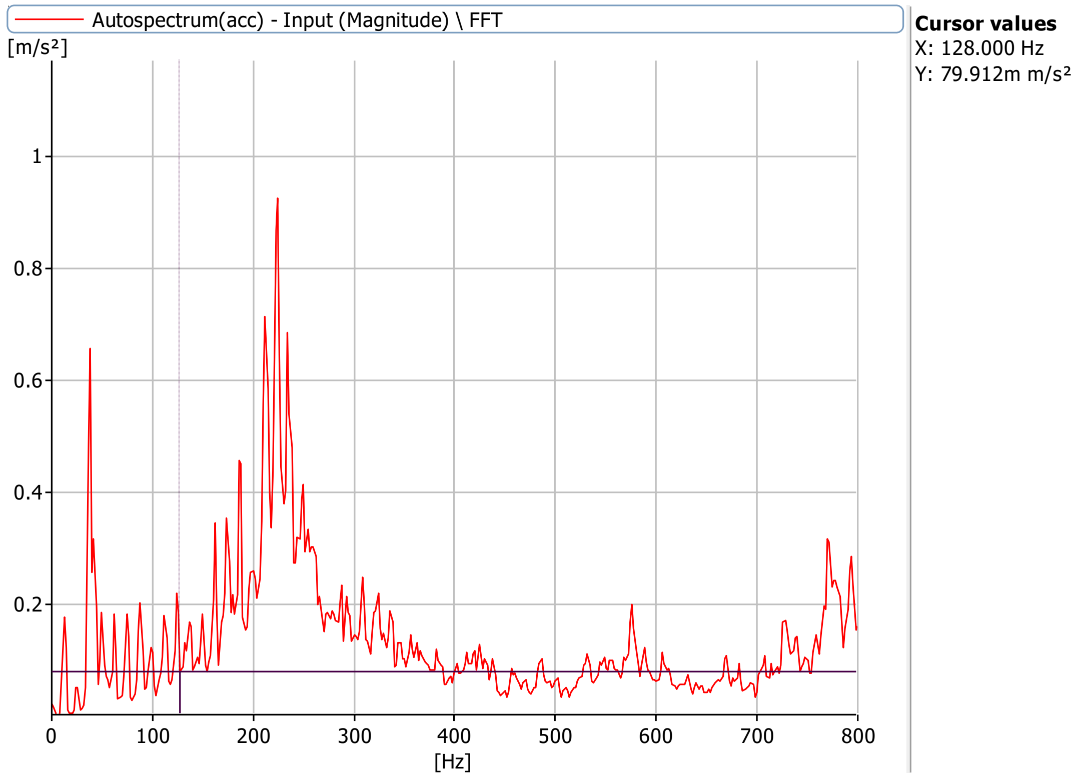


Figure 6.19.2: Frequency Spectrum at 1440 rpm with a defect size of 5.1 mm with no load with 0.75mm misalignment at X Axis

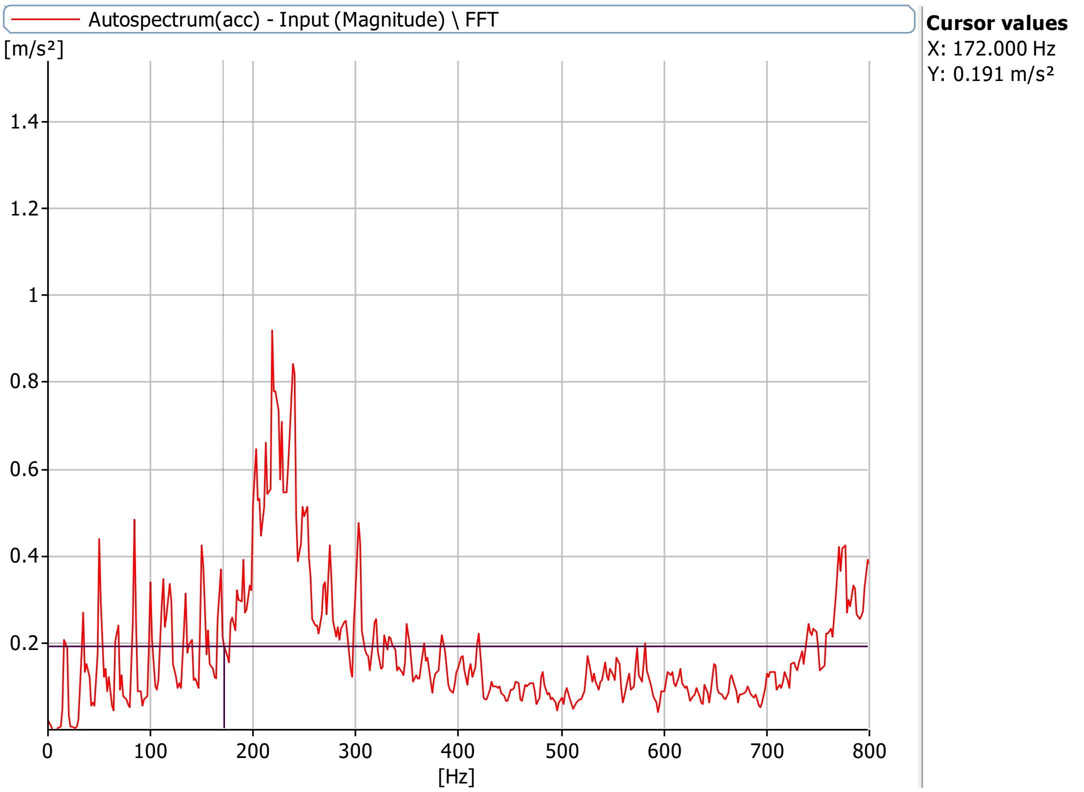


Figure 6.19.3: Frequency Spectrum at 720 rpm with a defect size of 5.1 mm with no load with 0.75mm misalignment at X Axis

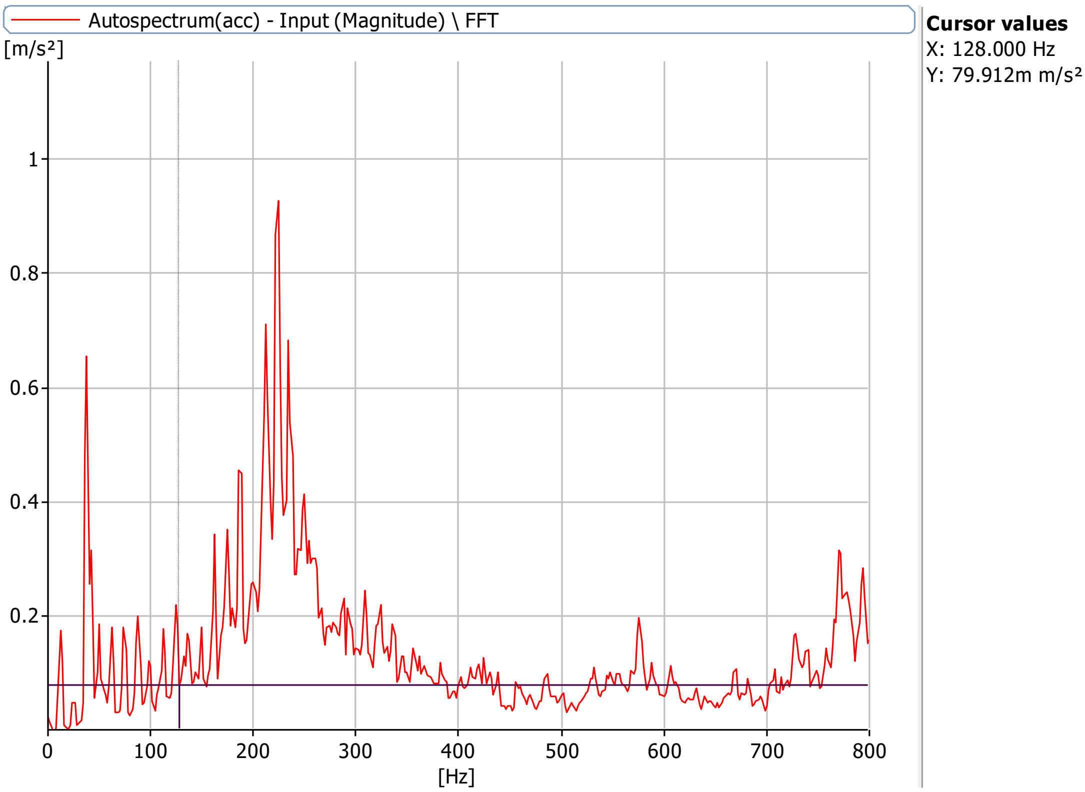


Figure 6.19.4: Frequency Spectrum at 1000 rpm with a defect size of 5.1 mm with no load with 0.75mm misalignment at X Axis

Frequencies at 720 rpm, 1000 rpm,1440 rpm and 2160 rpm for defect of 8 mm bearing and 0.75mm misalignment

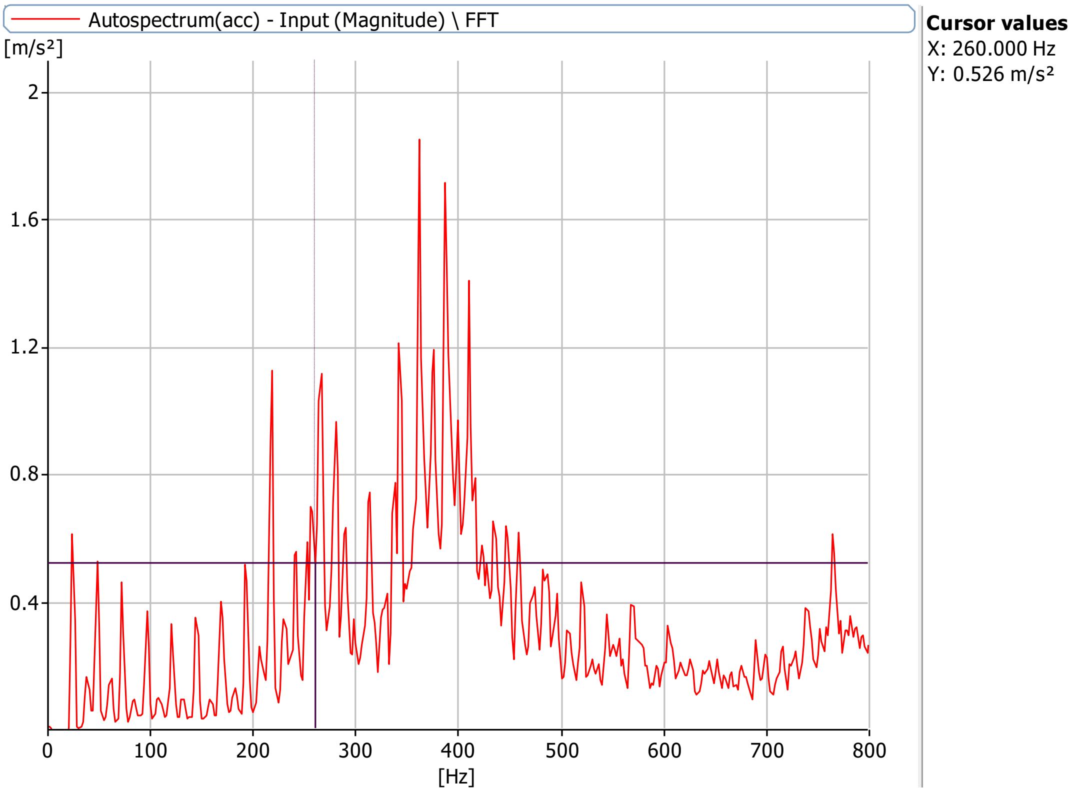


Figure 6.20.1: Frequency Spectrum at 1440 rpm with a defect size of 5.1 mm with no load with 0.75mm misalignment at X Axis

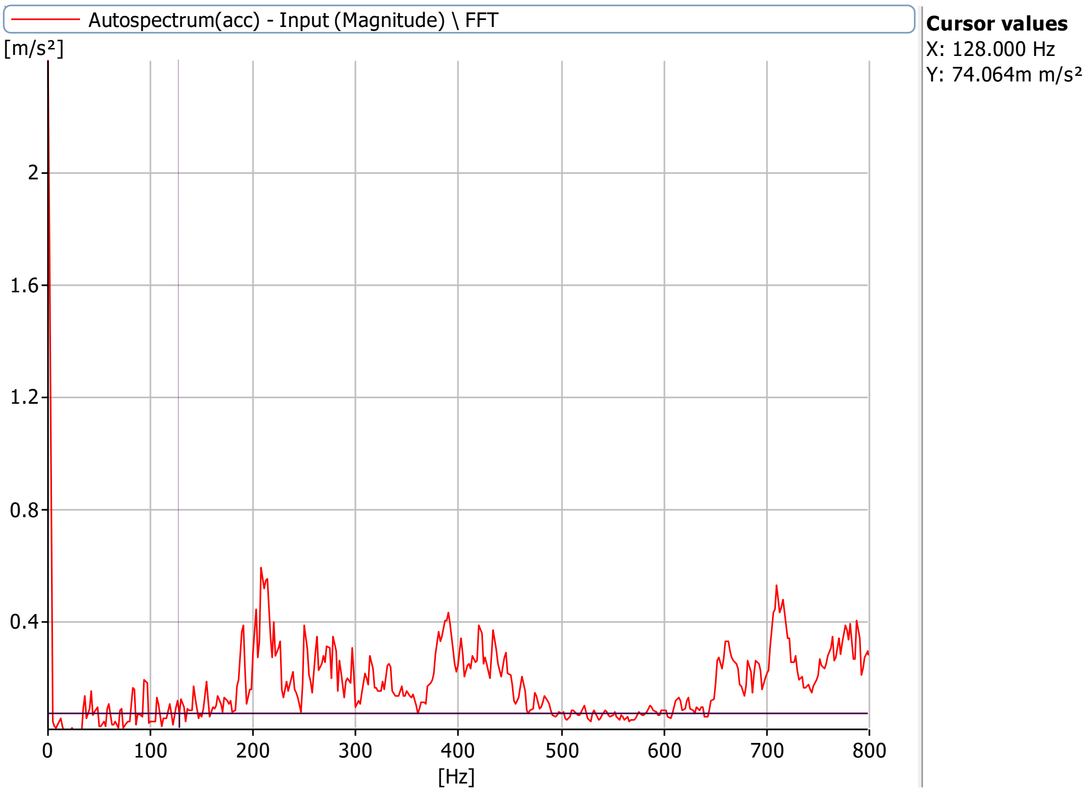


Figure 6.20.2: Frequency Spectrum at 720 rpm with a defect size of 5.1 mm with no load with 0.75mm misalignment at X Axis

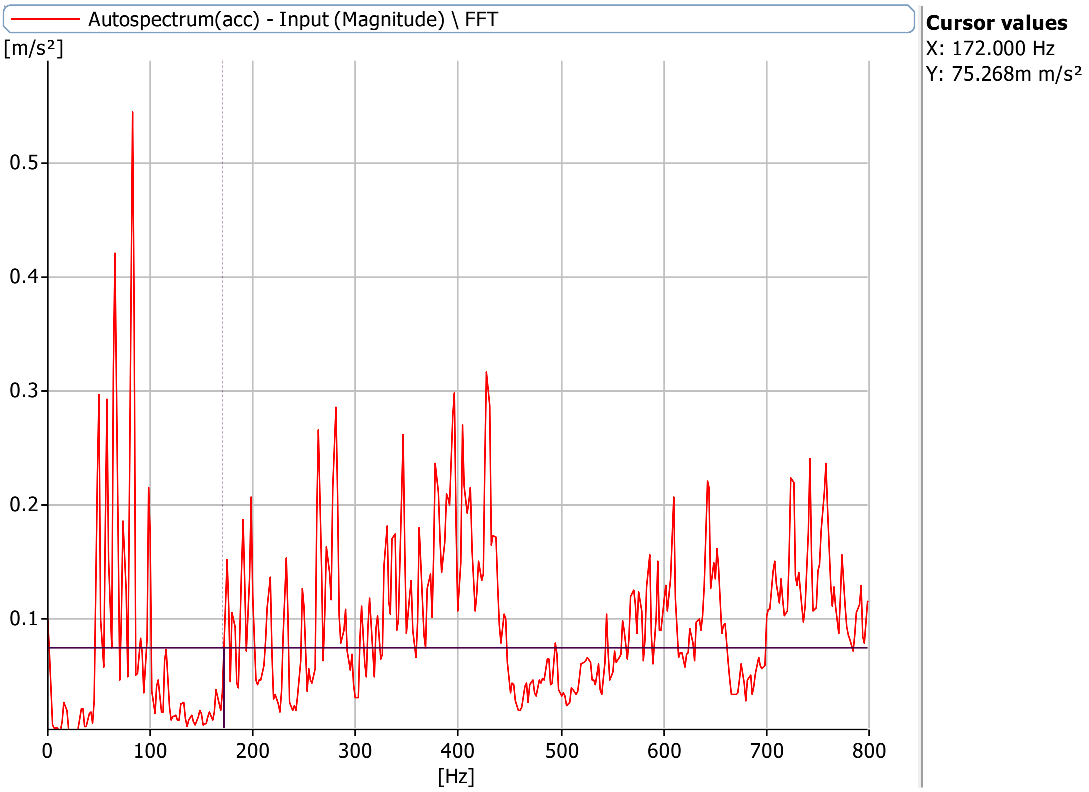


Figure 6.20.3: Frequency Spectrum at 1000 rpm with a defect size of 5.1 mm with no load with 0.75mm misalignment at X Axis

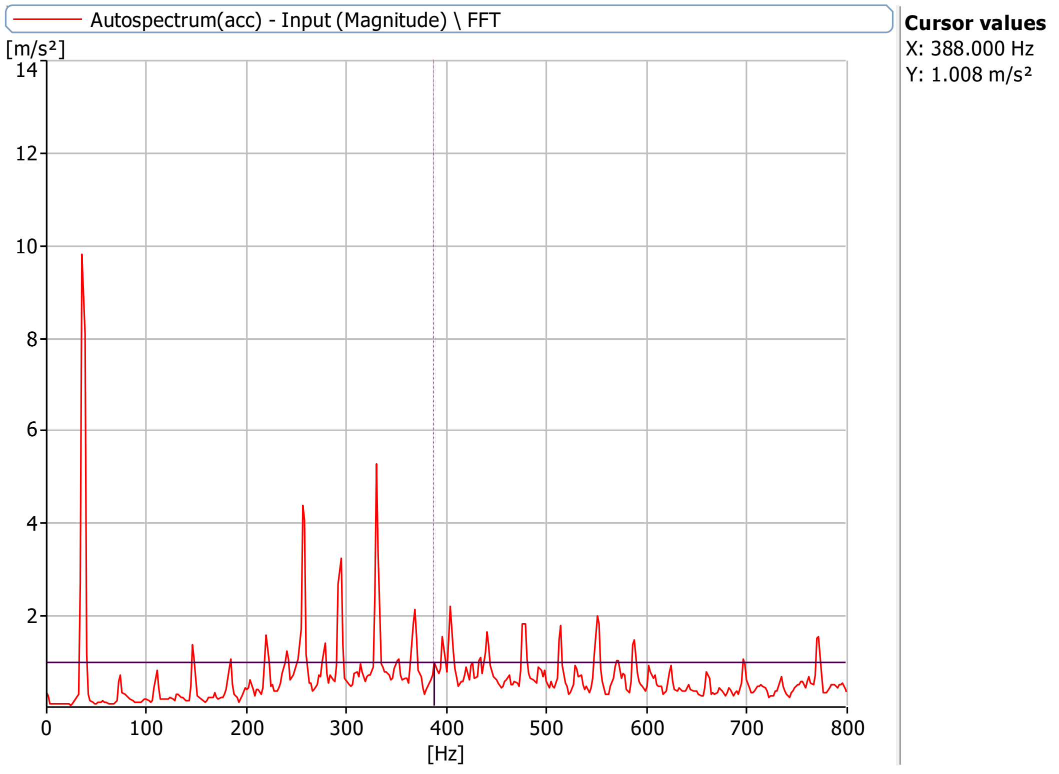


Figure 6.20.4.: Frequency Spectrum at 2160 rpm with a defect size of 5.1 mm with no load with 0.75mm misalignment at X Axis

Frequencies at 720 rpm, 1000 rpm,1440 rpm and 2160 rpm for defect of 0 mm bearing and 1mm misalignment

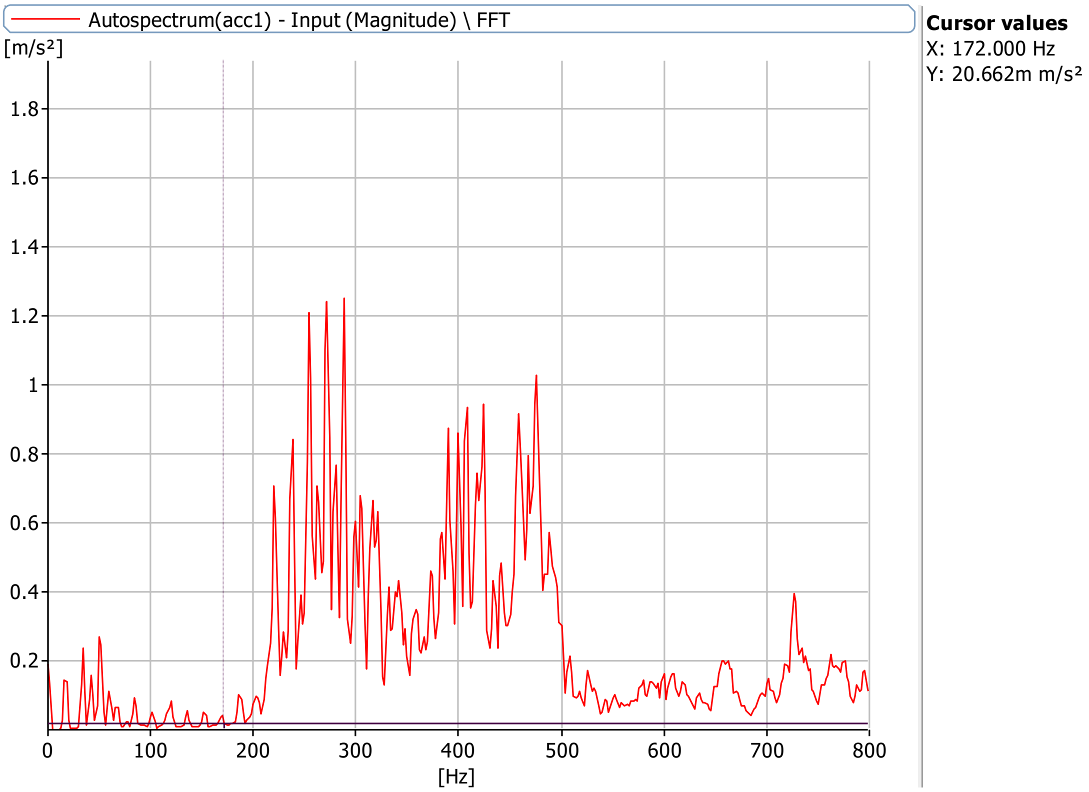


Figure 6.21.1: Frequency Spectrum at 1000 rpm with a defect size of 0 mm with no load with 1mm misalignment at X Axis

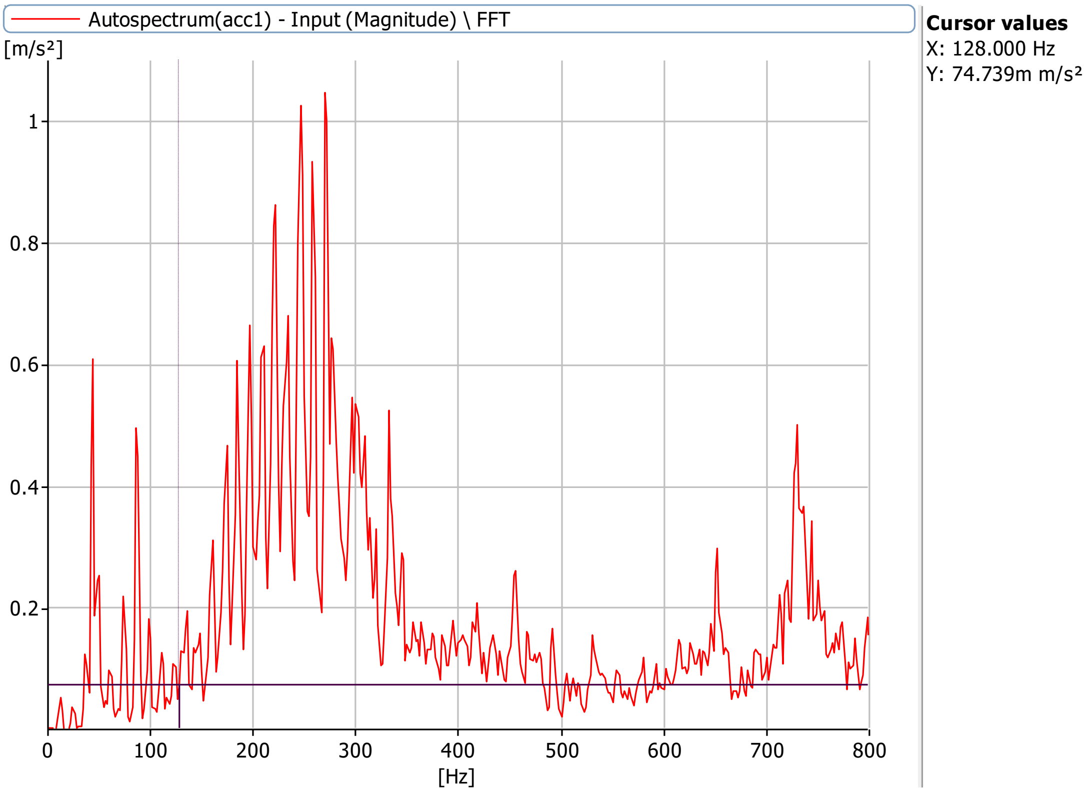


Figure 6.21.2: Frequency Spectrum at 720 rpm with a defect size of 0 mm with no load with 1mm misalignment at X Axis

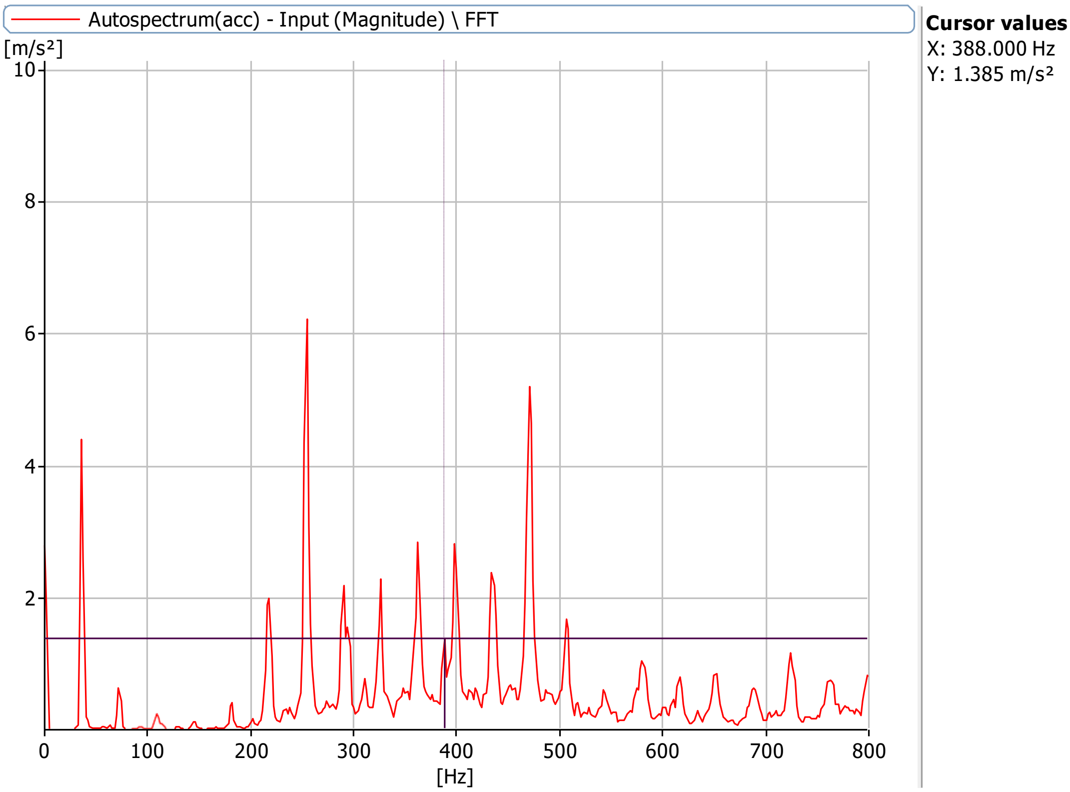


Figure 6.21.3: Frequency Spectrum at 2160 rpm with a defect size of 0 mm with no load with 1mm misalignment at X Axis

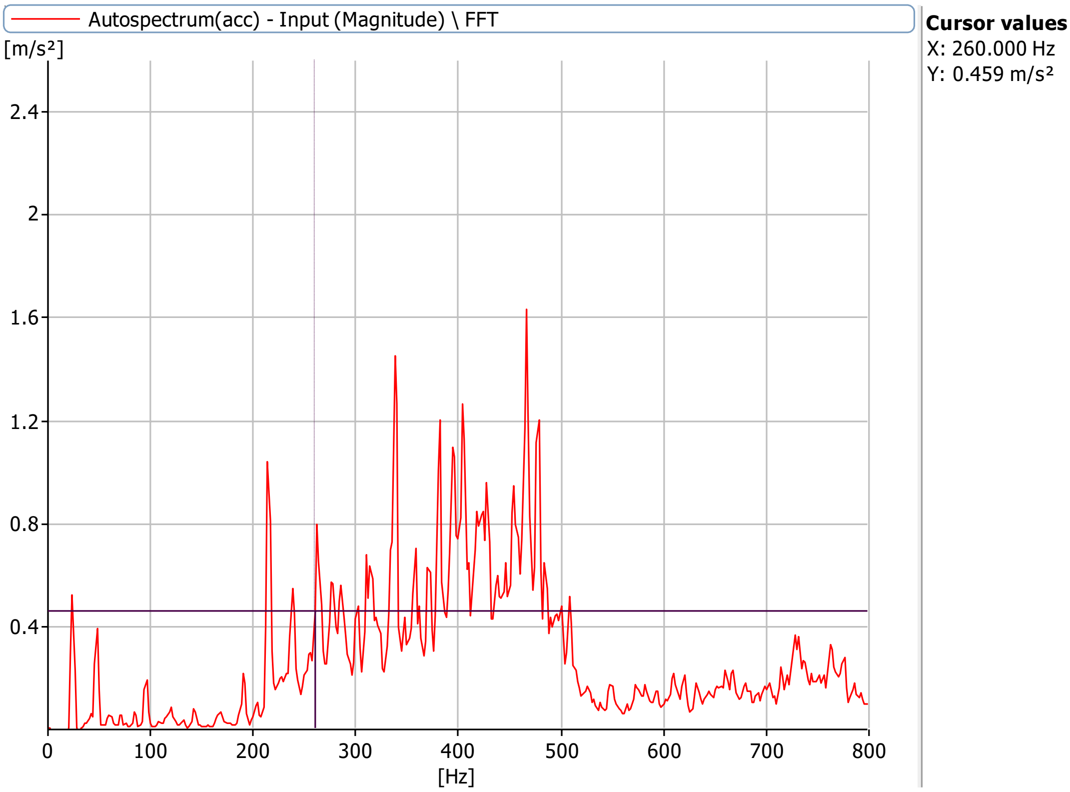


Figure 6.21.4: Frequency Spectrum at 1440 rpm with a defect size of 0 mm with no load with 1mm misalignment at X Axis

Frequencies at 720 rpm, 1000 rpm,1440 rpm and 2160 rpm for defect of 3.2 mm.5.1 mm bearing and 1 misalignment

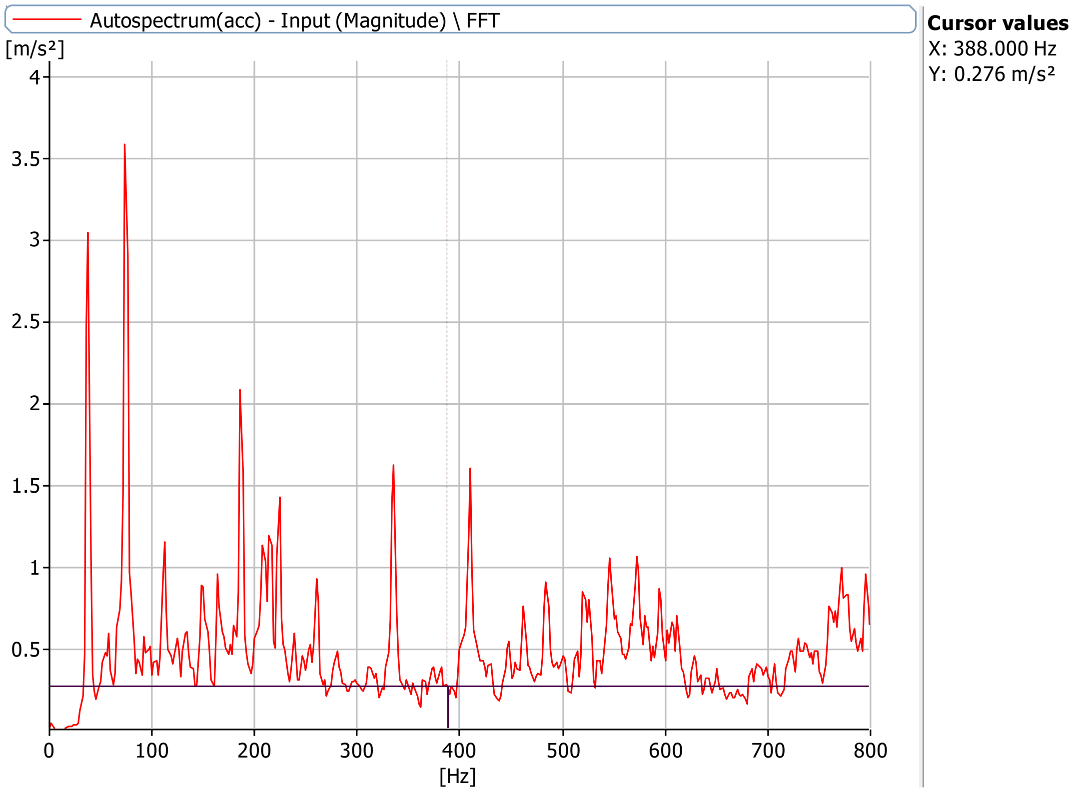


Figure 6.22.1: Frequency Spectrum at 2160 rpm with a defect size of 3.2 mm,5.1 mm with no load with 1mm misalignment at X Axis

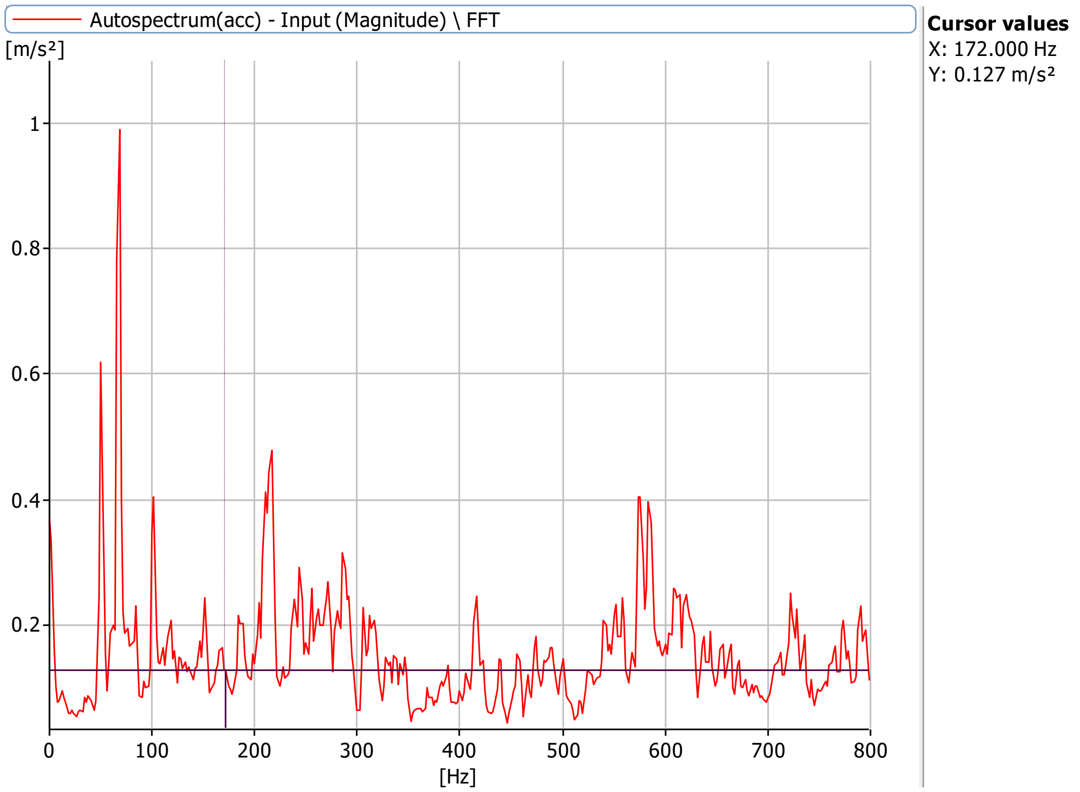


Figure 6.22.2: Frequency Spectrum at 1000 rpm with a defect size of 3.2 mm,5.1 mm with no load with 1mm misalignment at X Axis

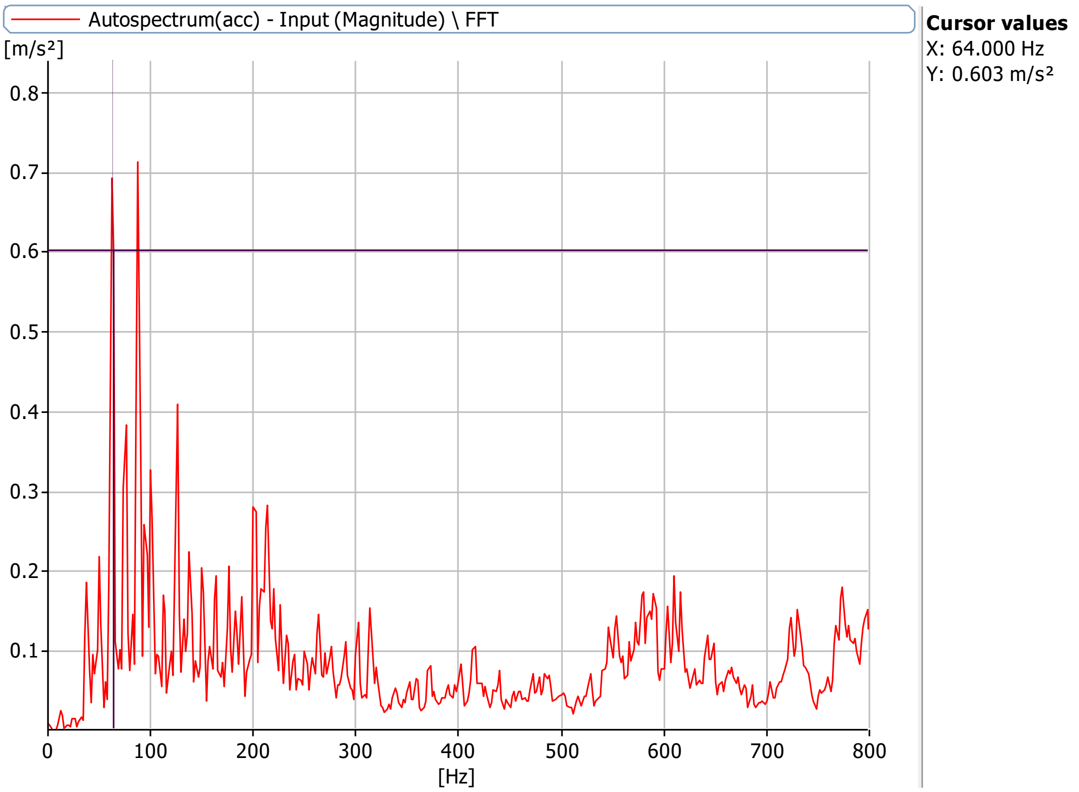


Figure 6.22.3: Frequency Spectrum at 720 rpm with a defect size of 3.2 mm,5.1 mm with no load with 1mm misalignment at X Axis

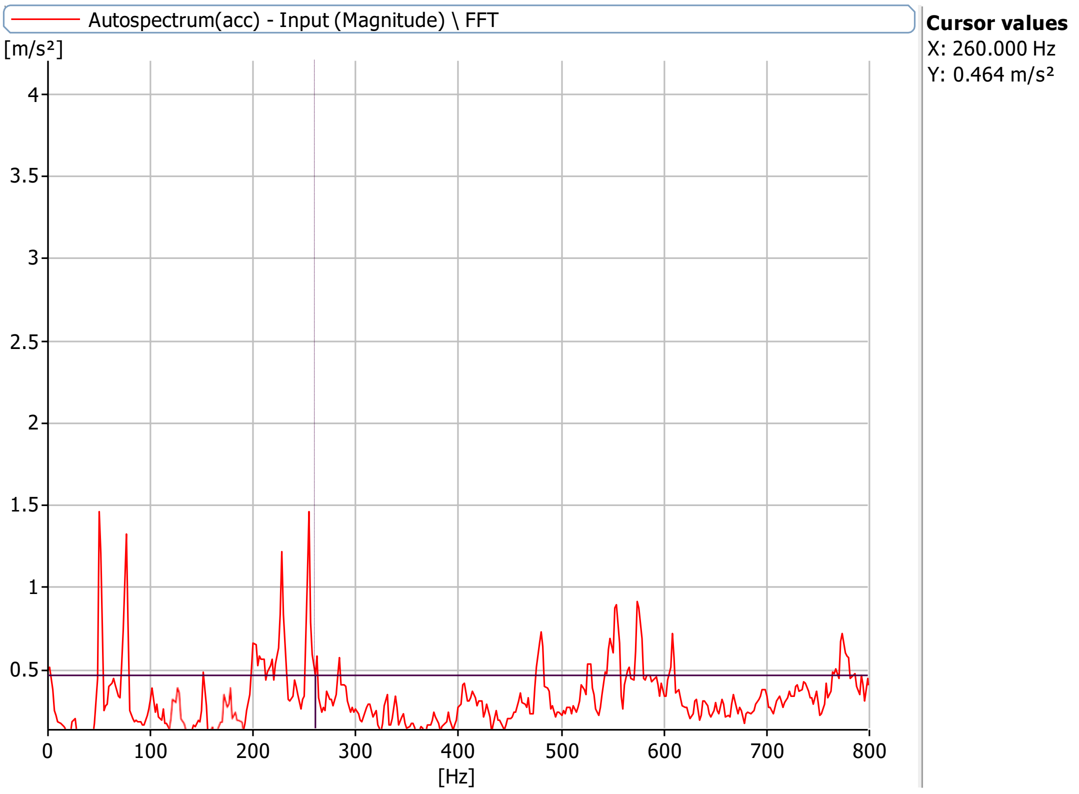


Figure 6.22.4: Frequency Spectrum at 1440 rpm with a defect size of 3.2 mm,5.1 mm with no load with 1mm misalignment at X Axis.

Frequencies at 720 rpm, 1000 rpm,1440 rpm and 2160 rpm for defect of 3.2 mm.8.2 mm bearing and 1 misalignment

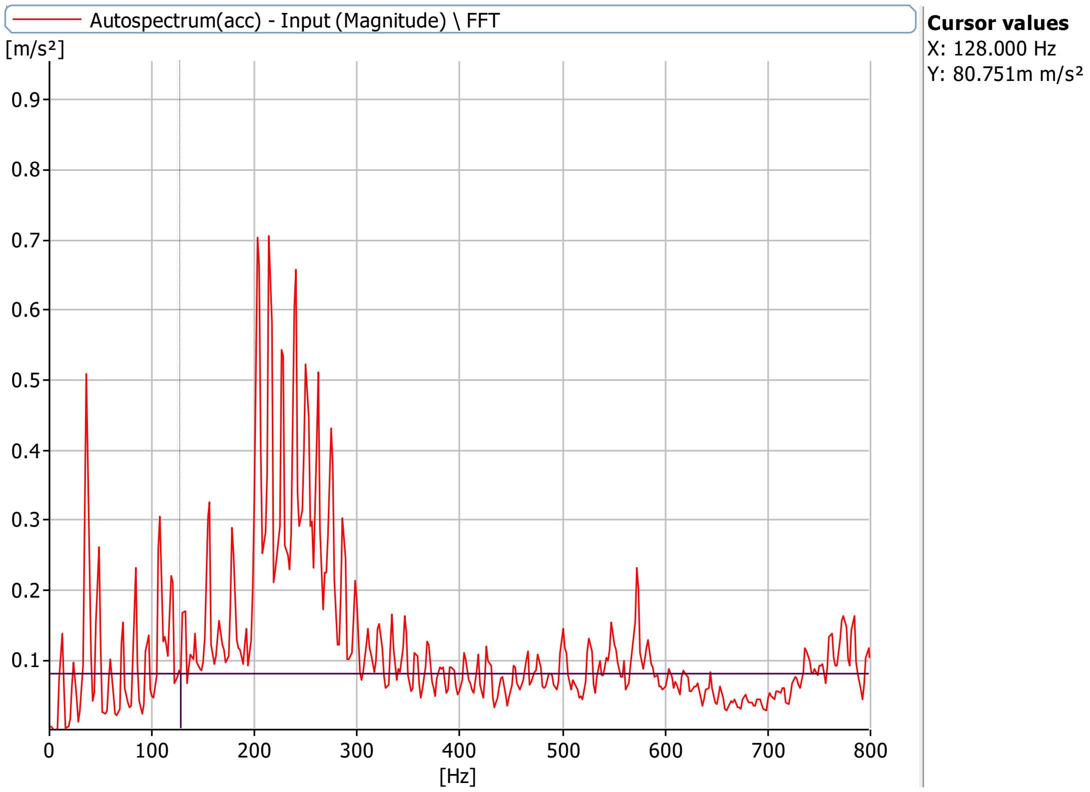


Figure 6.23.1: Frequency Spectrum at 720 rpm with a defect size of 3.2 mm,8.2 mm with no load with 1mm misalignment at X Axis

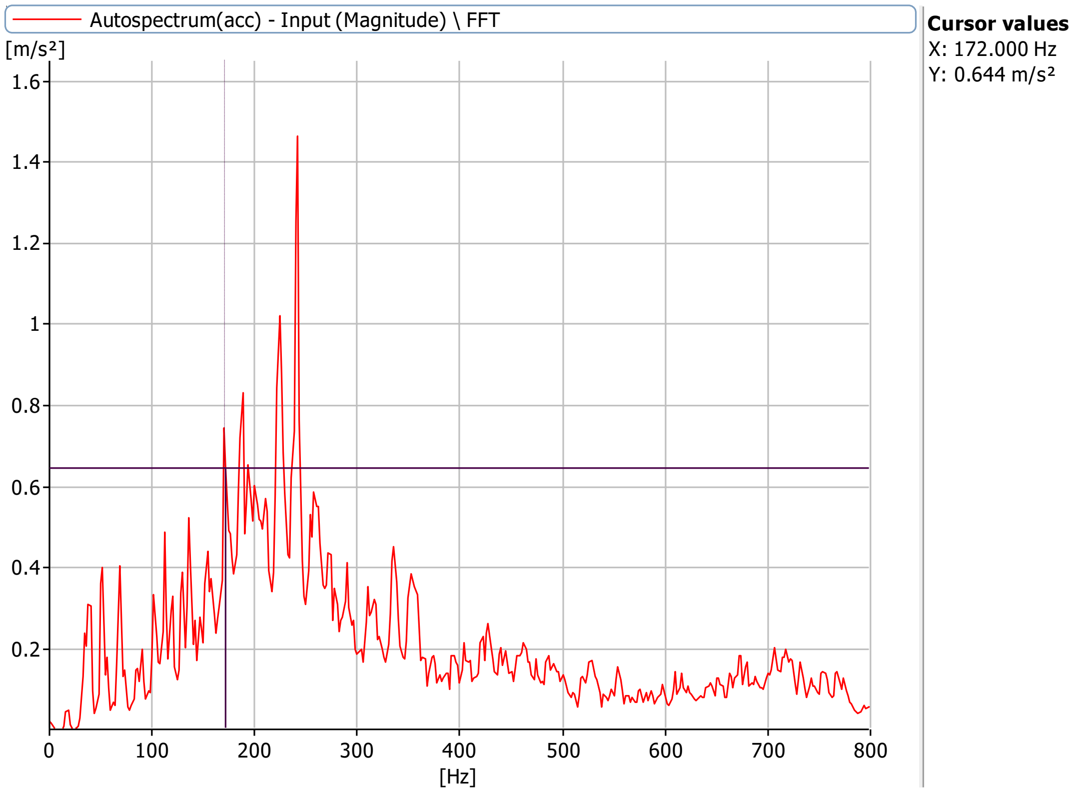


Figure 6.23.2: Frequency Spectrum at 1000 rpm with a defect size of 3.2 mm,8.2 mm with no load with 1mm misalignment at X Axis

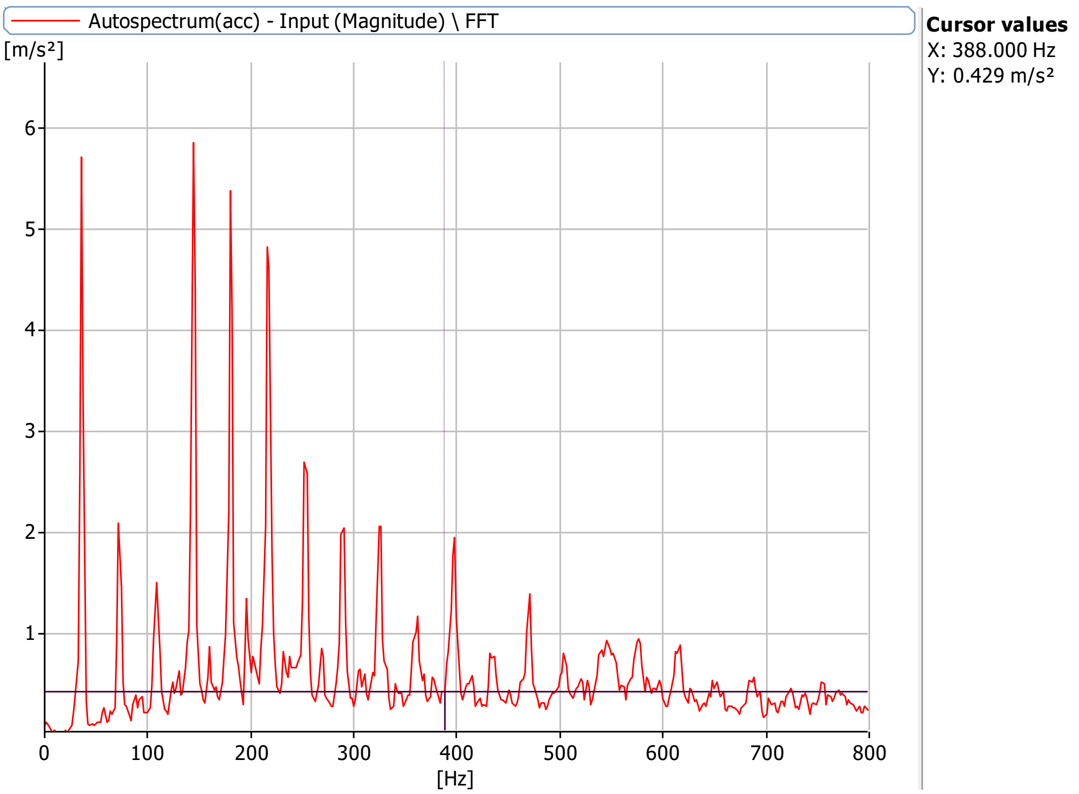


Figure 6.23.3: Frequency Spectrum at 2160 rpm with a defect size of 3.2 mm,8.2 mm with no load with 1mm misalignment at X Axis

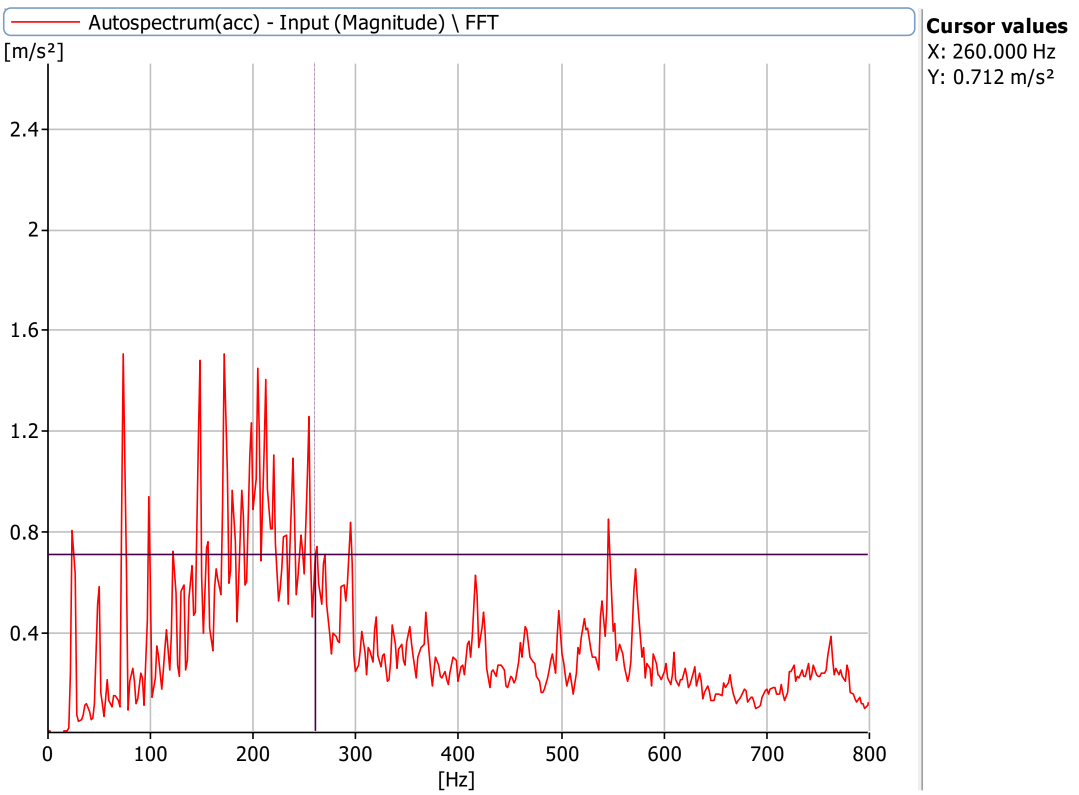


Figure 6.23.4: Frequency Spectrum at 1440 rpm with a defect size of 3.2 mm,8.2 mm with no load with 1mm misalignment at X Axis

Frequencies at 720 rpm, 1000 rpm,1440 rpm and 2160 rpm for defect of 5.1 mm bearing and 1mm misalignment

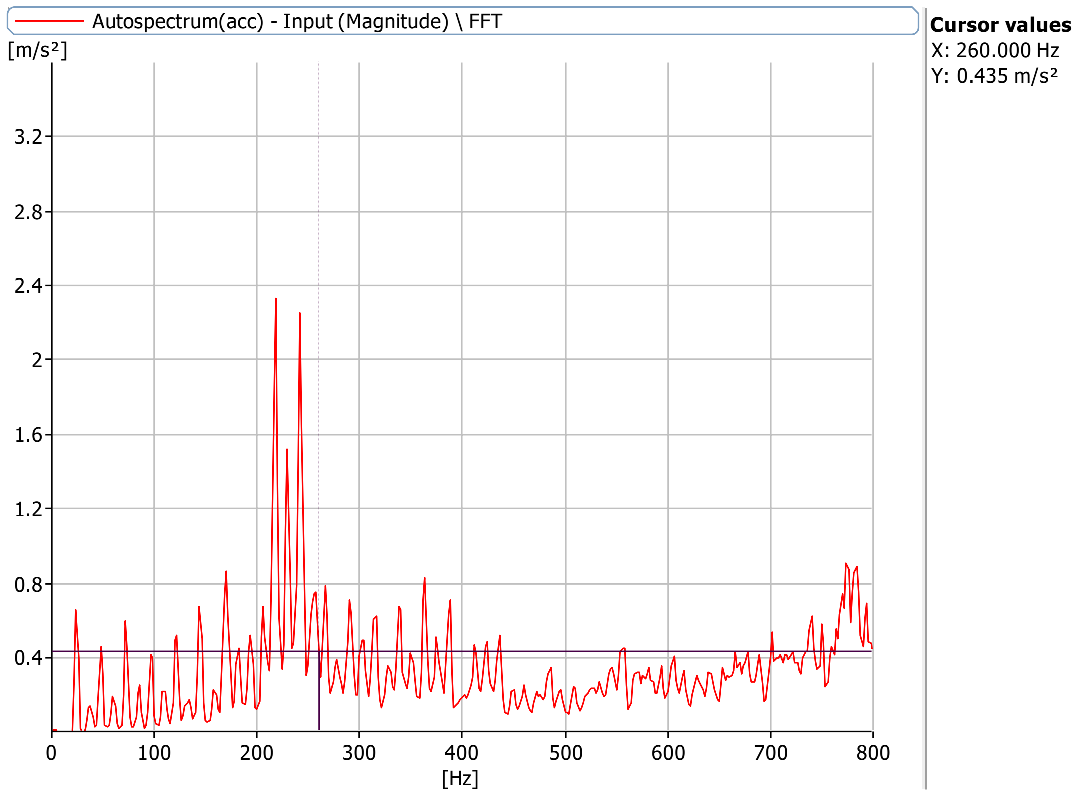


Figure 6.24.1: Frequency Spectrum at 1440 rpm with a defect size of 5.1 mm with no load with 1mm misalignment at X Axis

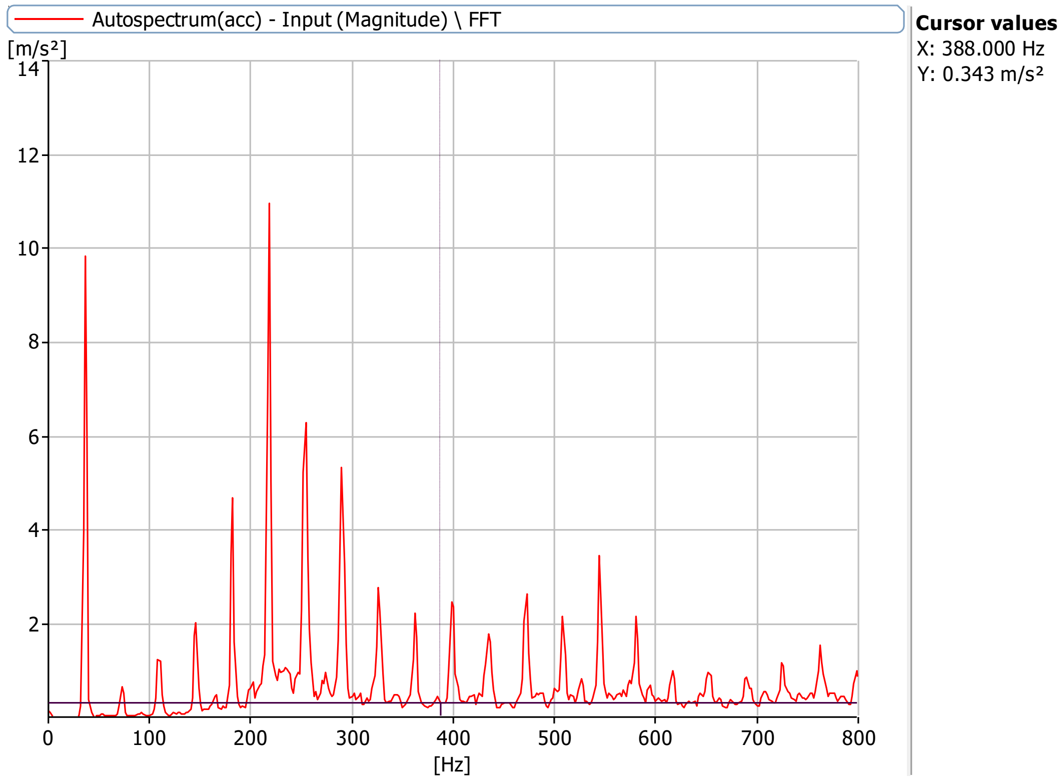


Figure 6.24.2: Frequency Spectrum at 1000 rpm with a defect size of 5.1 mm with no load with 1mm misalignment at X Axis

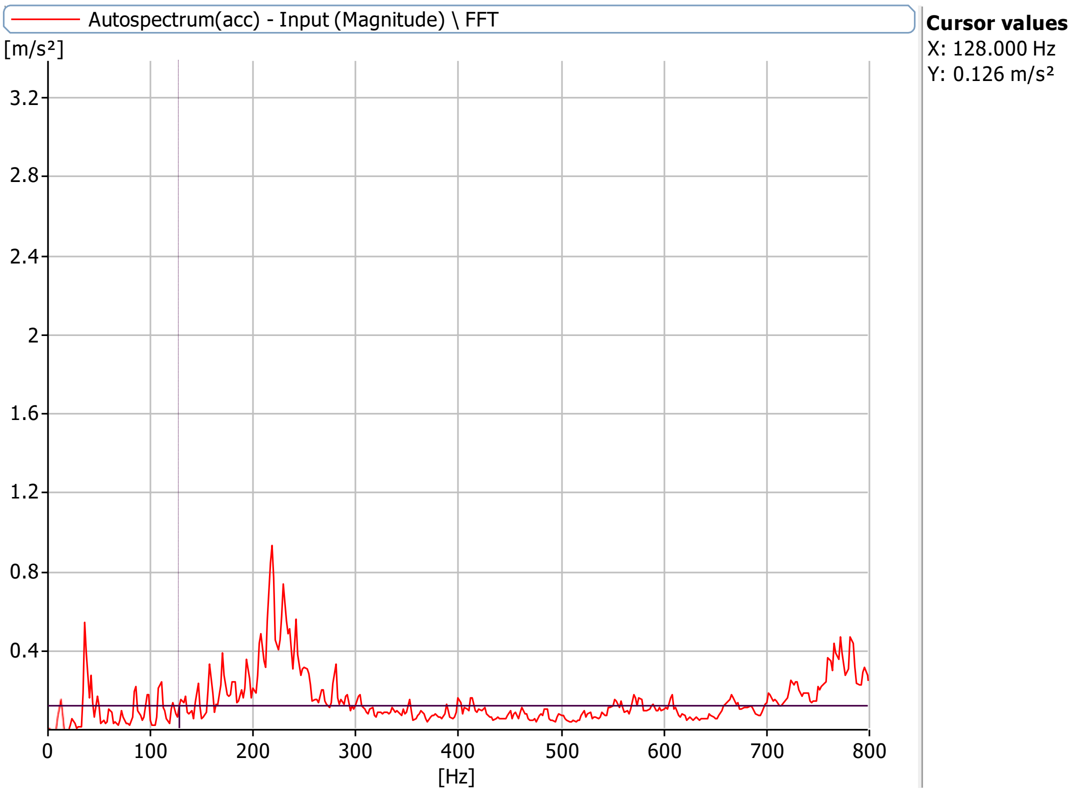


Figure 6.24.3: Frequency Spectrum at 2160 rpm with a defect size of 5.1 mm with no load with 1mm misalignment at X Axis

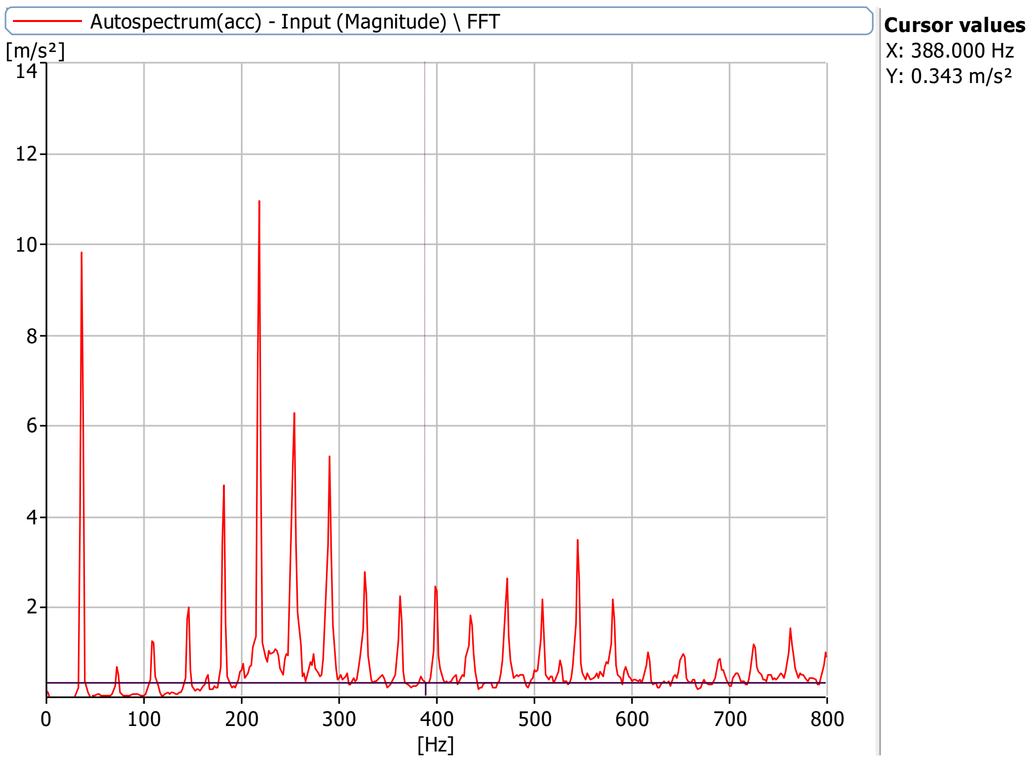


Figure 6.24.4: Frequency Spectrum at 720 rpm with a defect size of 5.1 mm with no load with 1mm misalignment at X Axis

Frequencies at 720 rpm, 1000 rpm,1440 rpm and 2160 rpm for defect of 8 mm bearing and 1mm misalignment

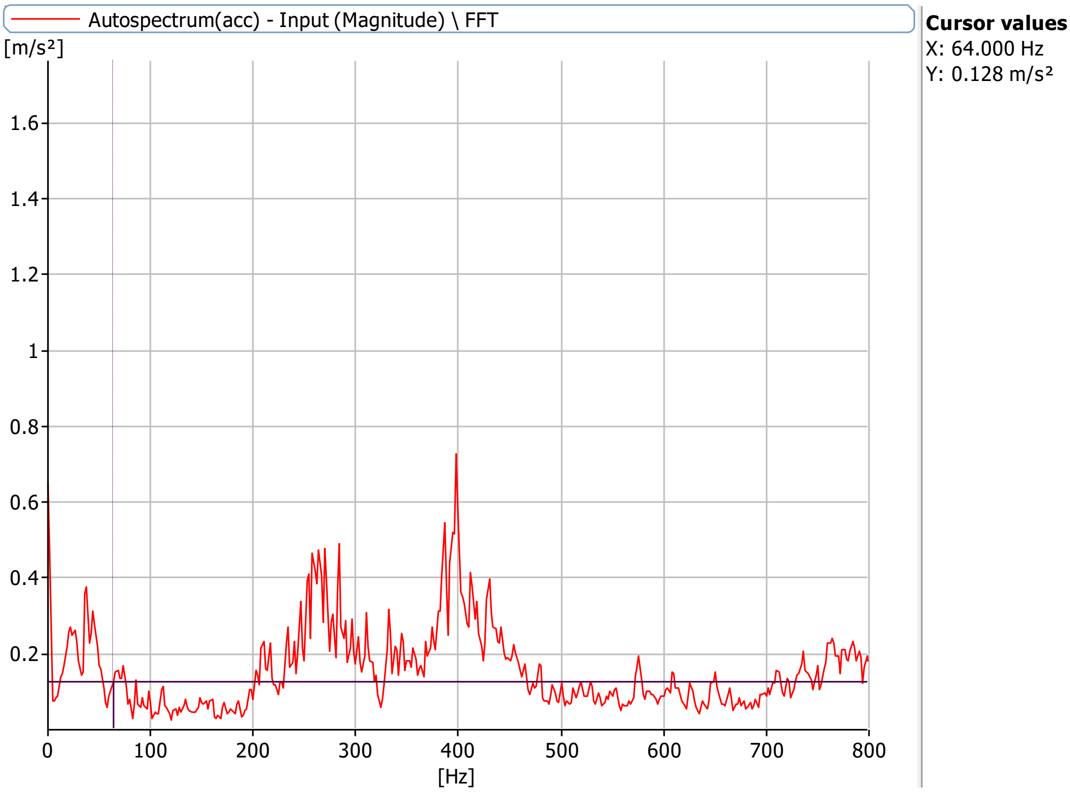


Figure 6.25.1: Frequency Spectrum at 720 rpm with a defect size of 8 mm with no load with 1mm misalignment at X Axis

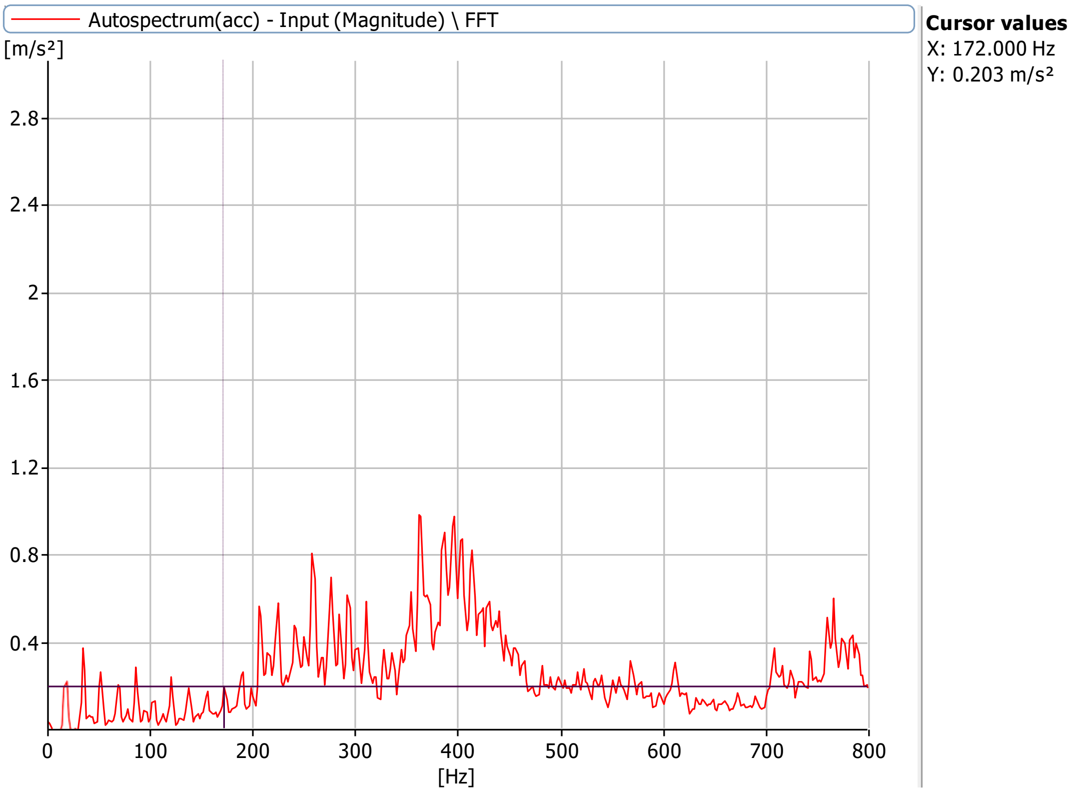


Figure 6.25.2: Frequency Spectrum at 1000 rpm with a defect size of 8 mm with no load with 1mm misalignment at X Axis

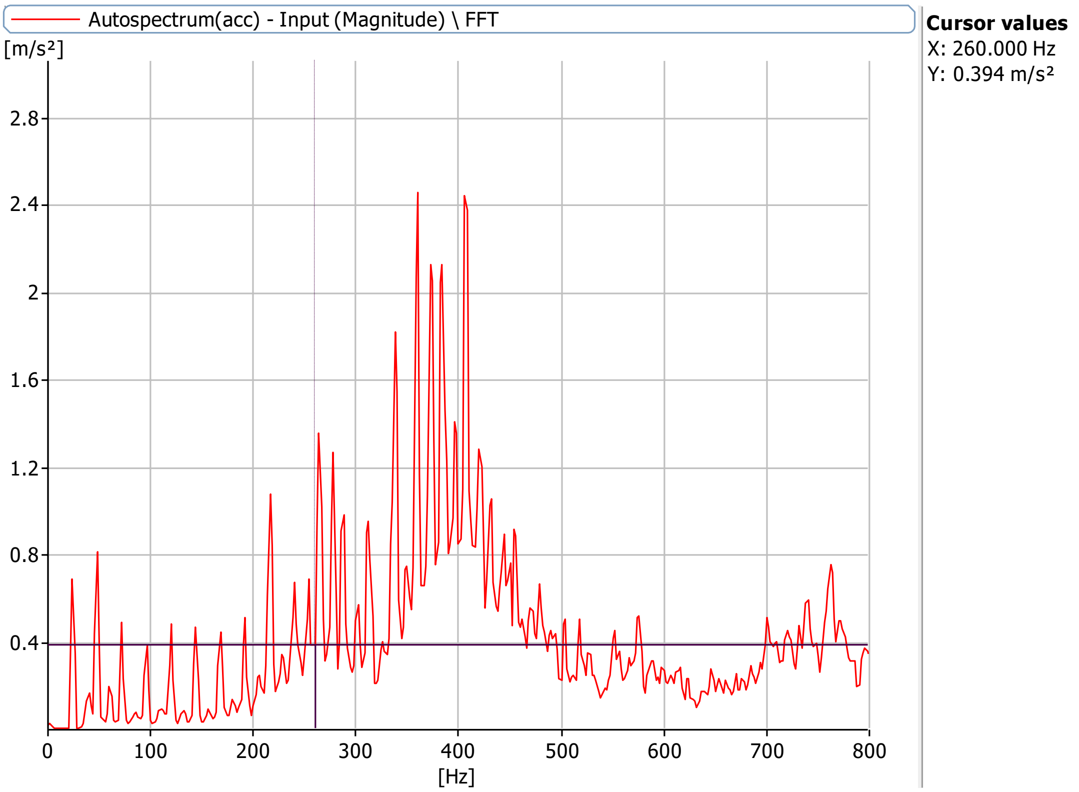


Figure 6.25.3: Frequency Spectrum at 1440 rpm with a defect size of 8 mm with no load with 1mm misalignment at X Axis

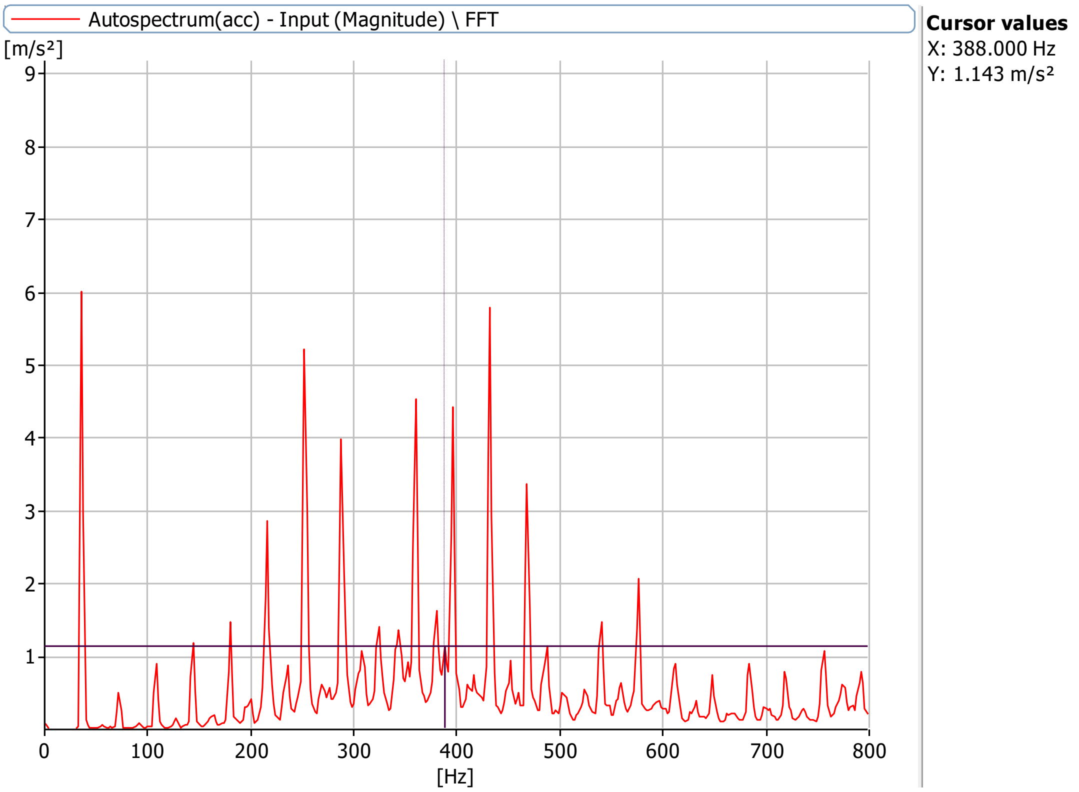


Figure 6.25.4: Frequency Spectrum at 2160 rpm with a defect size of 8 mm with no load with 1mm misalignment at X Axis

Closure

The work presented in this chapter may be summarized as follows.

1. Experimental setup and different defects are first discussed.
2. The bearing characteristics frequencies are computed for various defects.
3. The first study is on a healthy bearing of the rotor. The vibration spectra are recorded for various misalignment conditions and varying speeds for healthy bearing. It showed the peaks much smaller than the acceptable limits. This study enables to identify whether a bearing is healthy.
4. In the same manner study of five other bearings is conducted, each with different defects, and vibration spectra is obtained for an individual case.
5. The dataset is fed to a support vector machine which in turn predicts the defect for given defect frequency and amplitude.

CONCLUSION AND FUTURE SCOPE

Summary of the work

Support vector machine which is part of Machine learning is very useful technique for the detection of rotor unbalance and bearing defect. As it requires only vibration signature of that rotor bearing system which is used as the input data to the support vector machine hence defects can be detected without interrupting manufacturing or production process of rotary machine. The input data of vibration signature different combinations of bearing defect size, unbalance and speed are given to the SVM model training. Because of the variation of signature due to defect and misalignment it is possible to the support vector machine to detect the bearing defect and misalignment of unknown signature of that rotor bearing system.

After successful completion of the project, the conclusion drawn are as follows:

1. The experiment setup was developed and assembled successfully, demonstrating reliable performance.

2. Misalignment in bearings typically produces vibration amplitudes at 1X and 2X shaft rotational frequencies Slot or raceway defects in bearings produce characteristic frequencies, like BPFO, BPFI, and Cage Frequency, increasing with severity.

3. The amplitude at the defect frequency increases or decreases based on the defect size.
4. Experiments are designed and carried out with different parameter values, measuring the vibration amplitude in each one.

5. After training, the accuracies for SVM, DT and RF algorithms are found to be 98.43%, 97.12% and 96.76% respectively. After comparing accuracies of these three algorithms, it can be concluded that the SVM algorithm gives better accuracy than DT and RF algorithms hence SVMs can be preferred when there are complex data relationships.

Scope for Future Work

After seeing the outcome of the present project work. It is felt that further work may be done in the following directions.

1. This method we can use for experimental study of various rotor bearing system

2. We can also predict the faults in applications like Pump, Turbine by developing and training similar SVM model
3. Bearing with multiple defect and system with faults like misalignment is also scope for this project.
4. The input data can be increased with variation in rotor frequencies in order to increase the accuracy of the mode

Chapter 8

Reference

1. **Wang et al. (2018)** proposed a fault diagnosis method based on multi-scale permutation entropy (MSPE) and support vector machine (SVM). The authors applied MSPE to analyse vibrationsignals and extract informative features at different scales. These features were then fed intoan SVM classifier for fault diagnosis. The experimental results demonstrated the effectiveness of the proposed method in accurately identifying and classifying outer race defects.
2. **Li et al. (2019)** presented an intelligent fault diagnosis method for deep groove ball bearings based on time-frequency image analysis and extreme learning machine (ELM). The authors utilized the time-frequency image of vibration signals and employed ELM as the classification model.The study showed promising results in achieving high accuracy in diagnosing outer race defects.
3. **Li and Lim (2018)** proposed an enhanced fault diagnosis method for deep groove ball bearings based on kernel Fisher discriminant analysis (KFDA) and ensemble empirical mode decomposition (EEMD). The authors used KFDA to reduce the dimensionality of the extracted features obtained from EEMD. The reduced features were then employed for fault classification. The study demonstrated the effectiveness of the proposed method in achieving improved fault diagnosis performance for outer race defects.
4. **Yong Eun Lee, Bok-Kyung Kim, Jun-Hee Bae, Kyung Chun Kim (2021)**
Fault diagnosis For rotating machinery is important for reliability and safety, and shaft misalignment is one of the most \common causes of mechanical vibration or shaft failure. In this study, we detected defects caused by misalignment of the shaft axis in rotating machinery using an artificial intelligence machine learning technique for fault recognition.
5. **Youcef Khodja, A., Guersi, N., Saadi, M. N., & Boutasseta, N. (2020).**
In this paper, we propose a novel method for the classification of bearing faults using a convolutional neural network (CNN) and vibration spectrum imaging (VSI).The normalized amplitudes of the spectralcontent extracted from segmented temporal vibratory signals using a time-moving segmentation window are transformed into spectral images for training and testing of the CNN classifier.
6. **N. Bachschmid P. Peennacchi and A. Vania *et al*** identified the different types of faults in the rotor shaft and analyzed by frequency domain and also by numerical simulation.
7. **Pennacchi et al.** gave the use of a modal model for fault identification in the large rotating machinery.

8.Lees et al. estimated the rotor unbalance and misalignment which is mainly focused on the parallelland angular misalignment.

9.Kumbhar S. G., et al present a study on defects in taper roller bearing i.e rolling element bearingbyusing vibration signature analysis. It further explains the types of defects and causes of defects. They carried out experimental as well as a theoretical study on defected bearings.

10.Desavale R. G., et al has presented an effective empirical model for fault diagnosis of roller bearing by using the Buckingham p-theorem using FLT0 system has been derived.

11.Desavale R. G., et al presented an experimental and numerical study on spherical roller bearingby using the multivariable regression analysis method.

12.Patel V. N., et al have explained the nature of vibrations generated by rolling element bearing having multiple localized defects on bearing races. The study was carried out in Time and Frequency Domain. They also performed the experimental study on vibration behaviors of deep groove ball bearings under dynamic radial load for local defects.

13. Saruhan H., et al explained Vibration Analysis of Rolling Element Bearings Defects Sham Kulkarni and Wadkar S.B. performed Experimental Investigation for Distributed Defects in Ball Bearing using Vibration Signature Analysis.

14. Shrikant S. N., et al. [10] performed an Experimental Investigation of Rolling Element Bearing withand without Unbalanced Rotor by using an FFT analyzer.

15.Sakshi Kokil, et al. [11] performed an experimental study on the detection of a fault in rolling element bearing by using condition monitoring methodology. It also explains the numerical methodology for the same.

16.H. Arslan and N. Aktürk et. Al [12] carried out an Investigation of vibration in Rolling Element bearings caused due to local defects on different parts of bearing. They experimented on angular contact ball bearing with and without defects to obtain some mathematical relations.

17.Zeki Kiral and Hira Karagulle et al [13] performed a Vibration analysis of rolling element bearingswith various defects under the action of an unbalanced force created during the working of the system. They performed the finite element vibration analysis on roller element bearings for theoretical results.

18.Desavale et al [14] presented the MVRA and ANN model to describe the vibration characteristics of rolling element bearings. The vibration signals collected from rolling element bearings were usedto demonstrate the proposed EDBM method compared with the MVRA. The results showed the effectiveness of the EDBM method in extracting fault characteristics and diagnosing faults of rollingelement bearings. Good correlations between EDBM, MVRA, and ANN have been observed.

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